Vehicle Crash Reconstruction: Principles and Technology - Volume 2

Instructors: Neal Carter & Nathan Rose

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Vehicle Crash Reconstruction: Principles and Technology

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MODULE 5 - EVENT DATA RECORDERS

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Module 5 – Event Data Recorders

Vehicle Crash Reconstruction: Principles and Technology
SAE International Course #1728

Neal Carter
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Passenger Vehicle EDRs

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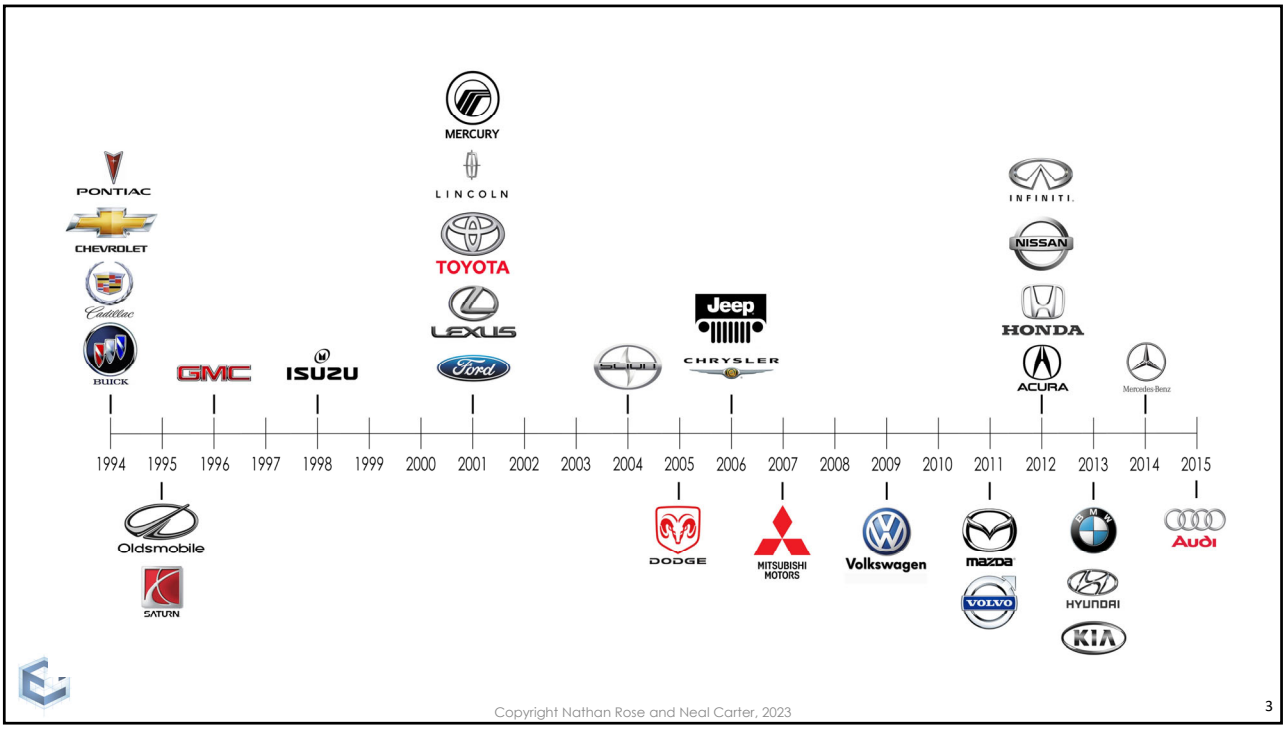

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Acronyms

- EDR = Event Data Recorder
- ACM = Airbag Control Module
- SDM = Sensing and Diagnostic Module (GM)
- RCM = Restraint Control Module (Ford)
- PCM = Powertrain Control Module
- CDR = Crash Data Retrieval
- SNA = Signal Not Available

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Passenger Vehicle Download Equipment

Bosch CDR

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GIT

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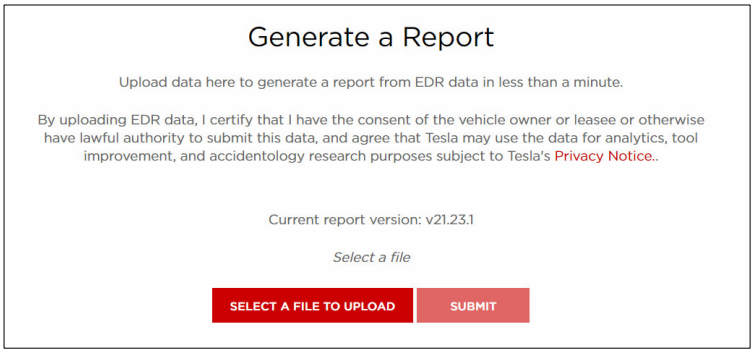
Tesla Translation

https://edr.tesla.com/



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Accessing Electronic Crash Data - Methods

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DLC

- Electrical system in working condition
- Key/fob present
  - Some vehicles do not require key
- Power

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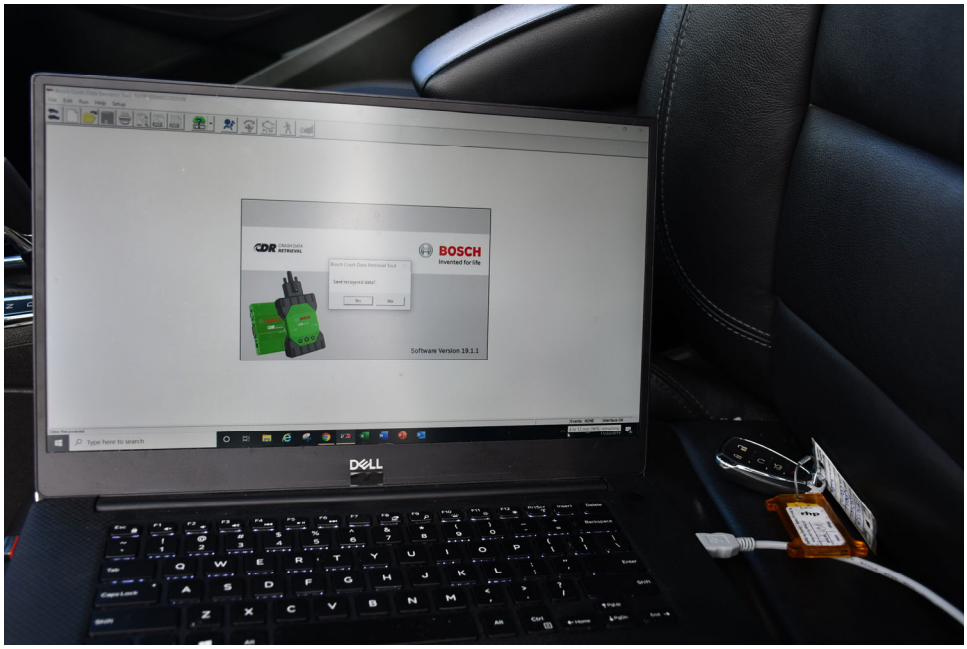

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Back-Powering

- Key missing?
- No power?
- Minor electrical issue?

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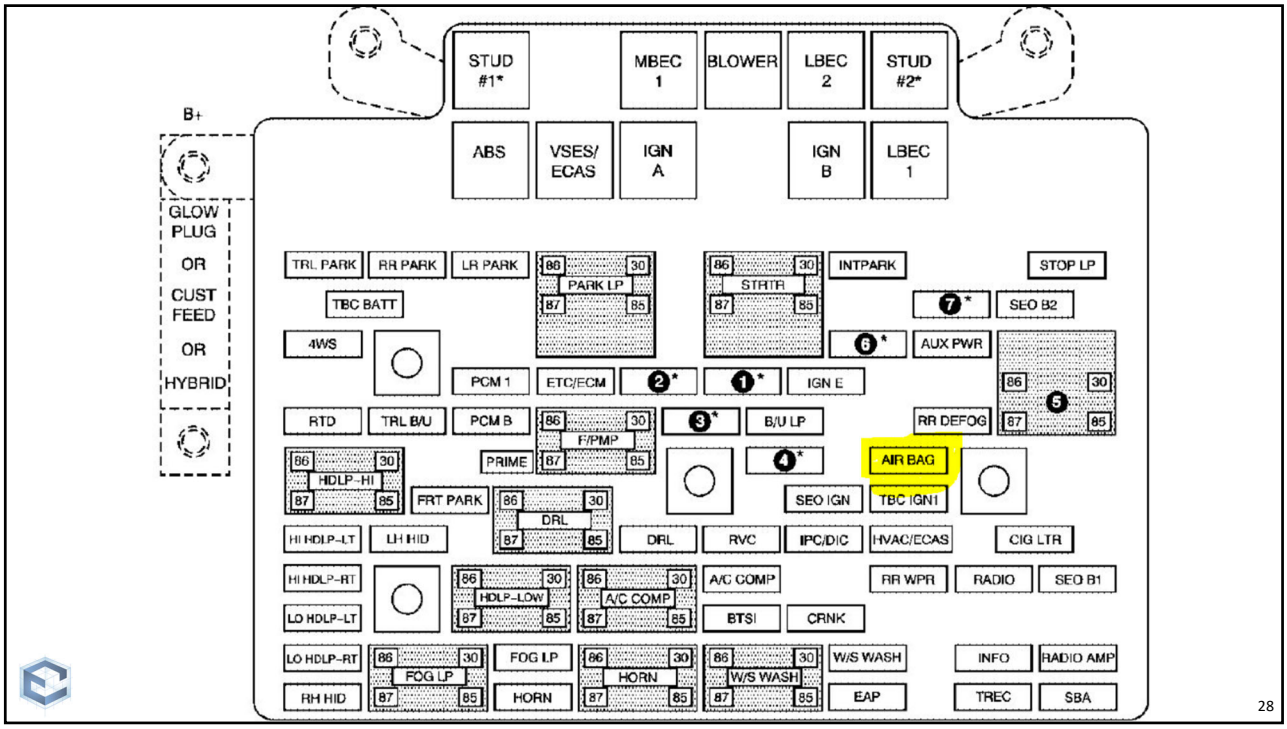

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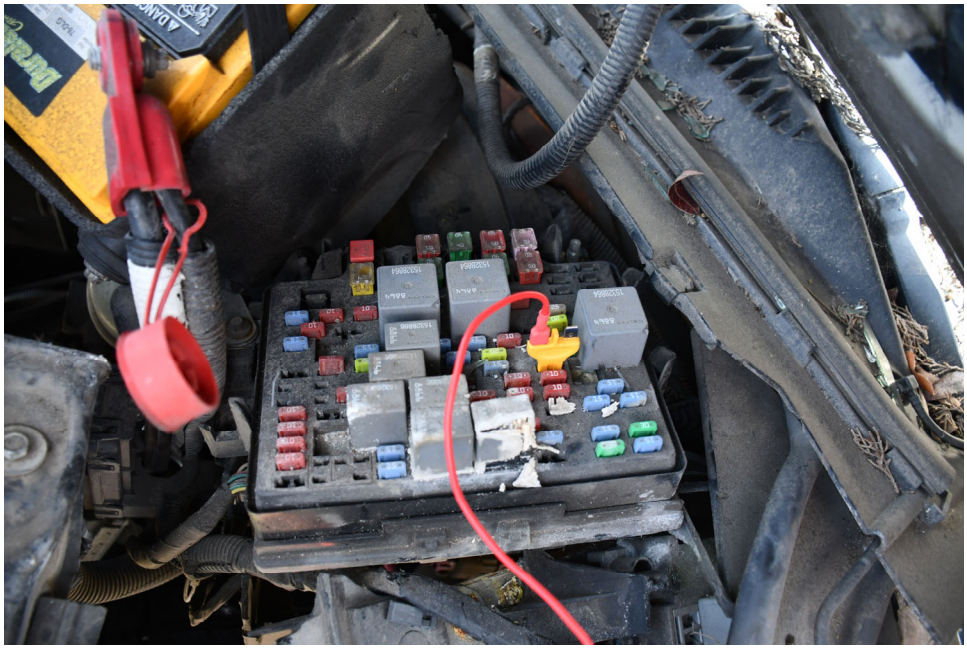

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Direct-to-Module

- Vehicle is too damaged
- Electrical system is compromised
- Unable to back-power or download through the DLC

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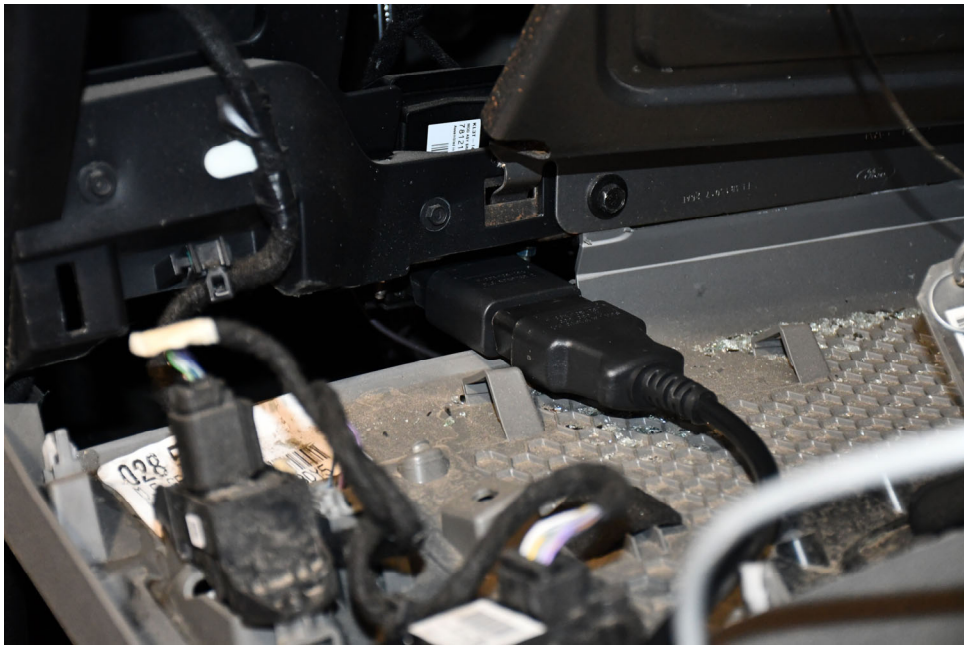

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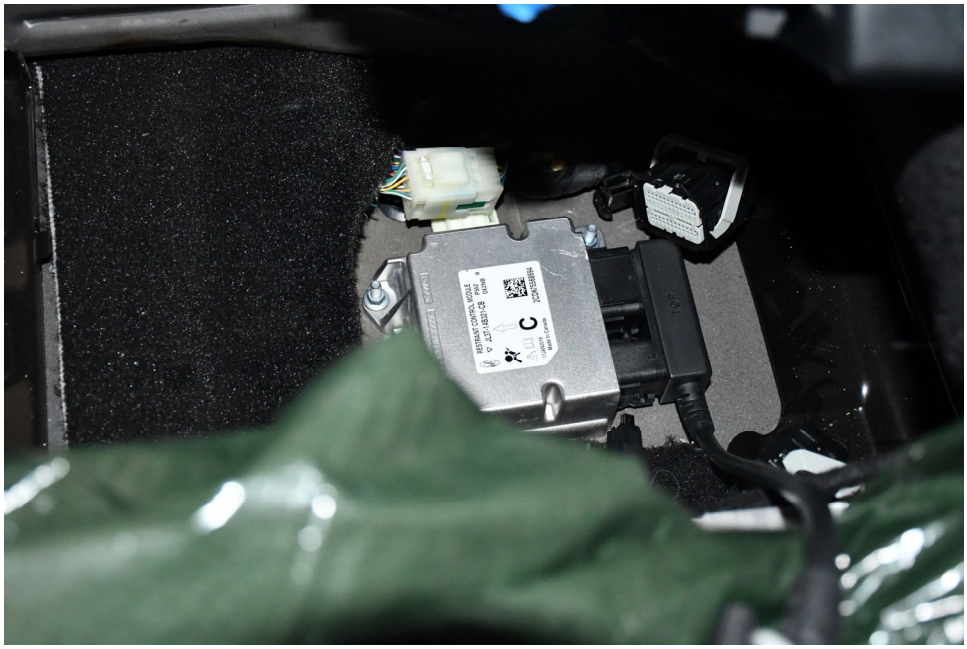

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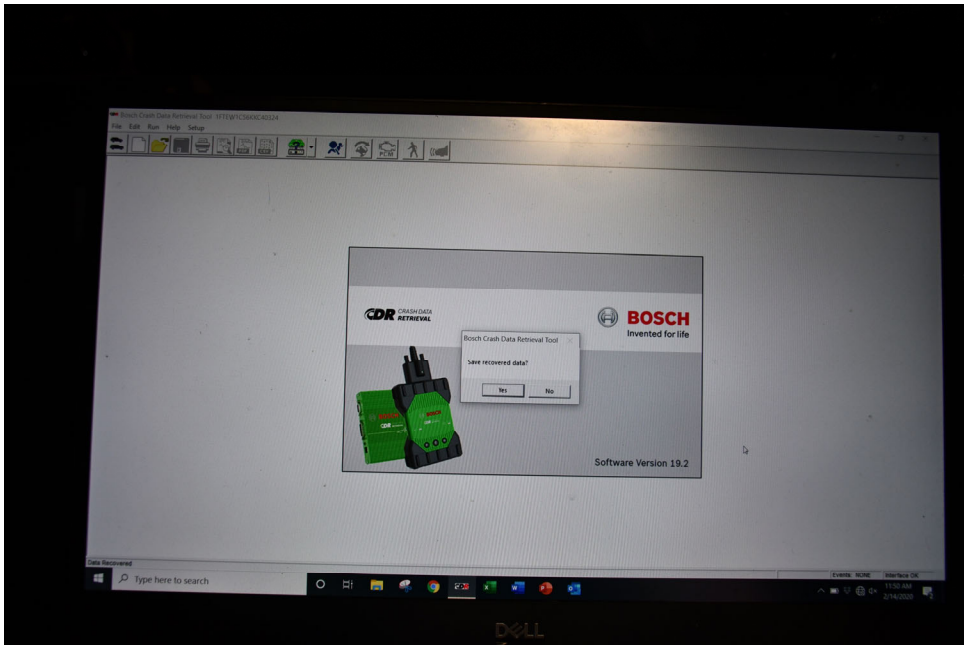

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Module Too Damaged?

- Chip swap?
  - First, is a chip swap necessary?
  - Back-probe?

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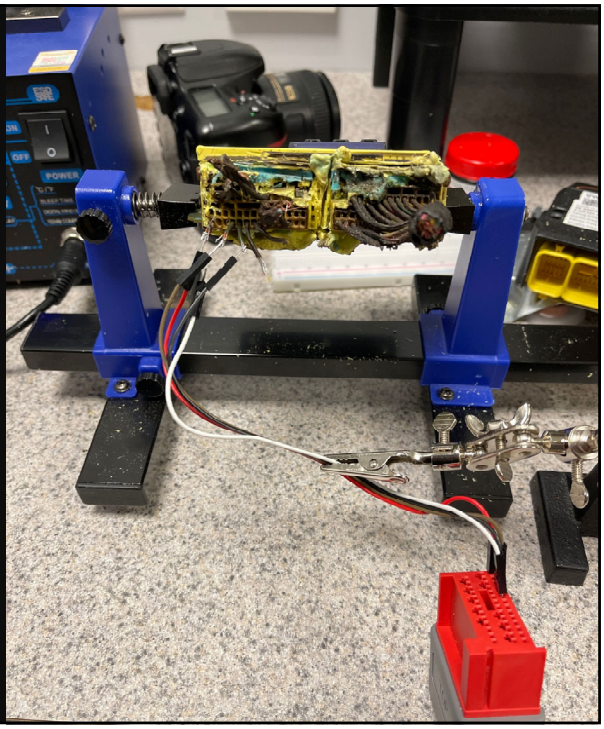
Back-Probe



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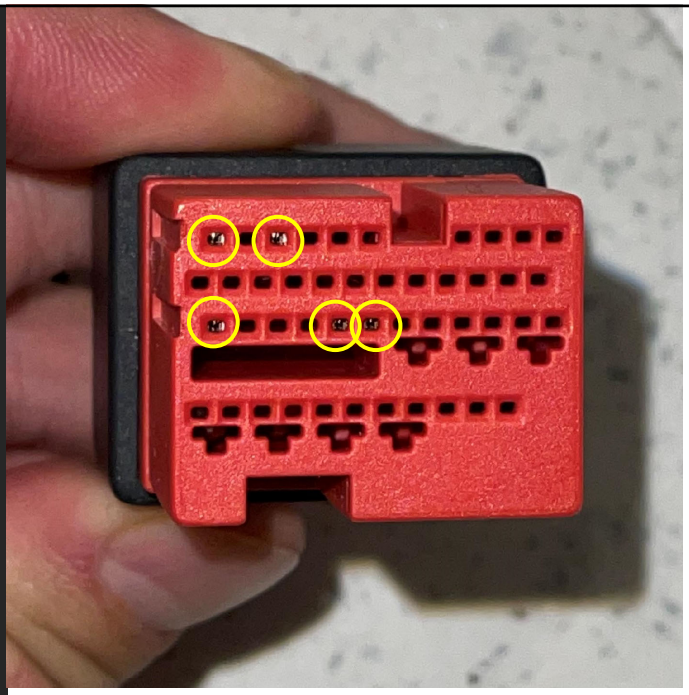
Back-Probe



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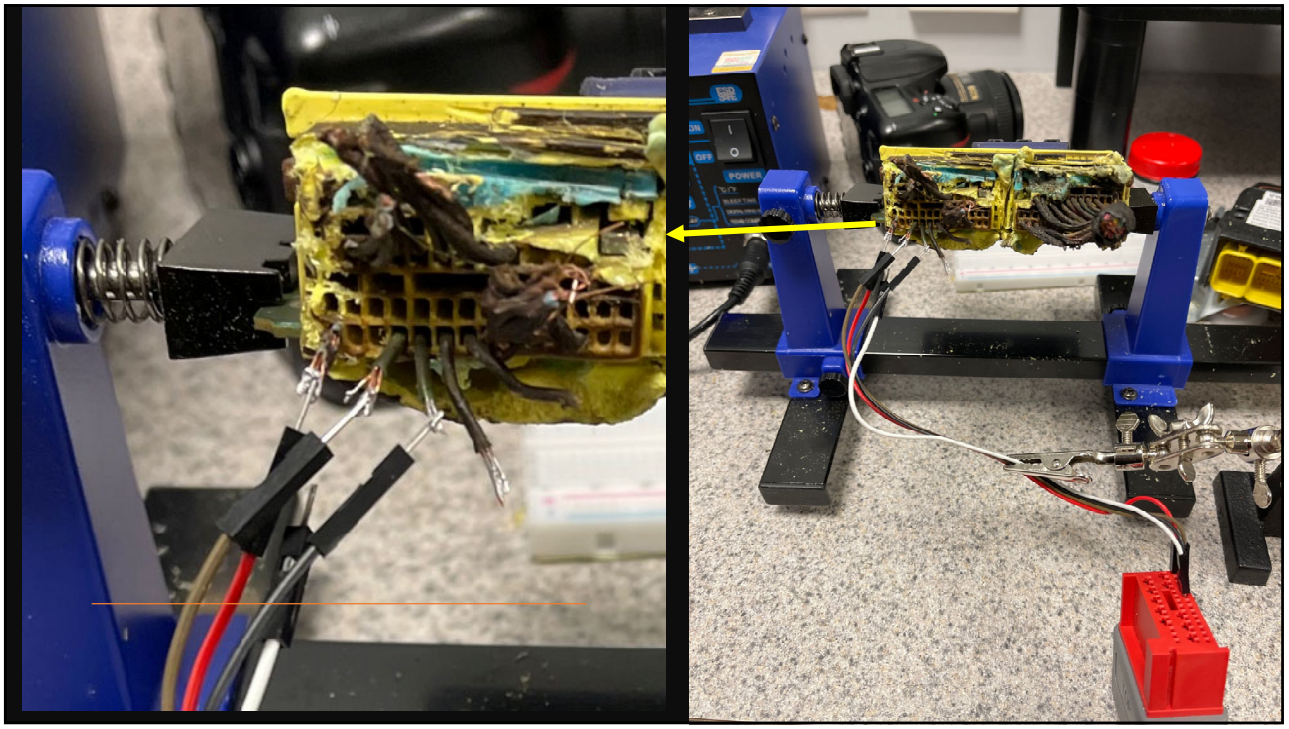

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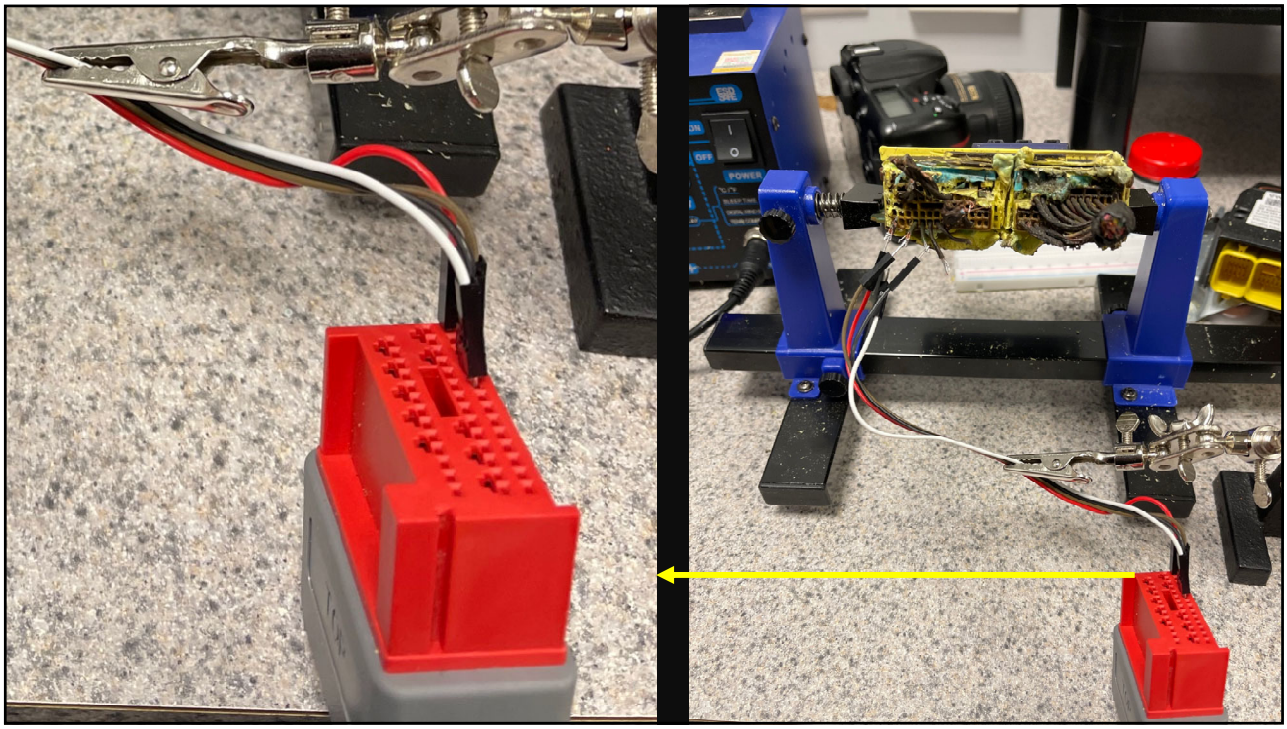

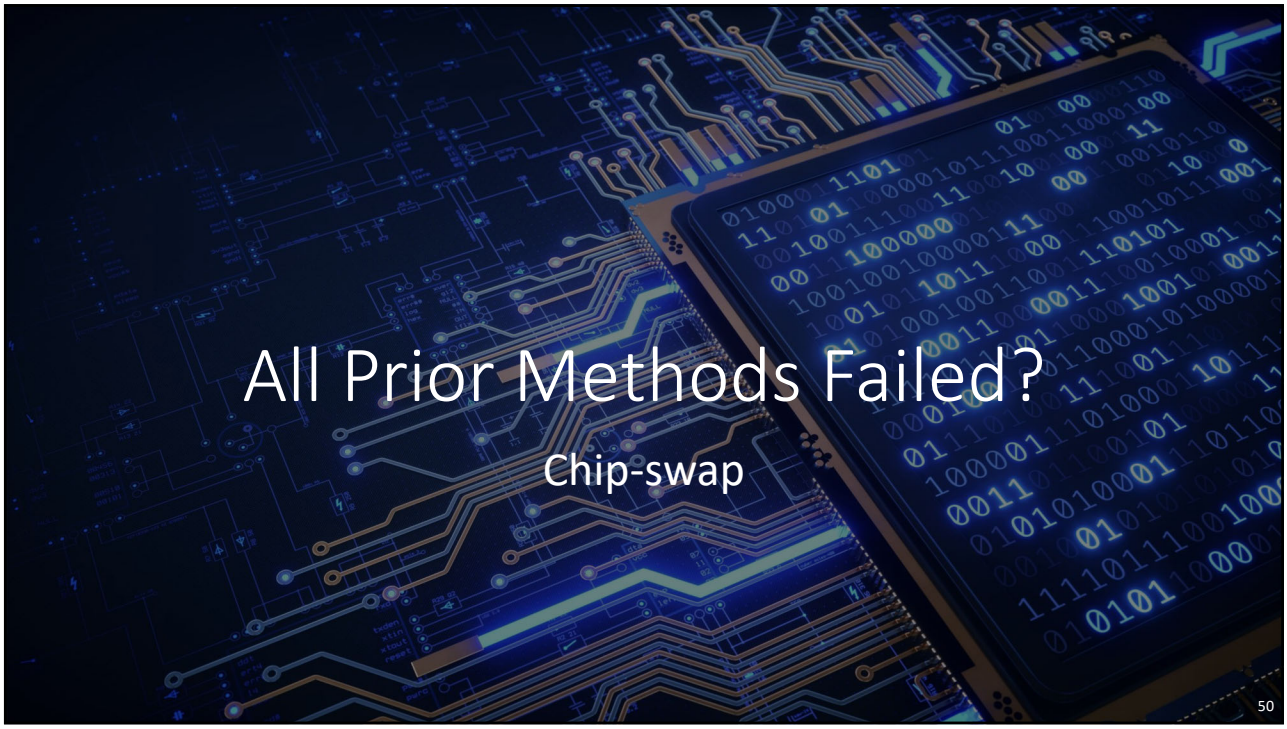

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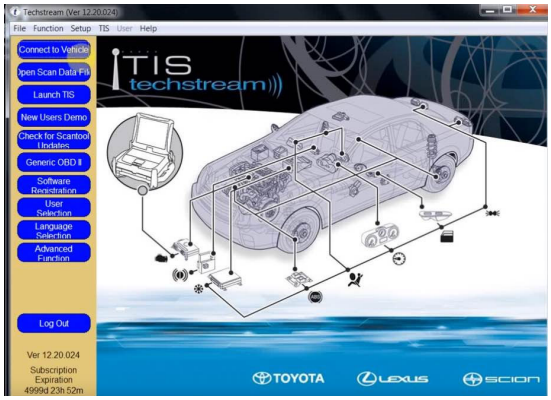
Toyota Techstream

- Debut 2013+ (Rav 4)
- Vehicle Control History (VCH)
  - 26 triggers
  - Pre-Collision System (PCS)
- SAE 2019-01-0632
- SAE Accident Reconstruction, the Autonomous Vehicle and ADAS



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Active Safety Control Module (GM)

- ASCM
  - Alerts and driver assistance
  - Can be triggered by hard braking, or
  - Airbag deployment
  - Data can also be recorded such as auto-braking
  - Data may not be recorded if a power loss occurs
  - Check CDR Supported Vehicle file for RPO and required cable, if DTM

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Subaru EyeSight

- Stereo cameras
- Driver Assist Technology
- 22 seconds of recording
- Subaru must perform extraction
  - Requires owners consent

www.Subaru.com

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Subaru EyeSight

- https://www.subaru.com/engineering/eyesight.html

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Motorcycle EDRs

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Can-Am Spyder

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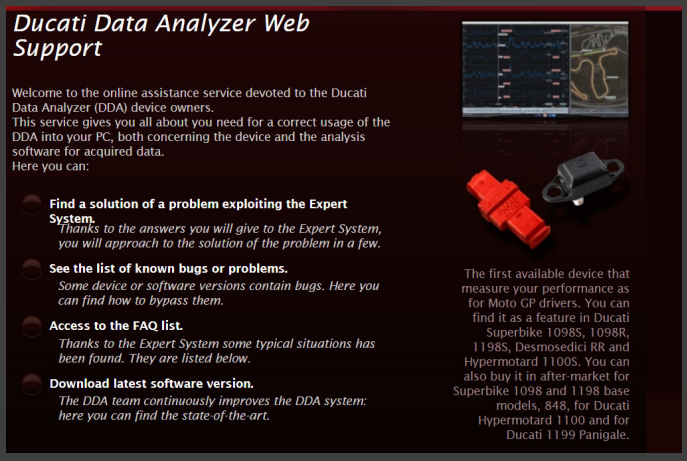
Ducati Data Analyzer

http://dda.prosa.com/



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Kawasaki

An Analysis of EDR Data in Kawasaki Ninja 300 (EX300) Motorcycles

2017-01-1436
Published 03/28/2017

Edward Fatzinger and Jon Landerville
Momentum Engineering Corp.

CITATION: Fatzinger, E. and Landerville, J., "An Analysis of EDR Data in Kawasaki Ninja 300 (EX300) Motorcycles," SAE Technical Paper 2017-01-1436, 2017, doi:10.4271/2017-01-1436.

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Kawasaki

- Limited coverage beginning 2013
- Denso ECUs
- Three event storage
- Power loss can interrupt events from being recorded
- Check owner's manual

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Harley Davidson

- Beginning in 2014
- Boom! Box Infotainment System
- 6.5gt version
- "Off" by default

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Pre-Crash Data

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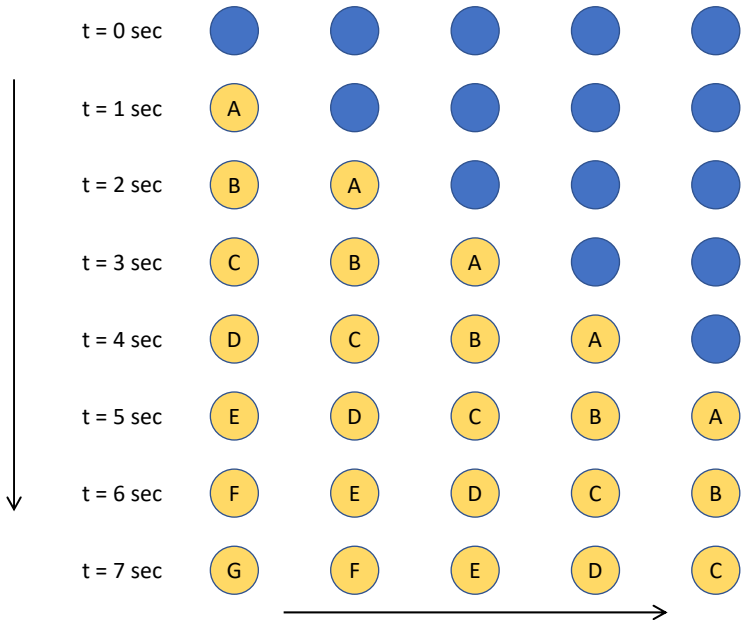
Pre-Crash Data Elements Flowing Into and Through the Buffer



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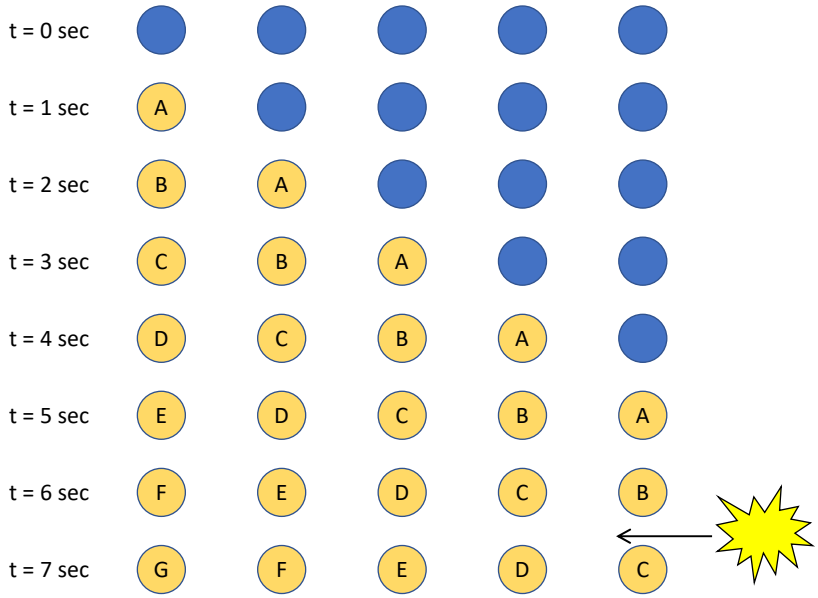

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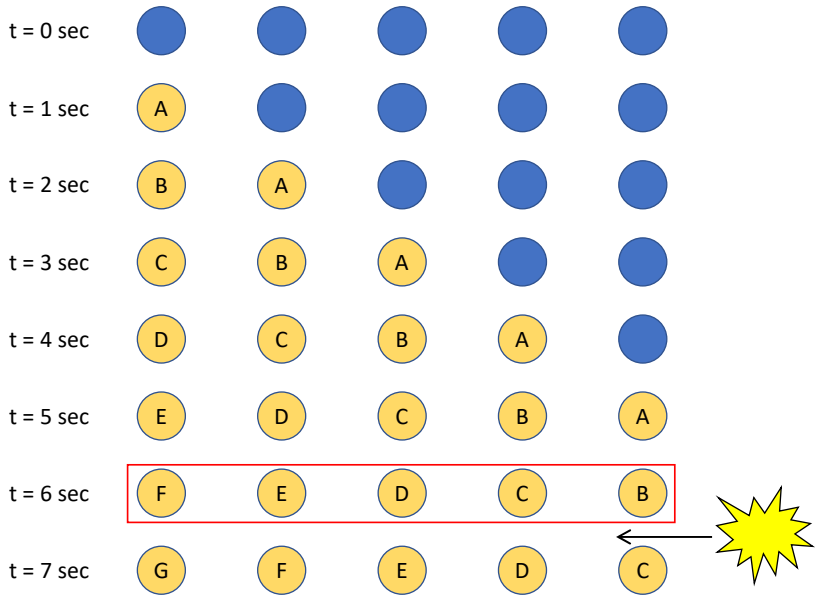

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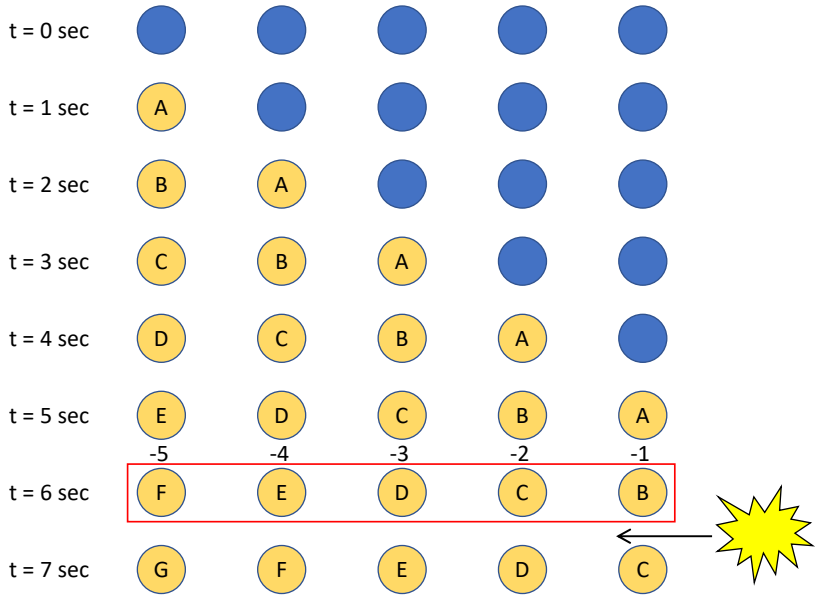

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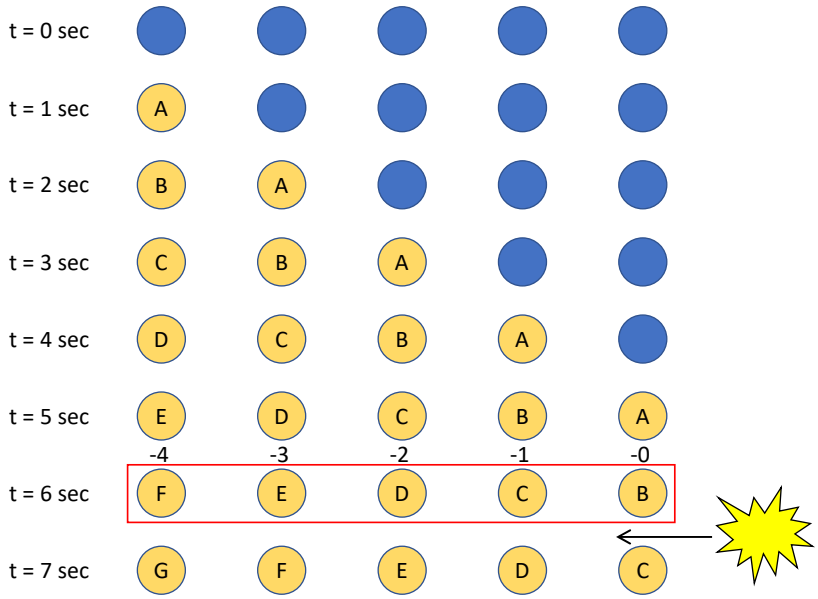
Pre-Crash Data Elements Flowing Into and Through the Buffer



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Principle

There is typically uncertainty related to the exact timing of the last set of pre-crash values. They could have been recorded immediately prior to the collision, or as much as one sample period prior to the collision. This uncertainty can be accounted for within your reconstruction.

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Types of Errors in EDR Data

Random – These errors are caused by unknown and unpredictable changes in the experiment. These changes may occur in the measuring instruments or in the environmental conditions. These errors affect PRECISION. These cannot usually be eliminated, but perhaps can be minimized.

Systematic – These errors are usually caused by a problem with the measuring instrument (i.e., an accelerometer or steering angle sensor with a limited full-scale value) or an error in the use of the instrument. These errors affect ACCURACY. These can be identified and eliminated, potentially.

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Types of Errors in EDR Data

Accurate, but Not Applicable Measurements – These are not actually errors but sometimes are treated as errors in the accident reconstruction literature. An example would be when a reconstructionist wants the overall speed of the vehicle, but the EDR is reporting only the longitudinal speed of the vehicle.

Clearly Anomalous Errors – These are reported values that are obviously wrong but may or may not have a clear physical explanation and may or may not be correctible.

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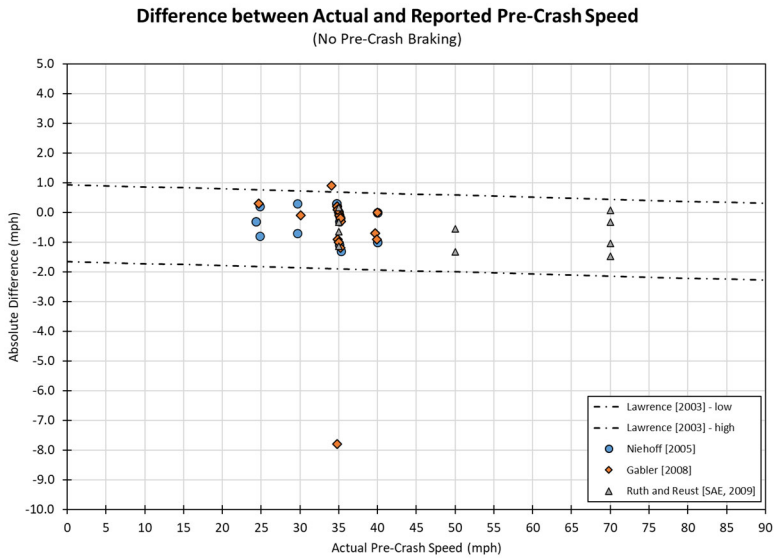
Random Errors



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Potential Sources of "Error" in the EDR-Reported Pre-Crash Speeds

1. Longitudinal tire slip due to acceleration or braking.
2. Lateral slip due to yawing of the vehicle.
3. A difference between the rolling radius of the tires on the vehicle and the tires for which the speed sensor is calibrated.
4. A difference between the final drive ratio on the vehicle and the drive ratio for which the speed sensor is calibrated.

These are not measurement errors, nor are they systematic errors.

They are differences between what the sensor is measuring and the numerical value you are wanting as an accident reconstructionist.

To "correct" these "errors," we need models that help us relate what the sensor is measuring to the value we are interested in.

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Longitudinal tire slip due to acceleration or braking

$$s_{long} = \frac{V - r\omega}{V} = \frac{V - V_{rolling}}{V}$$

In this equation, V is the travel speed of the wheel center which, for a heavy braking scenario in which the vehicle continues traveling in a straight line, would coincide with the over-the-ground travel speed of the vehicle. The variable r is the effective rolling radius of the tire and ω is the angular velocity of the wheel. The quantity $r\omega$ is the rolling speed of the tire ($V_{rolling}$). This equation can be rearranged for application to EDR data, as follows:

$$V = V_{rolling}\,\frac{1}{1 - s_{long}}$$

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Lateral slip due to yawing of the vehicle

$$S = \frac{S_{reported}}{\cos \beta}$$

In this equation, β is the vehicle sideslip angle, $S_{reported}$ is the EDR-reported speed, and S is the vehicle's over-the-ground speed.

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A difference between the rolling radius of the tires on the vehicle and the tires for which the speed sensor is calibrated

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NHTSA Test #4955

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NHTSA Test #4955

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NHTSA Test #4955

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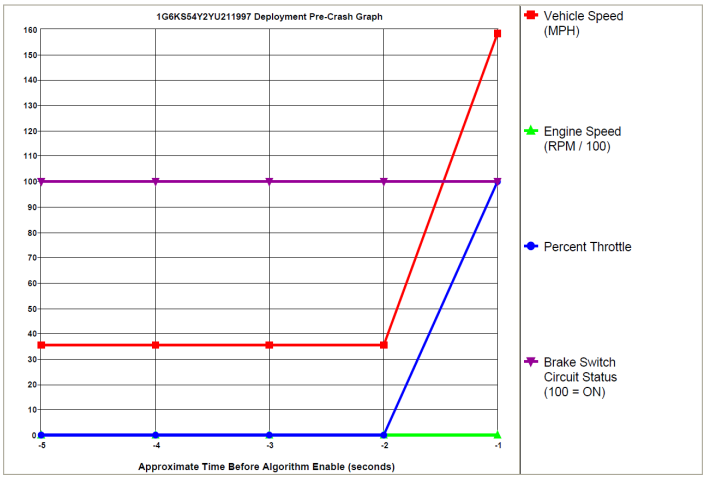
NHTSA Test #4955



Seconds Before AE	Vehicle Speed (MPH)	Engine Speed (RPM)	Percent Throttle	Brake Switch Circuit Status
-5	35	0	0	ON
-4	35	0	0	ON
-3	35	0	0	ON
-2	35	0	0	ON
-1	158	0	100	ON

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Crash Data

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Potential Sources of Error in the EDR-Reported ΔV

1. A ΔV can occur prior to the threshold acceleration being reached.
2. The recording window could be too short.
3. The recording window could be too long.
4. There could be accelerations that exceed the capabilities of the accelerometer in the airbag control module (ACM).
5. The ACM may reside somewhere other than the center of gravity (CG) and experience a different ΔV than the CG.
6. Damage to or displacement of the ACM could occur during the collision.
7. There could be power loss to the ACM prior to the collision being complete.
8. The ACM may reside somewhere other than the center of gravity (CG) and experience a different ΔV than the CG.
9. Tire forces can influence the ΔV measured by the ACM.

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Potential Sources of Error in the EDR-Reported ΔV

1. [Systematic] A ΔV can occur prior to the threshold acceleration being reached.
2. [Systematic] The recording window could be too short.
3. [Systematic] The recording window could be too long.
4. [Systematic] There could be accelerations that exceed the capabilities of the accelerometer in the airbag control module (ACM).
5. [Systematic] The ACM may reside somewhere other than the center of gravity (CG) and experience a different ΔV than the CG.
6. [Systematic] Damage to or displacement of the ACM could occur during the collision.
7. [Systematic] There could be power loss to the ACM prior to the collision being complete.
8. [This is not an error source but needs to be considered.] The ACM may reside somewhere other than the center of gravity (CG) and experience a different ΔV than the CG.
9. [This is not an error source but needs to be considered.] Tire forces can influence the ΔV measured by the ACM.

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Fractured ACM Mount

Exponent, "Testing and Analysis of Toyota Event Data Recorders," October 2011.

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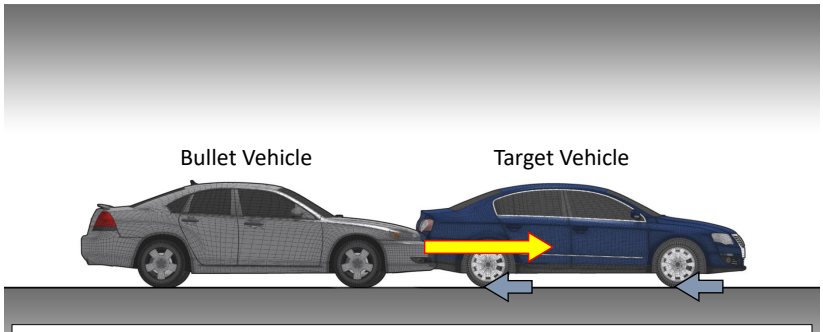
The Effect of Tire Forces



Assume the drivers of both vehicles are braking during the collision…

For the target vehicle in the depicted collision, the tire forces act opposite the collision force. For this vehicle, the tire forces will decrease the ΔV sensed by the ACM.

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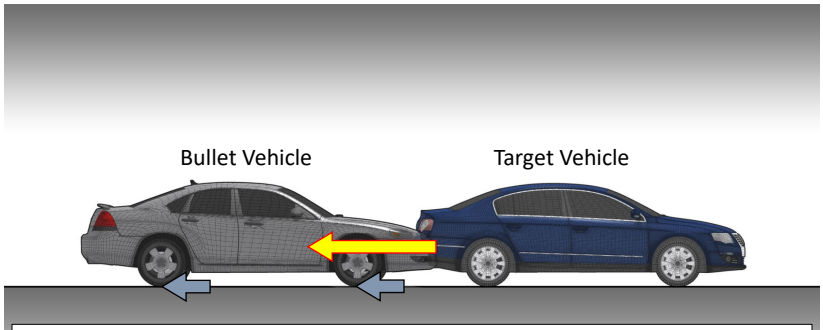
The Effect of Tire Forces



Assume the drivers of both vehicles are braking during the collision…

For the bullet vehicle in the depicted collision, the tire forces act in the same direction as the collision force. For this vehicle, the tire forces will increase the ΔV sensed by the ACM.

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Additional Classes to Consider

- SAE class Applying Automotive EDR Data to Traffic Crash Reconstruction
- SAE class Accessing and Interpreting Heavy Vehicle Event Data Recorders
- SAE class Applying Heavy Truck EDR Data to Crash Reconstruction Analysis
- NWCPS classes on heavy trucks & HVEDR
- University of Tulsa classes on heavy trucks & HVEDR

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In Module 5 We Did the Following

- Described types of EDRs in passenger vehicles and heavy trucks
- Discussed methods of extracting data from EDRs
- Discussed accuracy of speeds recorded in EDRs
- Described methods to confirm EDR data related
- Analyzed EDR reported delta-V data
- Discussed supplementing accident reconstruction with EDR data

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MODULE 6 - SIMULATIONS

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Module 6 – Simulations

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SAE International Course #1728

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Explico Inc.

Nathan Rose
Principal Accident Reconstructionist
Explico Inc.

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Crash Analysis Software

PC-Crash

HVE

Virtual Crash

MADYMO

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2018-01-0523 Published 03 Apr 2018

An Analytical Review and Extension of Two Decades of Research Related to PC-Crash Simulation Software

Nathan A. Rose and Neal Carter Kineticorp LLC

Citation: Rose, N.A. and Carter, N., "An Analytical Review and Extension of Two Decades of Research Related to PC-Crash Simulation Software," SAE Technical Paper 2018-01-0523, 2018, doi:10.4271/2018-01-0523.

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All simulations of real systems have their limitations…The key is for the user to recognize the limitations of a software, and then, to put in the effort to understand the physical models that software uses.

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Validation Concepts

Conceptual Model Validation: Validation of the component models of a software, separate from the software. Examples would be the collision model or the tire model.

Calibration: Once the validity of the component models has been demonstrated, testing and documenting to establish a range of reasonable inputs for the model.

Verification: Ensuring that the conceptual models are accurately coded in the software.

Operational Validation: Testing of the software as a complete package.

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Operational Validation Studies

Pay attention to how much information about the reference crash tests is given to the analyst. What is being tested? The software, the user skill level, the influence of uncertainties?

All of these are reasonable things to test. But, if the goal is to test and validate the software, it makes sense to minimize the influence of the other variables. So, validate with a highly-skilled user with access to all relevant information.

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Operational Validation Studies

Distinguish between:

The software's ability to produce a correct solution, and…

Your ability to discover a correct solution.

Take personal responsibility. Don't say: This crash cannot be simulated. Sometimes, that is true. But, more often, you should say: I cannot simulate this crash.

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Operational Validation Studies

From the PC-Crash website: "PC-Crash is technical software for serious users. It's powerful and gives you a lot of freedom, so that you can simulate unique events. It therefore necessarily leaves you the option to make a mess. Don't let yourself do this."

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Operational Validation Studies

Sufficient Accuracy

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Conceptual Models in PC-Crash

Collision Models

- Impulse-Momentum
- Ellipsoidal
- Multibody

Trajectory Model

- Tire Model (Linear or TM-Easy)
- Suspension Model

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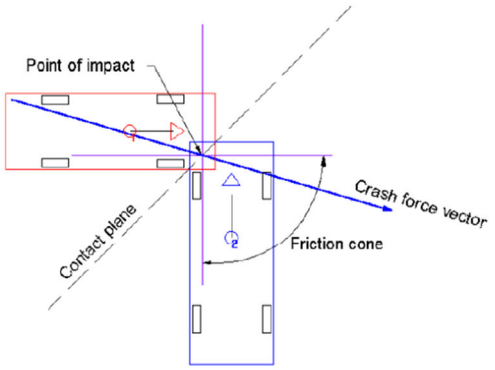
Impulse-Restitution Impact Model



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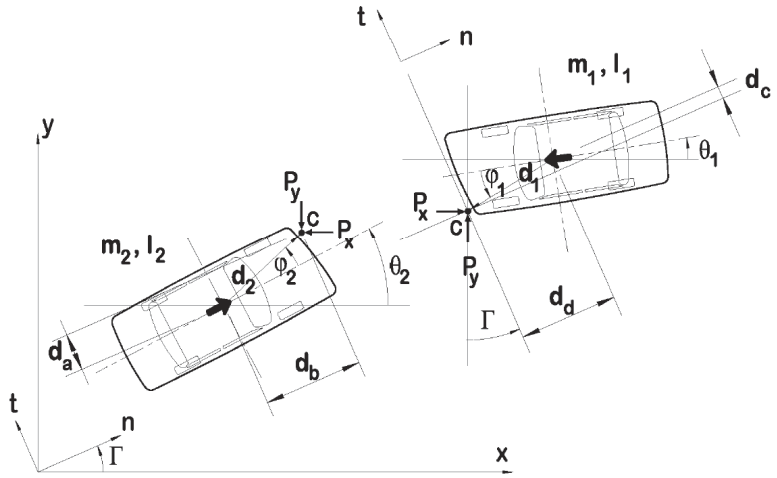
Planar Impulse-Momentum Model for Rigid Bodies

Figure 6.5
Free body diagram of the planar impact of two vehicles showing coordinates and dimensions.



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Planar Impulse-Momentum Model for Rigid Bodies

Assumptions

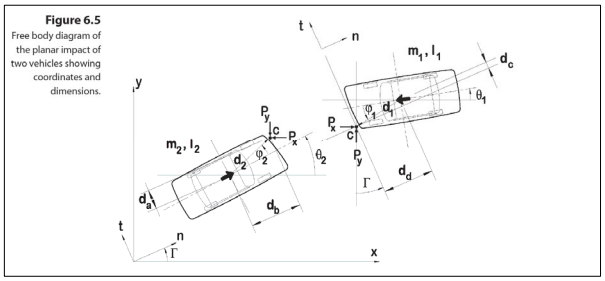

Figure 6.5 Free body diagram of the planar impact of two vehicles showing coordinates and dimensions.

(1) two-dimensional (planar) collision
(2) instantaneous collision (this equates to no change in position or orientation of the particles during the collision)
(3) the deforming region is assumed to be small relative to the size of the particle (no change in inertial properties due to the collision and collision force transfer occurs at a single point)
(4) forces other than the collision forces are assumed negligible or the collision force is assumed very large relative to other forces

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Data from Five Staged Car to Car Collisions and Comparison with Simulations

Mark N. Bailey, Jonathan M. Lawrence, Stuart J. Fowler, Peter B. Williamson, William E. Cliff and Jeff S. Nickel
MacInnis Engineering Associates, Ltd.

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Bailey et al.: “The five staged collisions were simulated by an engineer not directly involved in the staging of the collisions. Information on wheel-lockups and brake application time were not given to the analyst.”

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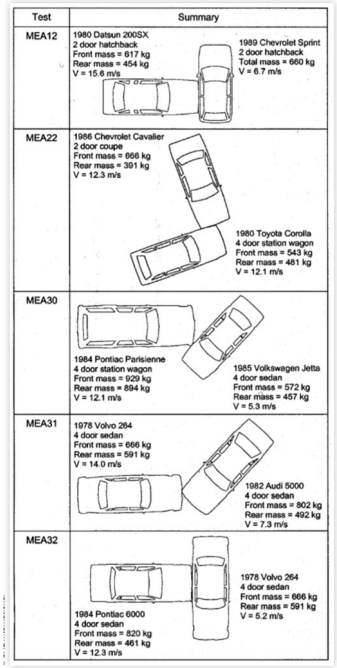

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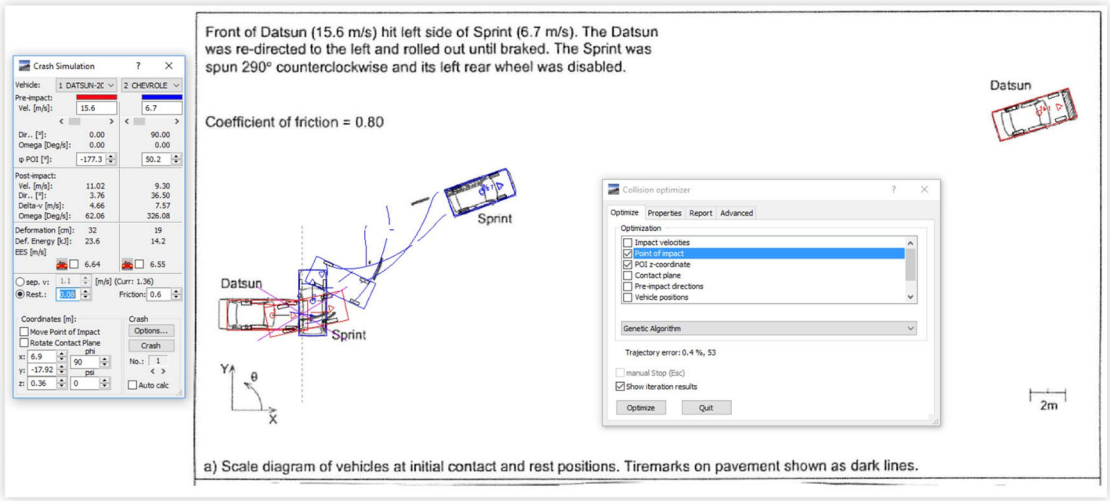
FIGURE A1 Optimization Results for MEA 12



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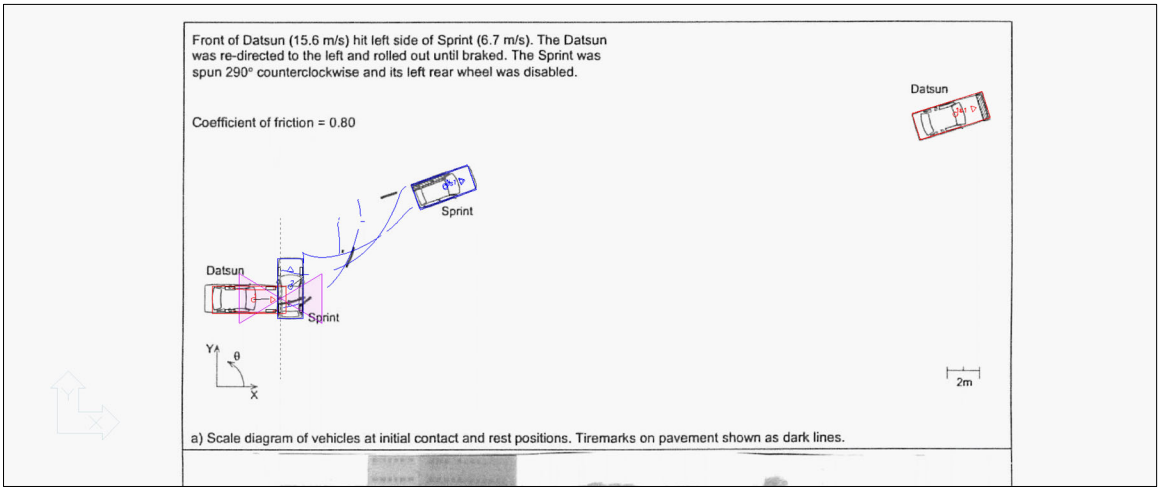
MEA 12



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	Initial contact					Full engagement			Speed change				Rest position			Restitution	
	X m	Y m	θ deg	V_{long} m/s	V_{lat} m/s	X m	Y m	θ deg	ΔV_{long} m/s	ΔV_{lat} m/s	ΔV_{tot} m/s	PDOF deg	X m	Y M	θ deg	Impulse	Relative speed
Datsun	1.6	5.2	0	15.6	0.1	2.1	5.2	0	-4.6	0.9	4.7	169	44.2	16.5	16.7	0.23	0.22
Sprint	4.4	6.0	90	6.7	0.2	4.5	6.2	90	-2.7	-6.4	6.9	-113	14.1	12.0	20.5	0.14	

i) Vehicle positions and headings at first contact, full engagement and rest as well as speed changes and collision restitution

Figure 3. MEA12 Test Results (continued)

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	Initial contact					Full engagement			Speed change				Rest position			Restitution	
	X m	Y m	θ deg	V_{long} m/s	V_{lat} m/s	X m	Y m	θ deg	ΔV_{long} m/s	ΔV_{lat} m/s	ΔV_{tot} m/s	PDOF 4.7 g	X 171	Y M	θ deg	Impulse	Relative speed
Datsun	1.6	5.2	0	15.6	0.1	2.1	5.2	0	-4.6	0.9	4.7	169	44.2	16.5	16.7	0.23	0.22
Sprint	4.4	6.0	90	6.7	0.2	4.5	6.2	90	-2.7	-6.4	6.9	-113	14.1	12.0	20.5	0.14	

i) Vehicle positions and headings at first contact, full engagement and rest as 7.6 s -99 changes and collision restitution

Figure 3. MEA12 Test Results (continued)

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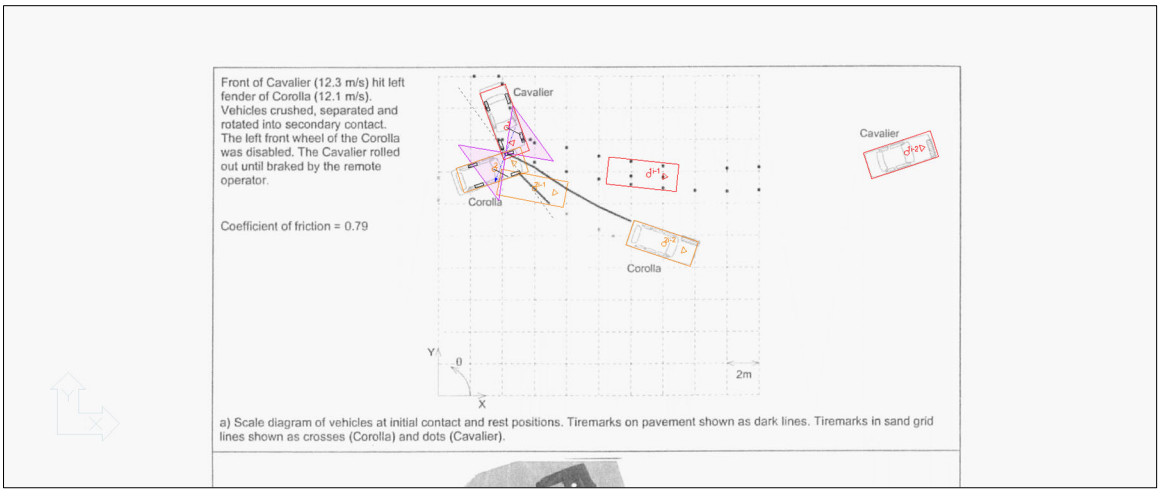
MEA 22



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	Initial contact					Full engagement			Speed change				Rest position			Restitution	
	X m	Y m	θ deg	V_{long} m/s	V_{lat} m/s	X m	Y m	θ deg	ΔV_{long} m/s	ΔV_{lat} m/s	ΔV_{tot} m/s	PDOF deg	X m	Y M	θ deg	Impulse	Relative speed
Cavalier	4.1	17.6	-69.9	12.3	0.5	4.3	16.9	-69.9	-6.1	6.9	9.3	132	29.1	15.4	18.2	0.13	0.05
Corolla	3.2	14.4	21.0	12.1	-0.5	3.5	14.4	21.0	-6.5	-7.3	9.8	-132	14.1	9.8	-18.8	0.05	

i) Vehicle positions and headings at first contact, full engagement and rest as well as speed changes and collision restitution

Figure 4. MEA22 Test Results (continued)

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	Initial contact					Full engagement			Speed change				Rest position			Restitution	
	X m	Y m	θ deg	V_{long} m/s	V_{lat} m/s	X m	Y m	θ deg	ΔV_{long} m/s	ΔV_{lat} m/s	ΔV_{tot} m/s	PDOF	X	Y M	θ deg	Impulse	Relative speed
Cavalier	4.1	17.6	-69.9	12.3	0.5	4.3	16.9	-69.9	-6.1	6.9	9.3	132	29.1	15.4	18.2	0.13	0.05
Corolla	3.2	14.4	21.0	12.1	-0.5	3.5	14.4	21.0	-6.5	-7.3	9.8	-132	14.1	9.8	-18.8	0.05	

i) Vehicle positions and headings at first contact, full engagement and rest as s s anges and collision restitution

Figure 4. MEA22 Test Results (continued)

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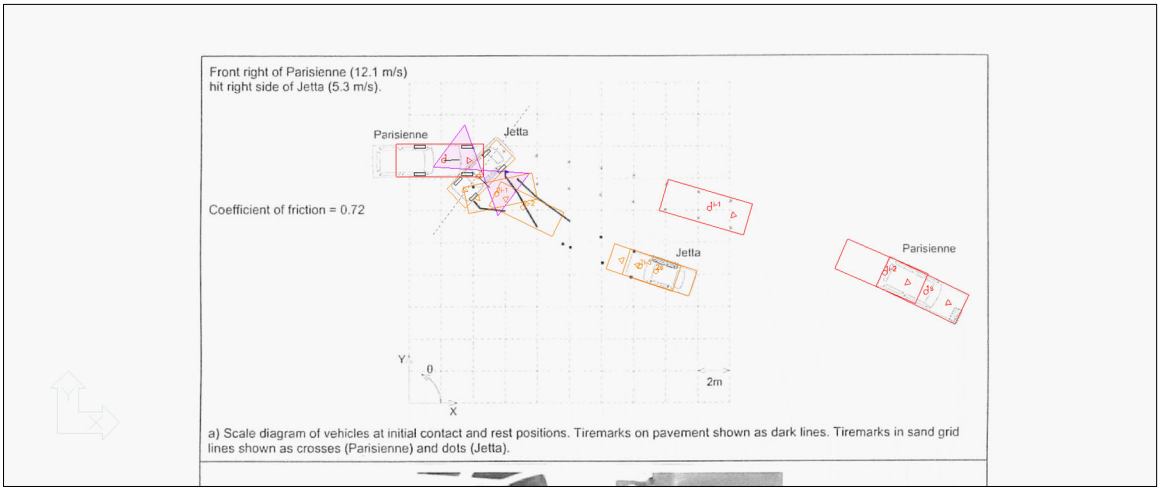
MEA 30



a) Scale diagram of vehicles at initial contact and rest positions. Tiremarks on pavement shown as dark lines. Tiremarks in sand grid lines shown as crosses (Parisienne) and dots (Jetta).

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	Initial contact					Full engagement			Speed change				Rest position			Restitution	
	X m	Y m	θ deg	V_{long} m/s	V_{lat} m/s	X m	Y m	θ deg	ΔV_{long} m/s	ΔV_{lat} m/s	ΔV_{tot} m/s	PDOF deg	X m	Y M	θ deg	Impulse	Relative speed
Jetta	4.3	14.1	-133.5	5.3	-0.2	4.1	13.9	-133.5	-5.4	6.8	8.7	128	15.6	7.9	162.1	0.02	0.02
Parisienne	0.5	14.9	1.0	12.1	1.2	1.7	14.9	1.0	-5.3	-0.3	5.3	-177	32.4	6.6	-25.9	0.01	

i) Vehicle positions and headings at first contact, full engagement and rest as well as speed changes and collision restitution

Figure 5. MEA30 Test Results (continued)

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	Initial contact					Full engagement			Speed change				Rest position			Restitution	
	X m	Y m	θ deg	V_{long} m/s	V_{lat} m/s	X m	Y m	θ deg	ΔV_{long} m/s	ΔV_{lat} m/s	ΔV_{tot} m/s	PDOF 8.2	X 132	Y M	θ deg	Impulse	Relative speed
Jetta	4.3	14.1	-133.5	5.3	-0.2	4.1	13.9	-133.5	-5.4	6.8	8.7	128	15.6	7.9	162.1	0.02	0.02
Parisienne	0.5	14.9	1.0	12.1	1.2	1.7	14.9	1.0	-5.3	-0.3	5.3	-177	32.4	6.6	-25.9	0.01	

i) Vehicle positions and headings at first contact, full engagement and rest as 4.7 s s -176 anges and collision restitution

Figure 5. MEA30 Test Results (continued)

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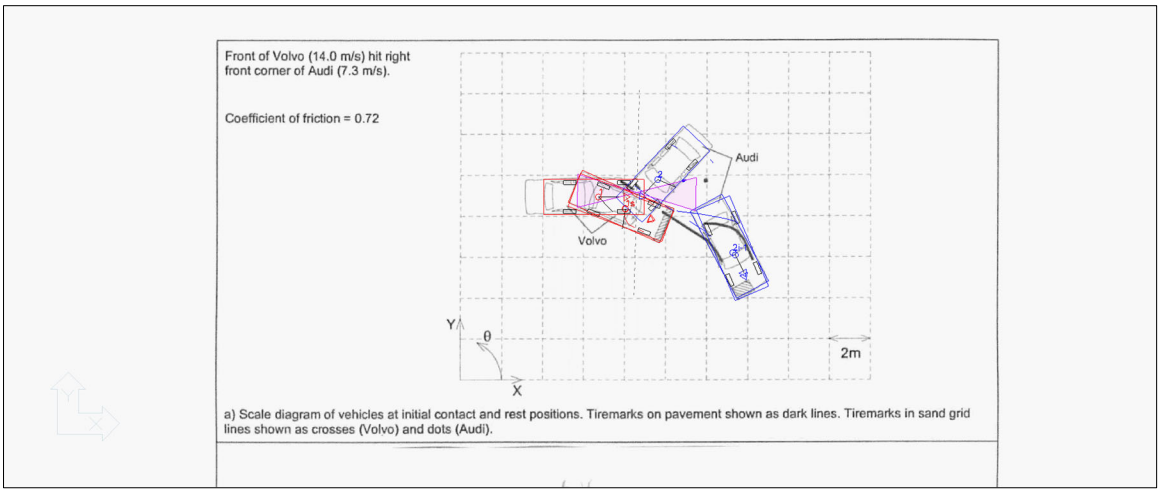
MEA 31



a) Scale diagram of vehicles at initial contact and rest positions. Tiremarks on pavement shown as dark lines. Tiremarks in sand grid lines shown as crosses (Volvo) and dots (Audi).

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	Initial contact					Full engagement			Speed change				Rest position			Restitution	
	X m	Y m	θ deg	V_{long} m/s	V_{lat} m/s	X m	Y m	θ deg	ΔV_{long} m/s	ΔV_{lat} m/s	ΔV_{tot} m/s	PDOF deg	X m	Y M	θ deg	Impulse	Relative speed
Audi	10.1	10.2	-134.1	7.3	-0.3	9.7	9.9	-134.1	-8.3	5.8	10.2	145	13.4	6.4	-60.5	0.15	0.19
Volvo	5.9	9.1	3.0	14.0	1.8	6.7	9.1	3.0	-10.9	-2.1	11.1	-169	8.2	8.6	-23.2	0.30	

i) Vehicle positions and headings at first contact, full engagement and rest as well as speed changes and collision restitution

Figure 6. MEA31 Test Results (continued)

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	Initial contact					Full engagement			Speed change				Rest position			Restitution	
	X m	Y m	θ deg	V_{long} m/s	V_{lat} m/s	X m	Y m	θ deg	ΔV_{long} m/s	ΔV_{lat} m/s	ΔV_{tot} m/s	PDOF	X	Y M	θ deg	Impulse	Relative speed
Audi	10.1	10.2	-134.1	7.3	-0.3	9.7	9.9	-134.1	-8.3	5.8	10.2	145	13.4	6.4	-60.5	0.15	0.19
Volvo	5.9	9.1	3.0	14.0	1.8	6.7	9.1	3.0	-10.9	-2.1	11.1	-169	8.2	8.6	-23.2	0.30	

11.2 148

i) Vehicle positions and headings at first contact, full engagement and rest as 11.6 -166 hanges and collision restitution

Figure 6. MEA31 Test Results (continued)

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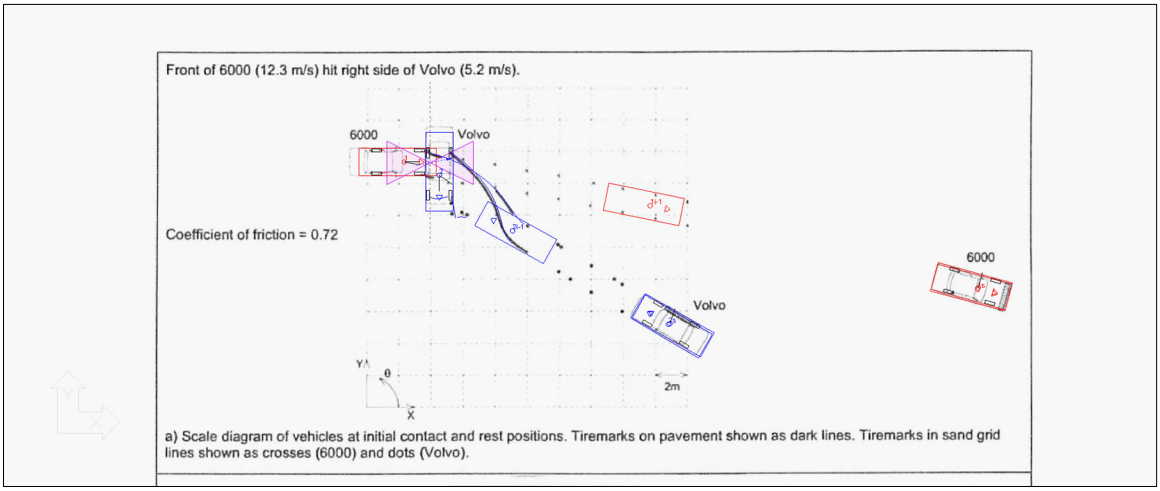
MEA 32



a) Scale diagram of vehicles at initial contact and rest positions. Tiremarks on pavement shown as dark lines. Tiremarks in sand grid lines shown as crosses (6000) and dots (Volvo).

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	Initial contact					Full engagement			Speed change				Rest position			Restitution	
	X m	Y m	θ deg	V_{long} m/s	V_{lat} m/s	X m	Y m	θ deg	ΔV_{long} m/s	ΔV_{lat} m/s	ΔV_{tot} m/s	PDOF deg	X m	Y m	θ deg	Impulse	Relative speed
6000	1.6	15.4	0.0	12.3	0.1	2.1	15.4	0.0	-6.4	-.56	7.1	-169	38.2	7.3	-16.5	0.15	0.13
Volvo	4.5	14.9	-90.0	5.2	-0.4	4.5	14.7	-90.0	-2.55	6.21	6.7	105	18.6	5.3	148.6	0.07	

i) Vehicle positions and headings at first contact, full engagement and rest as well as speed changes and collision restitution

Figure 7. MEA32 Test Results (continued)

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	Initial contact					Full engagement			Speed change				Rest position			Restitution	
	X m	Y m	θ deg	V_{long} m/s	V_{lat} m/s	X m	Y m	θ deg	ΔV_{long} m/s	ΔV_{lat} m/s	ΔV_{tot} m/s	PDOF 5.9	X -167	Y m	θ deg	Impulse	Relative speed
6000	1.6	15.4	0.0	12.3	0.1	2.1	15.4	0.0	-6.4	-.56	7.1	-169	38.2	7.3	-16.5	0.15	0.13
Volvo	4.5	14.9	-90.0	5.2	-0.4	4.5	14.7	-90.0	-2.55	6.21	6.7	105	18.6	5.3	148.6	0.07	

i) Vehicle positions and headings at first contact, full engagement and rest as 6.0 s s -101 anges and collision restitution

Figure 7. MEA32 Test Results (continued)

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Table 2. Input Values for Collision Simulations

Test #	Coefficient of Restitution		POI	
			X (m)	Y (m)
MEA12	0.19	0.08	3.82	5.25
MEA22	0.08	0.18	4.26	15.25
MEA30	0.02	0.05	3.99	14.30
MEA31	0.21	0.29	8.29	9.20
MEA32	0.12	0.13	3.82	15.40

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Optimizer

Quality versus Accuracy

Heinrichs showed that when the coefficient of friction was varied, a range of calculated impact speeds would be associated with a single optimizer error level. In other words, the optimizer error is an indicator of simulation quality (how well the simulation matches the actual vehicle positions), but not necessarily of simulation accuracy. If the analyst used the “correct” coefficient of friction or considered a range of friction coefficients, then the range of speeds obtained with a low optimizer error would be more accurate than those obtained with a high optimizer error.

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6.1.1 Ellipsoidal Model

This model uses same impact model as for the pedestrian multibody system to calculate contact forces and locations. It uses a linear stiffness model with restitution. Sliding impacts are taken into account, based on the specified contact friction. The friction force always acts in the opposite direction of relative velocities at the contact point.

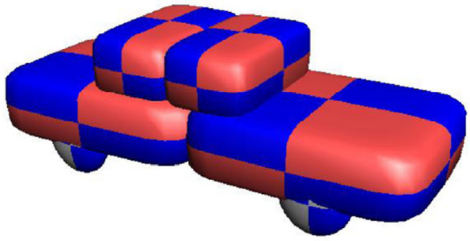

Figure 6.1:

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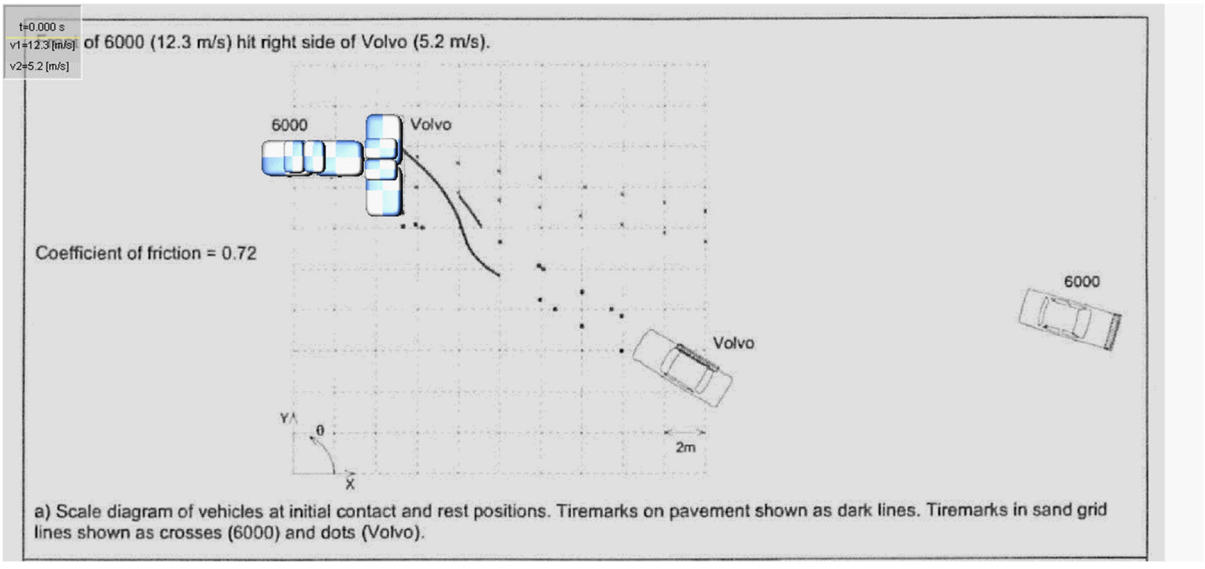

a) Scale diagram of vehicles at initial contact and rest positions. Tiremarks on pavement shown as dark lines. Tiremarks in sand grid lines shown as crosses (6000) and dots (Volvo).

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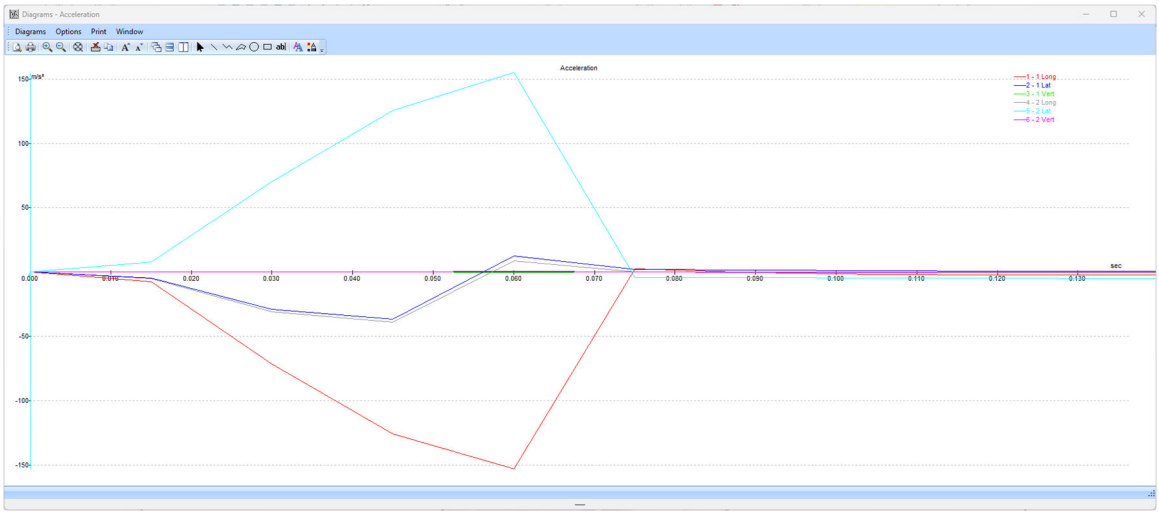

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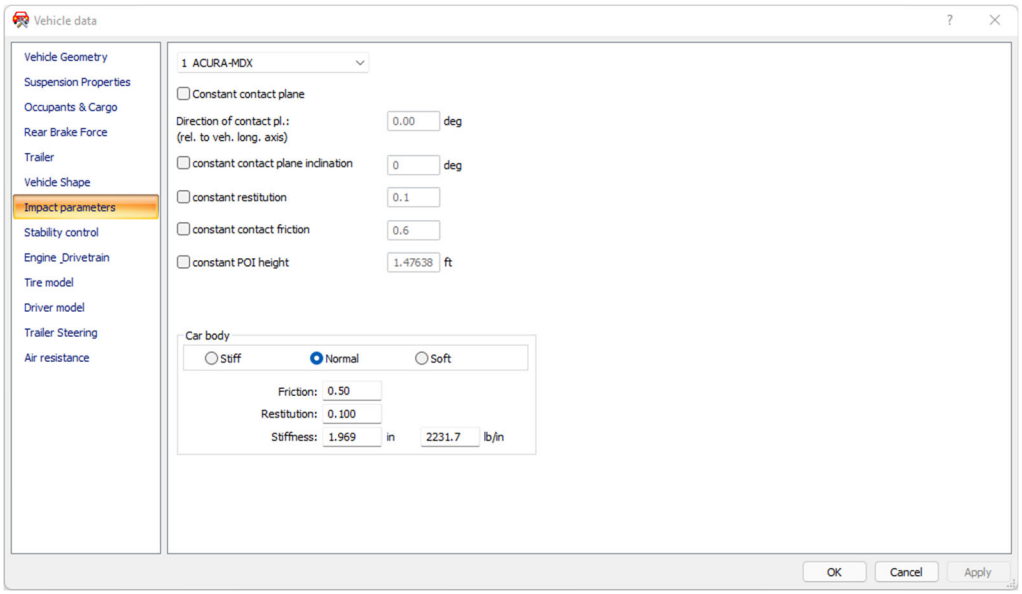

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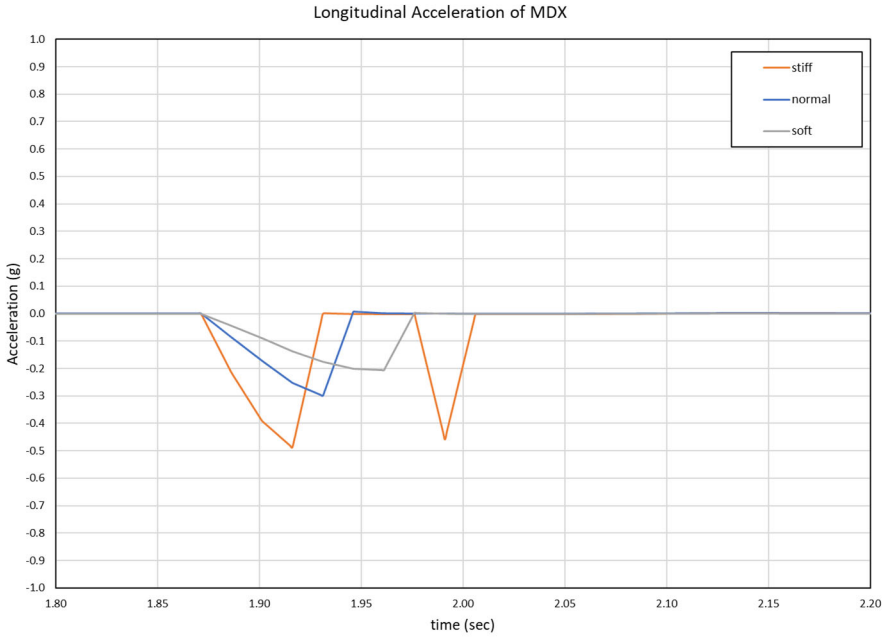

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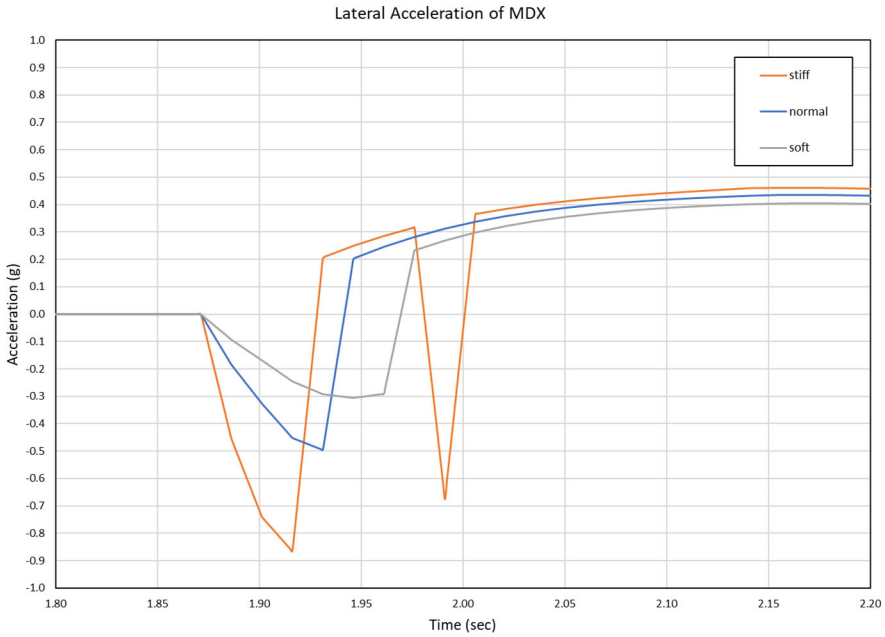

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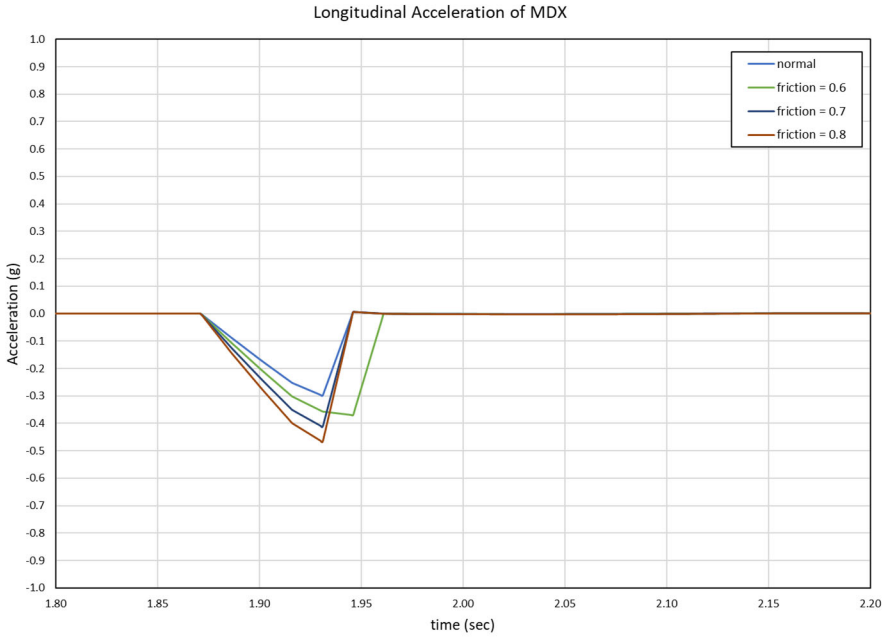

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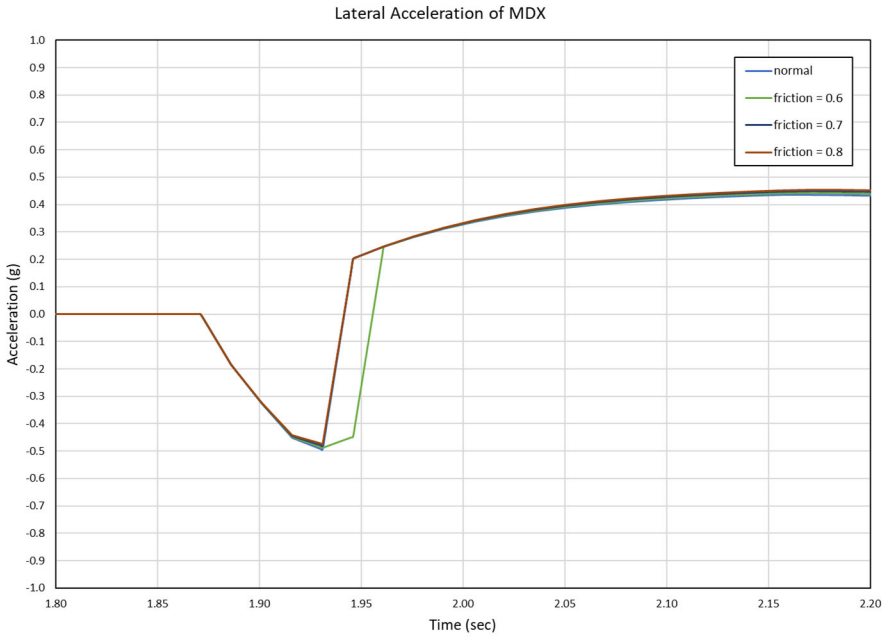

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2021-01-5043 Published 04 May 2021

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Human Subject Kinematic Response to Low-Speed Sideswipes Involving a Truck Tractor

Enrique Bonugli, Mathew Greenston, William Scott, Austin Folley, Rawson Wood, and Lisa Gwin
Biodynamic Research Corp.

Citation: Bonugli, E., Greenston, M., Scott, W., Folley, A. et al., "Human Subject Kinematic Response to Low-Speed Sideswipes Involving a Truck Tractor," SAE Technical Paper 2021-01-5043, 2021, doi:10.4271/2021-01-5043.

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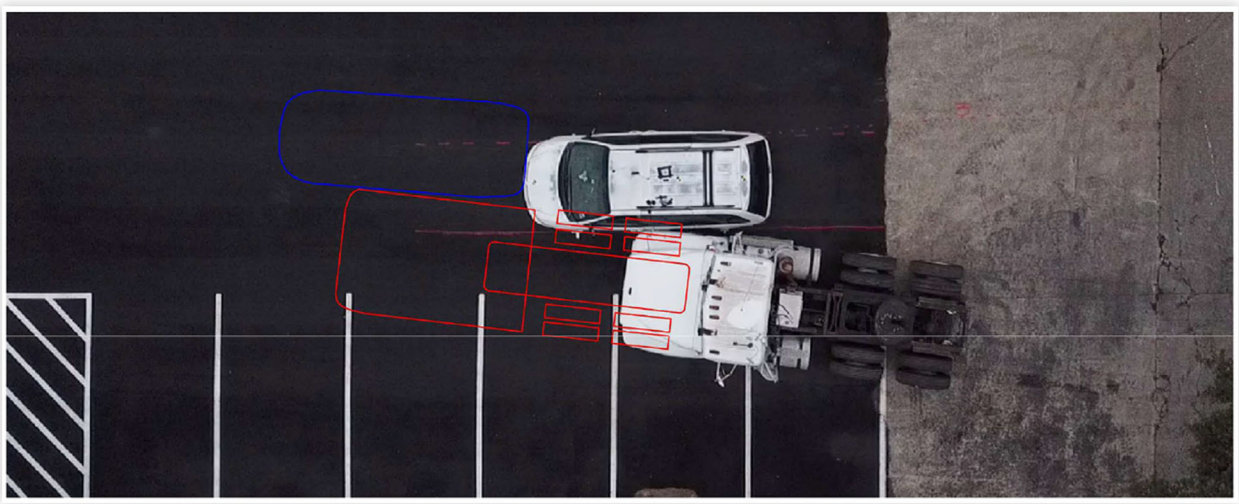

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Important Questions to Ask

Did snagging between the vehicles occur during the collision?

Was a common velocity reached along the contact surface?

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Important Questions to Ask

Did snagging between the vehicles occur during the collision?

Was a common velocity reached along the contact surface?

If NO, then inter-vehicular friction likely in the range of 0.5 to 0.6.

If YES, then inter-vehicular friction likely higher, perhaps 0.8 to 1.0.

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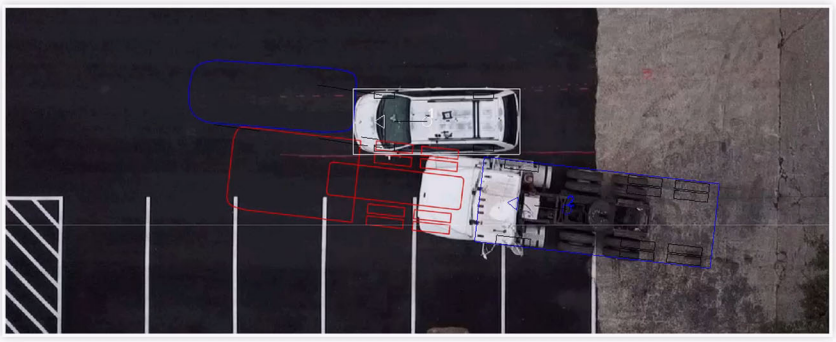

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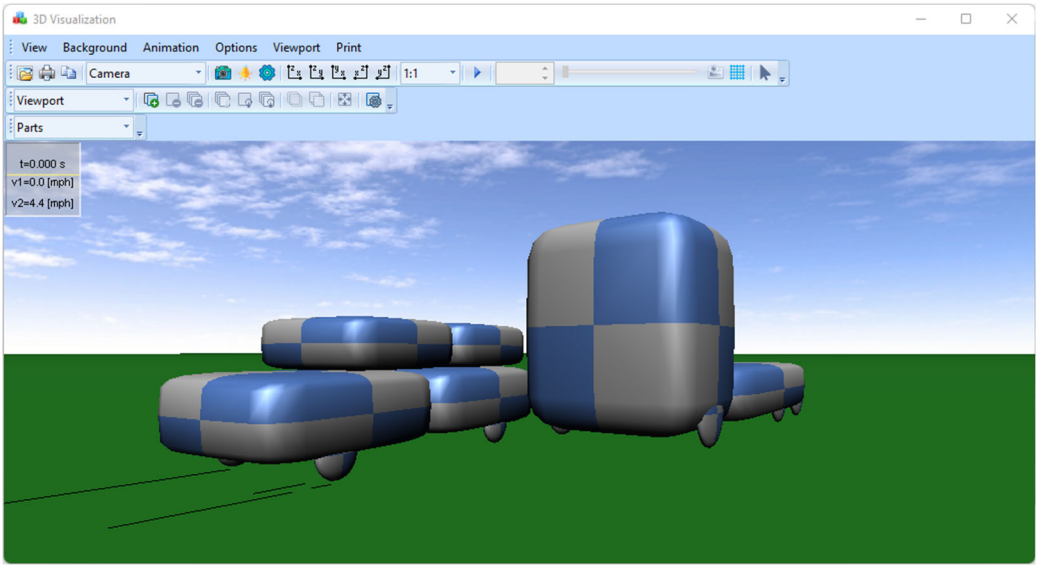

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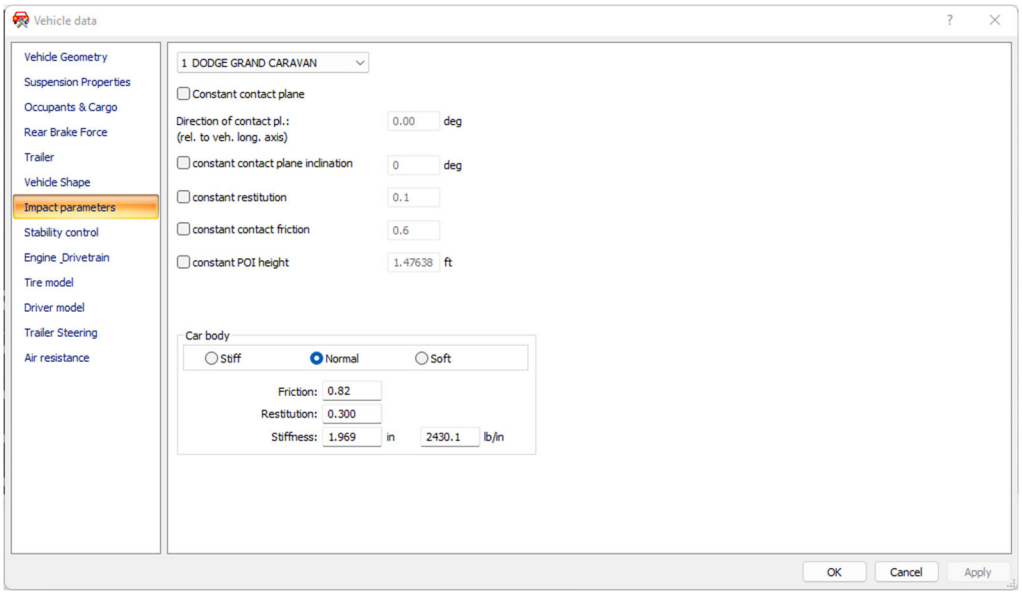

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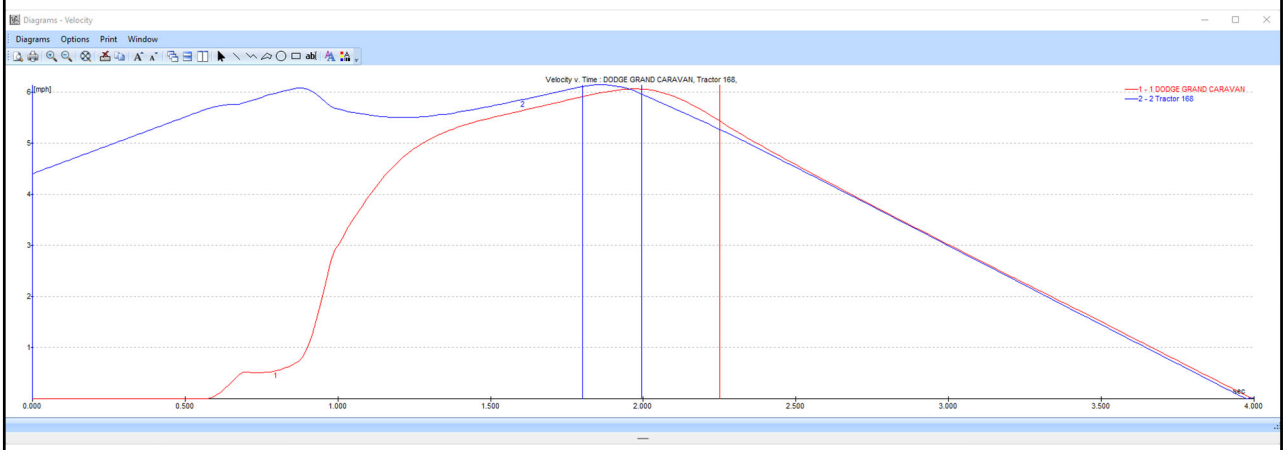

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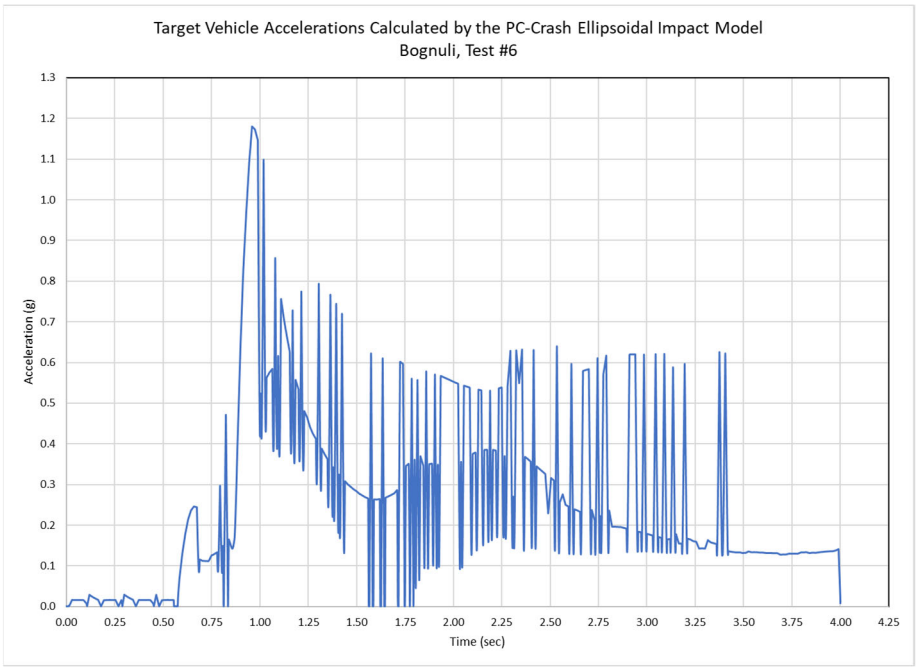

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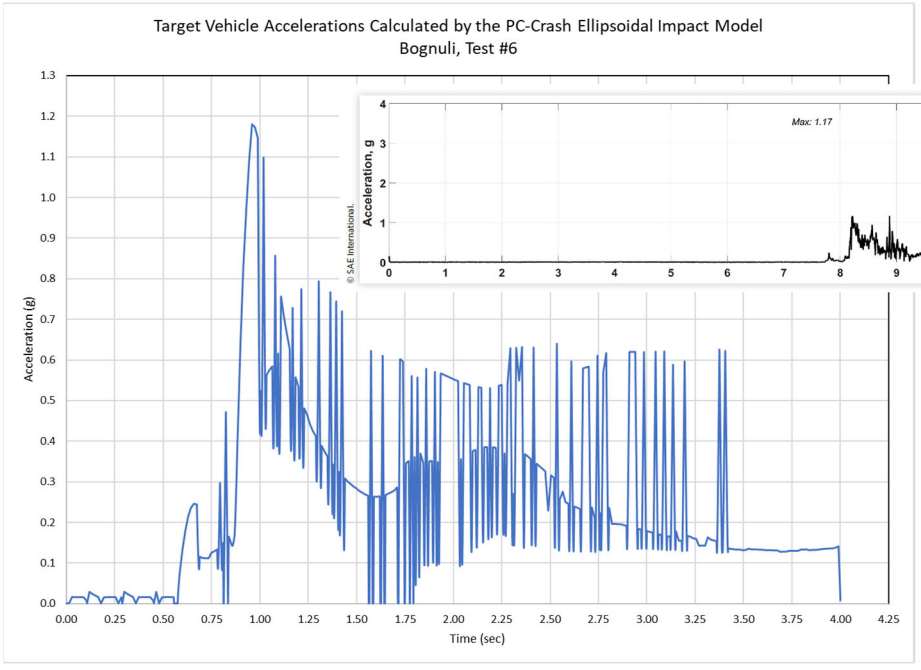

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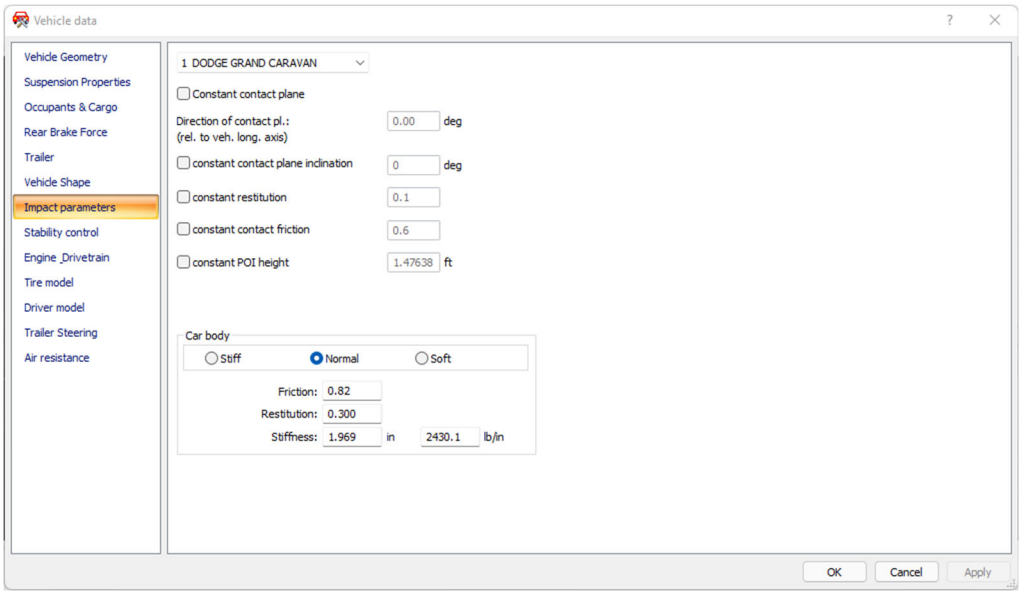

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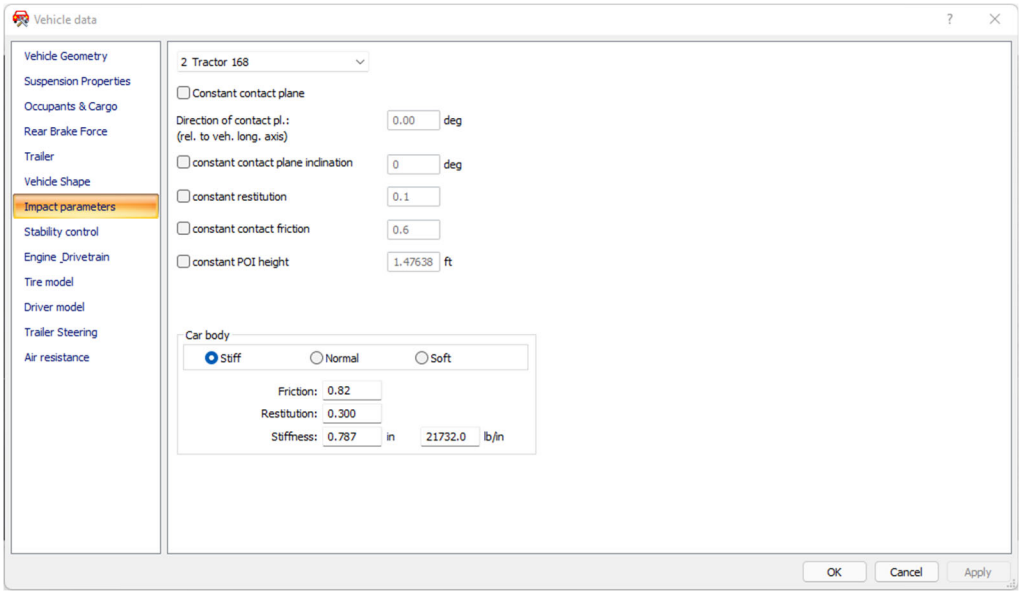

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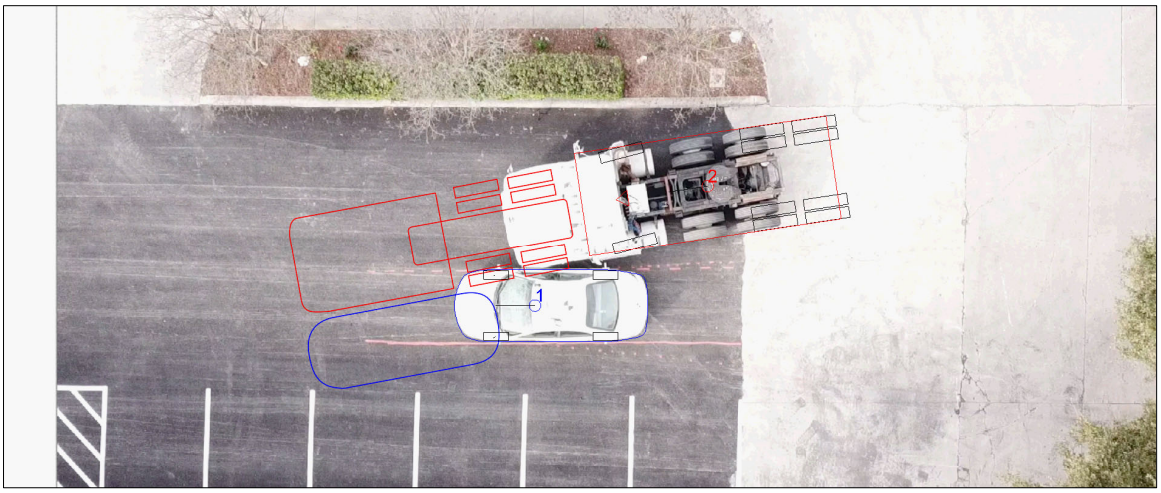
Test 2



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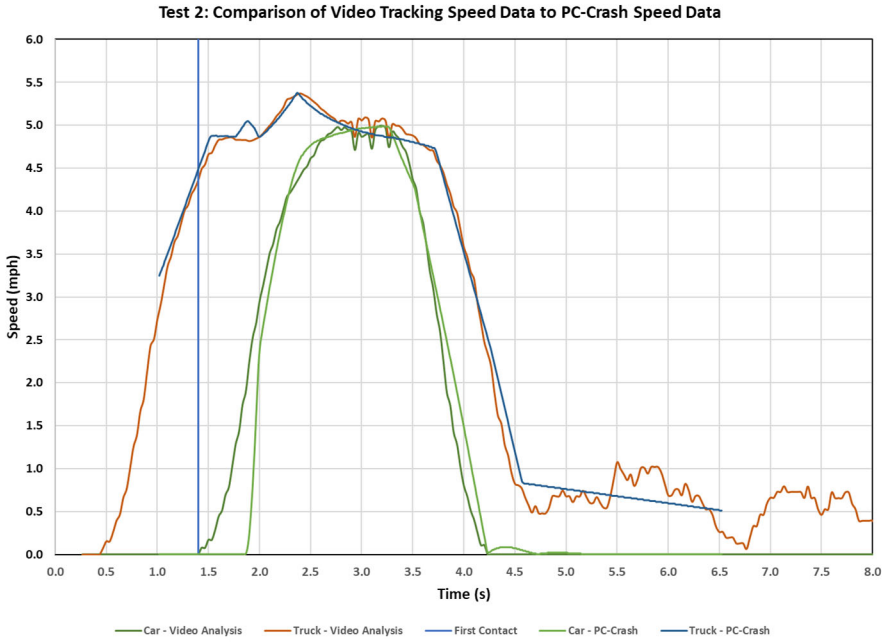

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MODULE 7 - MOTORCYCLE ACCIDENT RECONSTRUCTION

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Module 7 – Motorcycle Accident Reconstruction

Vehicle Crash Reconstruction: Principles and Technology
SAE International Course #1728

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Principal Accident Reconstructionist
Explico Inc.

Neal Carter
Principal Engineer
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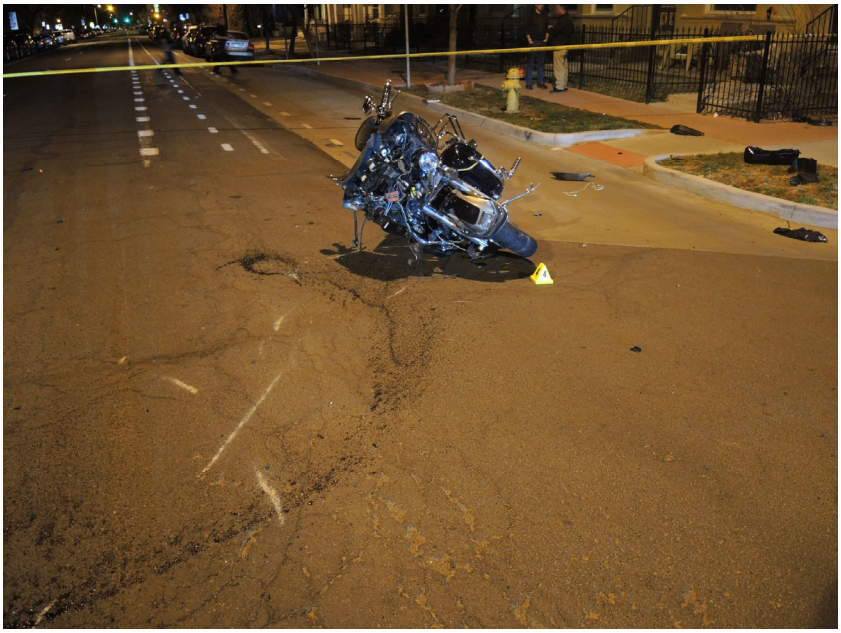

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Phases of a Motorcycle Crash (Accident)

- Pre-Collision (Pre-Impact)
  - Human Factors (Rider's Response)
    - Hazard Anticipation (Potential Hazards)
    - Hazard Avoidance
  - Pre-Collision Braking or Swerving
- Collision (Impact) with…
  - Vehicle
  - Ground
  - Fixed Object
- Post-Collision (Post-Impact)
  - Capsizing and Sliding
  - Motorcycle and Rider will Often Separate

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Pre-Collision

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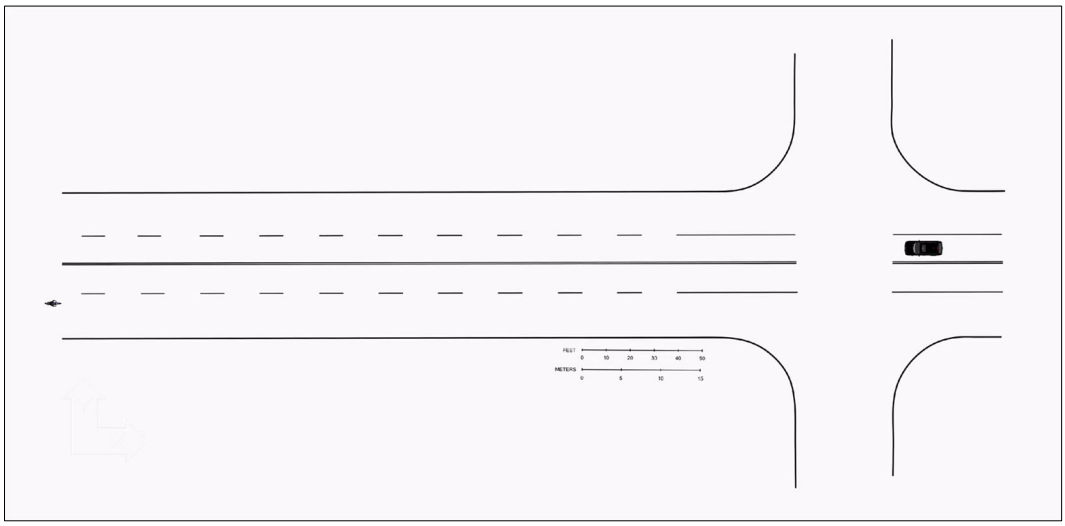
Classic Motorcycle Collision Scenario



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Classic Motorcycle Collision Scenario

Assume:

- Daytime
- Dry, Asphalt Roadway
- Conventional Motorcycle Braking System
  - Independent control of the front and rear brakes
  - No ABS
- No intervening traffic that could introduce a sight obstruction
- Both drivers have a green light

Focus on what the motorcyclist could know and when they could know it.

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Time-Limited Hazard

vs.

Distance Limited Hazard

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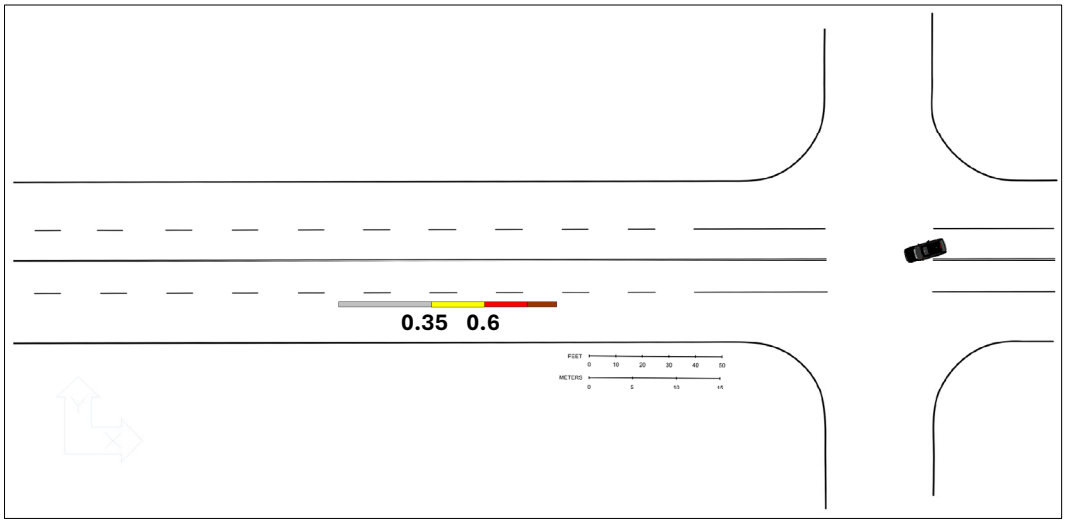
Time of Brake Application by Motorcyclist

(1.25 seconds after first lateral movement)



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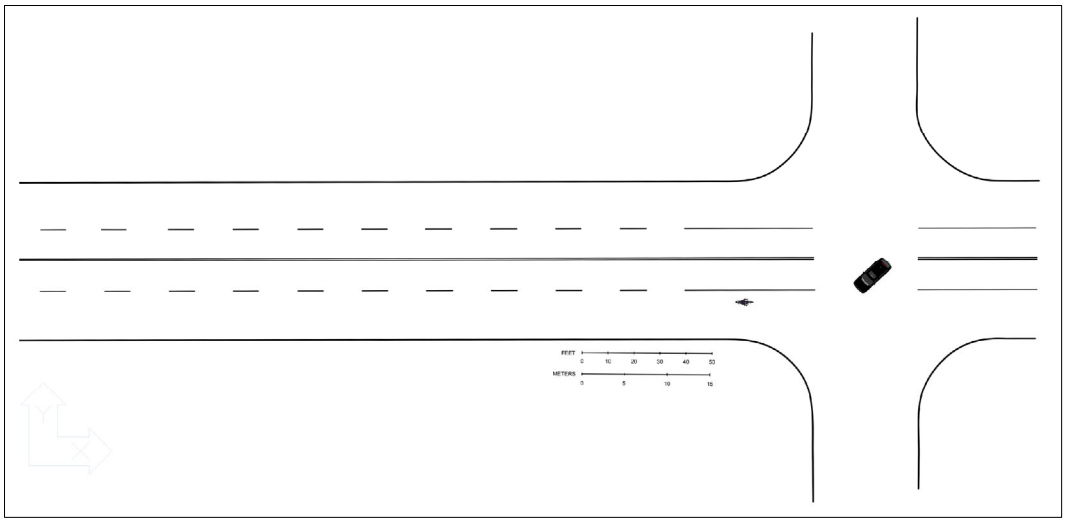

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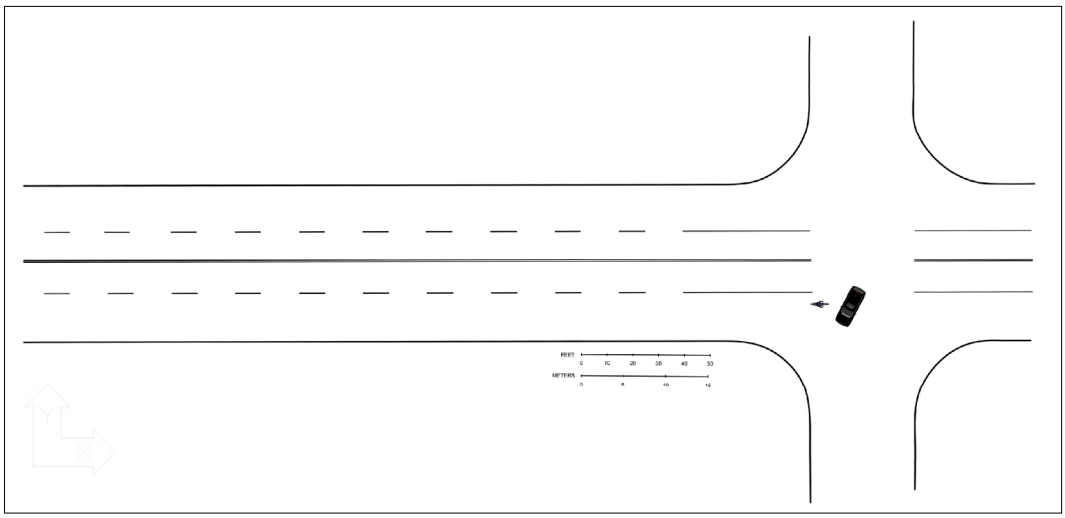

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There is initially no way for the motorcyclist to know if the left-turning driver will:

- Wait to turn until they pass.
- Turn in front of them.
- Begin to turn, but then stop.
  - ➢ Stop outside of the motorcyclist's lane.
  - ➢ Stop within the motorcyclist's lane.
- Begin to pull out intending to time a turn behind them.

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Hindsight Bias

Hindsight bias is the common tendency to believe that past events were more predictable than they were.

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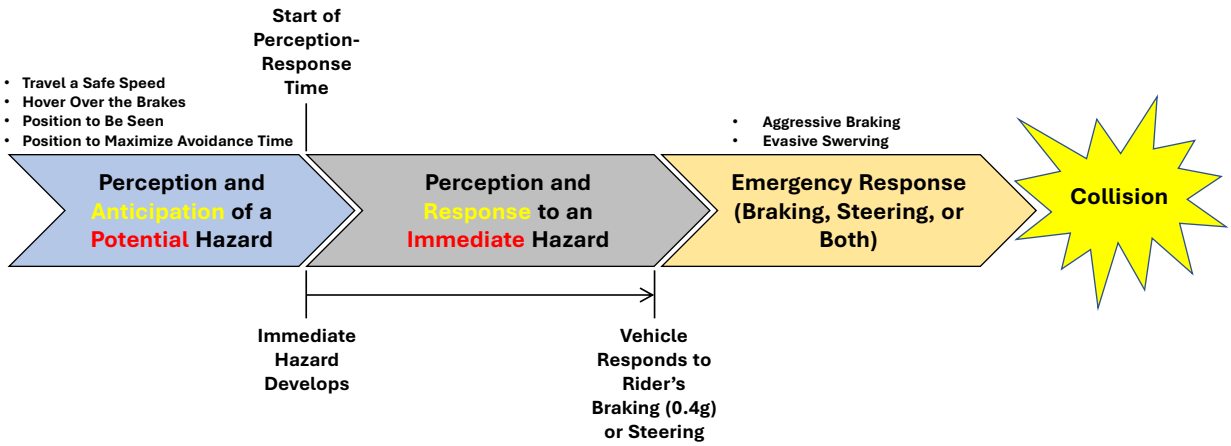
Correct Evaluation



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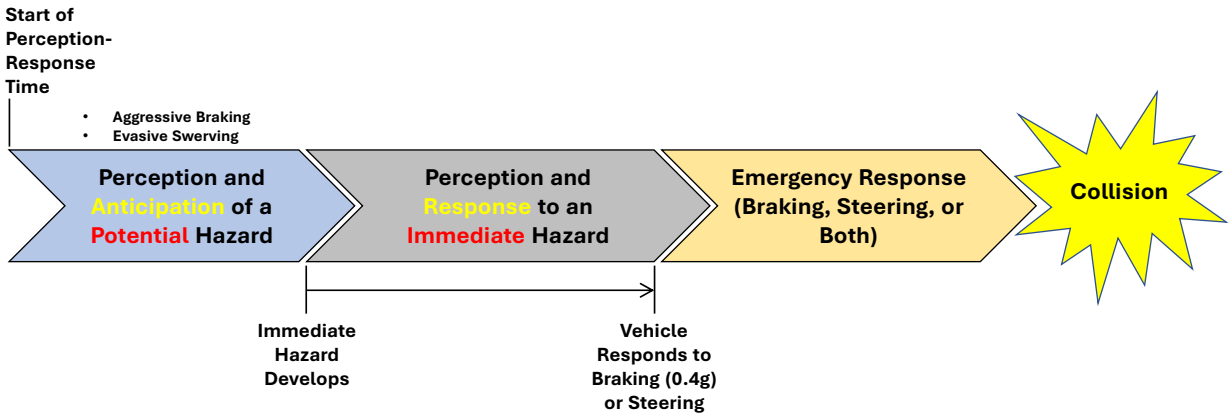
Incorrect Evaluation

The motorcyclist should have known…



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Principles

1. Distinguish between hazard anticipation and hazard avoidance.

2. The motorcyclist's actions should be judged based on what could be known at the time the response was required, not based on what the accident reconstructionist knows after the fact.

   - ➢ Did the motorcyclist engage in reasonable hazard anticipation?
     - o Often there will not be evidence one way or another, other than the motorcyclist's speed.
   - ➢ Did the motorcyclist have a reasonable perception-response time in response to the development of an actual hazard?

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Emergency Manuever Experimentation Using a Driving Simulator

Daniel Lechner
INRETS (M.A.)

Gilles Malaterre
INRETS (L.P.C.)

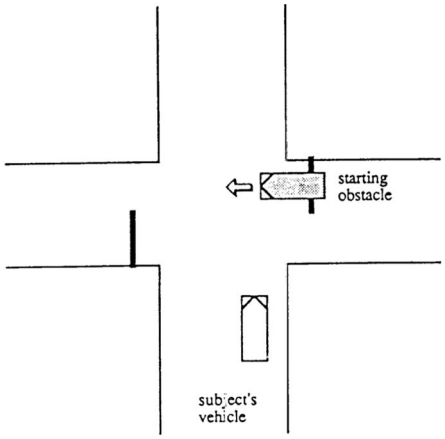

Figure 2 : Plan showing the emergency situation facing subjects

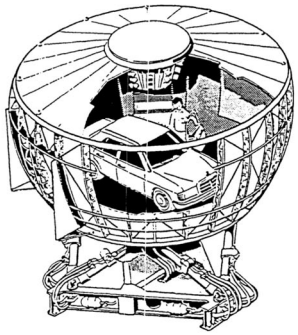

Figure 1 : View showing mobile simulator dome (Daimler-Benz document)

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Lechner and Malaterre

- Subjected 49 drivers to an unexpected intersection incursion.
- The incursion began when the approaching driver was 2.0, 2.4, or 2.8 seconds from the intersection (approaching vehicle traveling 90 to 100 km/h).
- 80% of the approaching drivers were unable to avoid.
- Intruding vehicle accelerated into the intersection for 1.9 seconds, and then braked to a stop.
- Average time for initial response, such as throttle release, was 0.8 seconds.

Therefore, the approaching driver begins responding when they could not know that the intruding vehicle would stop.

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Lechner and Malaterre

- 39% of subjects attempted to avoid by braking only.
- 67% of subjects began their avoidance maneuver with braking.
- 88% of subjects braked and swerved.

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Lechner and Malaterre

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- 67% of subjects began their avoidance maneuver with braking.
- 88% of subjects braked and swerved.

"Only 1 out of 5 drivers succeeded in avoiding the collision, whereas an appropriate and correctly performed maneuver, in particular a wider use of lateral swerving, would have allowed a higher success rate." The authors state that a driver using a combination of steering and braking in this situation represents them using "two totally conflicting strategies."

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Lechner and Malaterre

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These statements assume the approaching drivers could know more than they could know when the response was required.

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Lechner and Malaterre

Ultimately based on additional analysis, these authors conclude that "it is clear that the choice of a given maneuver by the driver is not influenced by the sudden stopping of the obstacle…the result of an emergency situation is completely uncertain." (emphasis in original)

"The present experiment shows clearly that the automatic reaction of drivers faced with an emergency situation is to brake, as this is the initial action for 2/3 of the subjects involved. The fact that braking always reduces the speed of a possible collision, whatever the obstacle does, is the probable reason…" (emphasis added)

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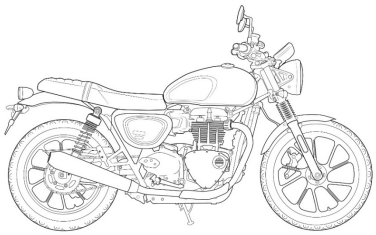
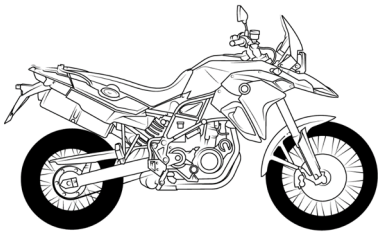
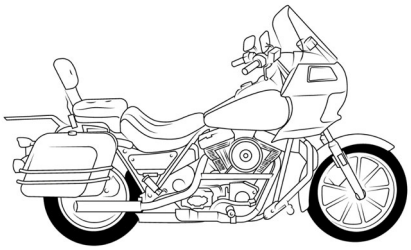
Motorcycle Types



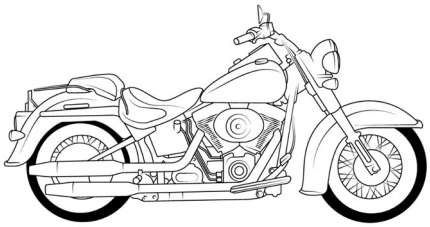
Standard



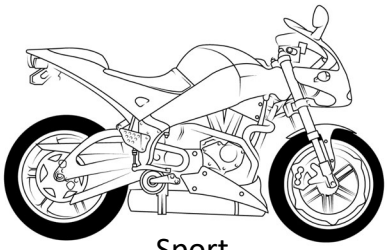
Dual-Sport/Adventure



Touring



Cruiser



Sport

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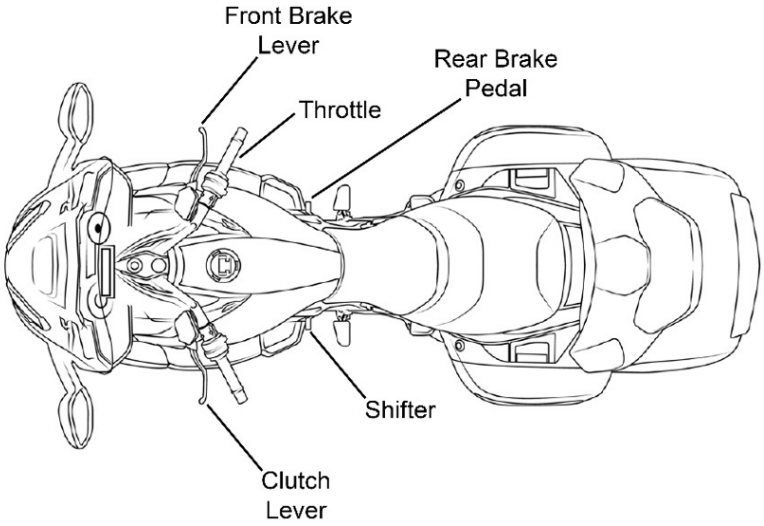
Motorcycle Controls



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Pre-Collision "Physical" Evidence

- Skid Mark(s)
  - Front Tire
  - Rear Tire
- Scrapes and Scuffs on the Road Surface
- EDR Data from the Motorcycle
- Video from a Camera on the Motorcycle or Rider
- GPS Data from GoPro Footage or a Navigation Device

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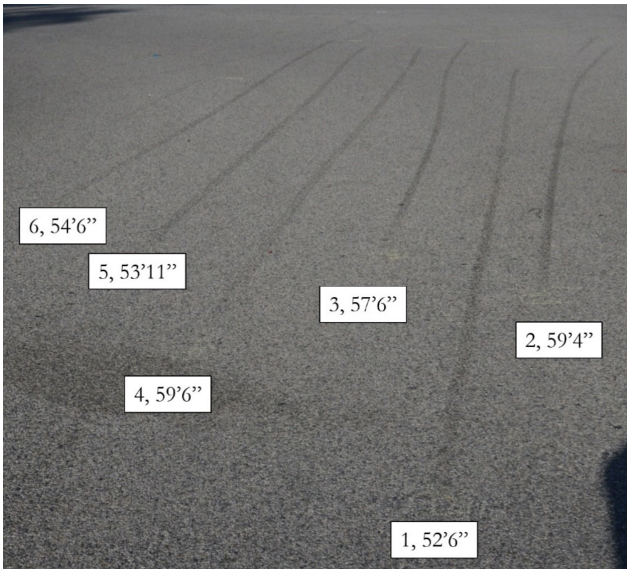
Skid Marks



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Oval-Shaped, Concentrated Abrasions from Skidding

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"Best Effort" Braking

But…in a parking lot, under ideal circumstances, without an actual impending collision.

Study	Best Effort Braking Decelerations on Dry Roadway (g)			
	Rear Brake Only (No ABS)	Front Brake Only (No ABS)	Front and Rear Combined (No ABS)	Front and Rear with ABS
Tolhurst and McKnight [6] (expert riders)		0.77	0.97	
Fries, Smith, and Cronrath [9]	0.31 to 0.52		0.54 to 0.88	
Hunter [10] (experienced riders)	0.35 to 0.36	0.64 to 0.74	0.63 to 0.96	
Hugemann and Lange [11]				
Bartlett [12]	0.38 to 0.46	0.88 to 0.89	0.96	
Ecker [13] (mixed experience levels)			0.63 ± 0.12	
Vavryn [14]			0.67	0.8
Bartlett, Baxter, Robar [15] (motor officers)	0.37 ± 0.06	0.60 ± 0.16	0.74 ± 0.15	
Bartlett and Greear [16] (students)			0.62 ± 0.15	
Dunn, et al [17]	0.345 to 0.386	0.518 to 0.709	0.612 to 0.708	0.642 to 0.842
Fatzinger, et al [19]		0.34 to 0.87	0.39 to 0.98	
Peck and Deyerl [20]	0.321 to 0.341			
Rose (adventure motorcycle, experienced rider, knobby tires)	0.31 to 0.37			

Table 3-2 – Summary of braking decelerations from various studies (dry pavement)

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Engine Braking

	N	1	2	3	4	5	6
2015 BMW R nineT (standard)	0.10	0.24	0.22	0.18	0.13	0.15	0.15
2003 Buell XB9S Lightning (standard/naked)	0.08	0.18	0.17	0.13	0.13	0.12	n/a
2015 Harley-Davidson XL1200X Forty-Eight (cruiser)	0.09	0.15	0.15	0.15	0.15	0.15	n/a
2006 Honda CBR1000RR (sport)	0.09	0.16	0.15	0.16	0.16	0.14	0.12
2013 Kawasaki KLR650 (dual-sport)	0.12	0.21	0.19	0.15	0.14	0.15	n/a
2019 Yamaha R3 (sport)	0.10	0.18	0.18	0.18	0.17	0.15	0.13
2017 Yamaha XSR900 (standard)	0.11	0.22	0.21	0.17	0.16	0.14	0.13
2013 Yamaha FZ6R (sport)	0.07	0.18	0.17	0.15	0.13	0.13	0.12
Average	0.09	0.19	0.18	0.16	0.15	0.14	0.13

Table 3-5 – Average deceleration in normalized gravitational constant (g) by forward gear for each motorcycle

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Avoidance Swerving

$$d_{turn-away} = 0.366 \cdot S \sqrt{\frac{L}{f_y}}$$

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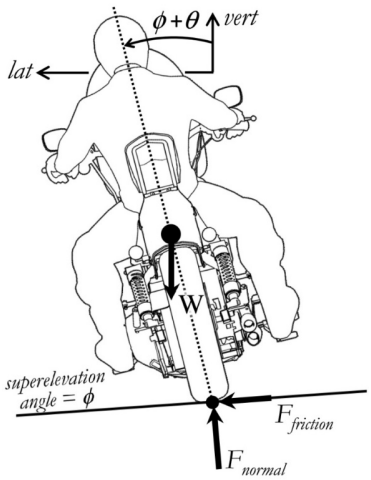
Lean Angle Equation



Lean Angle relative to Vertical

$$\theta = tan^{-1}\left(\frac{v_{mc}^2}{g \cdot r}\right)$$

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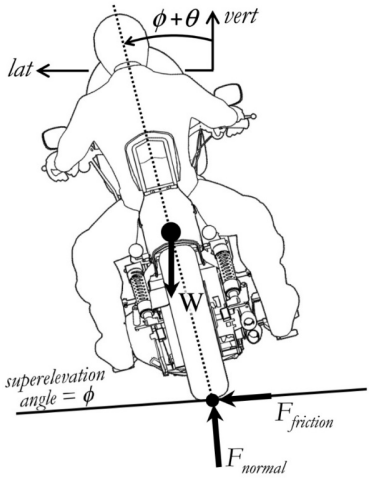
Lean Angle Equation



Lean Angle relative to Road Normal

$$\theta = tan^{-1}\left(\frac{v_{mc}^2}{g \cdot r}\right) - \phi$$

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Lean Angle Equation

Assumptions

1. Constant Velocity over the distance the radius is measured.
2. Motorcycle and Rider have the same lean angle.
3. No width to the motorcycle tires…the part of the tire contacting road doesn't change.

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Adjustment to Lean Angle Equation for Tire Width

$$\theta = \theta_{Equation\ (1)} + \Delta\theta_{tire\ width}$$

$$\Delta\theta_{tire\ width} = \sin^{-1}\left(\frac{\frac{t}{2} \times \sin\left(\theta_{Equation\ (1)}\right)}{h - \frac{t}{2}}\right)$$

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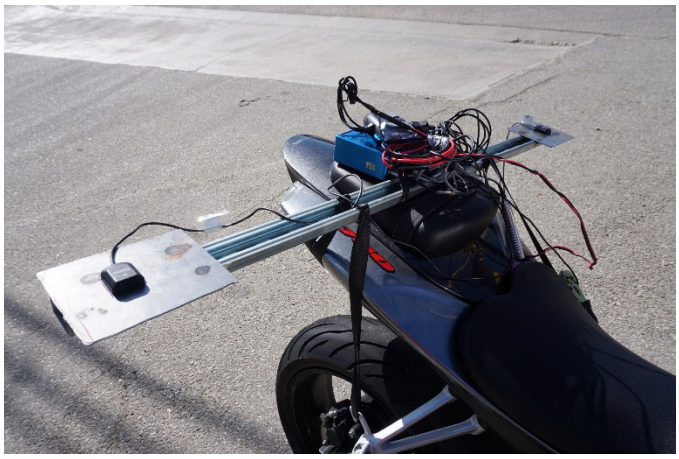

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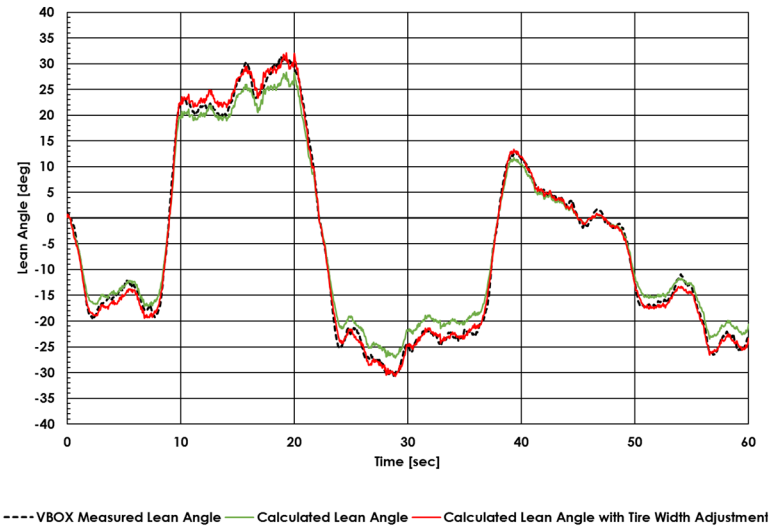
Turn 4



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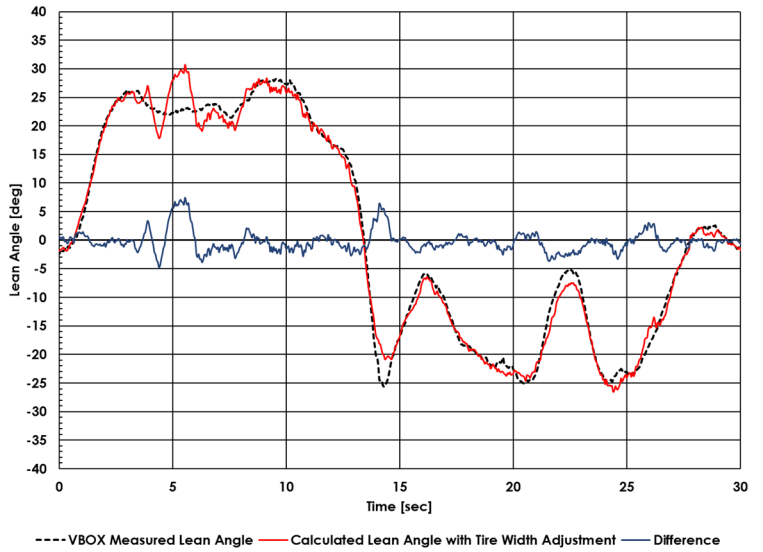
Turn 2



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What Limits a Motorcycle on a Curve?

Friction

Geometry

Rider's Psychological Limits

Friction-Limited Speed

$$v_{mc,max} = \sqrt{\frac{\mu + \tan\phi}{1 - \mu\tan\phi} gr}$$

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Geometry Limited Speed

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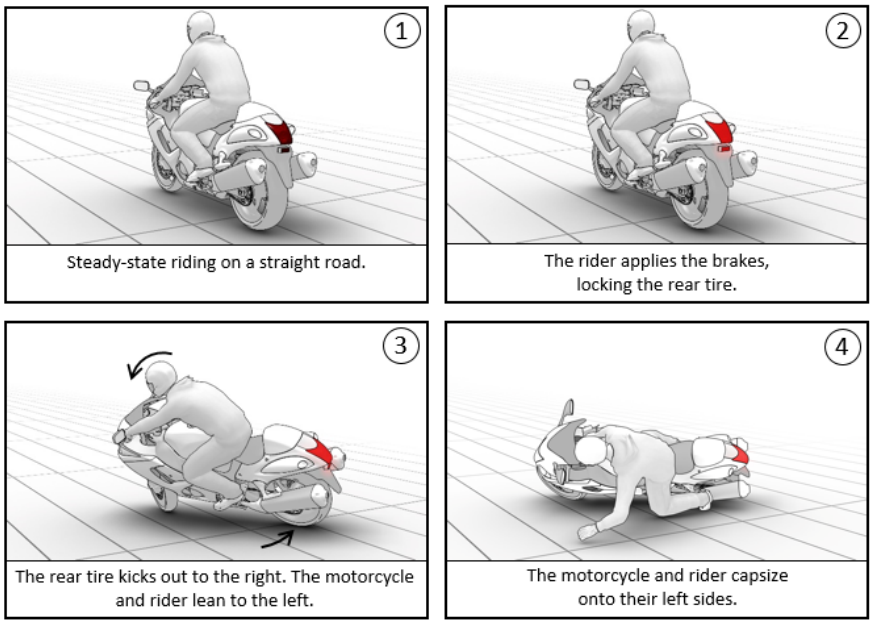
Low Side Fall



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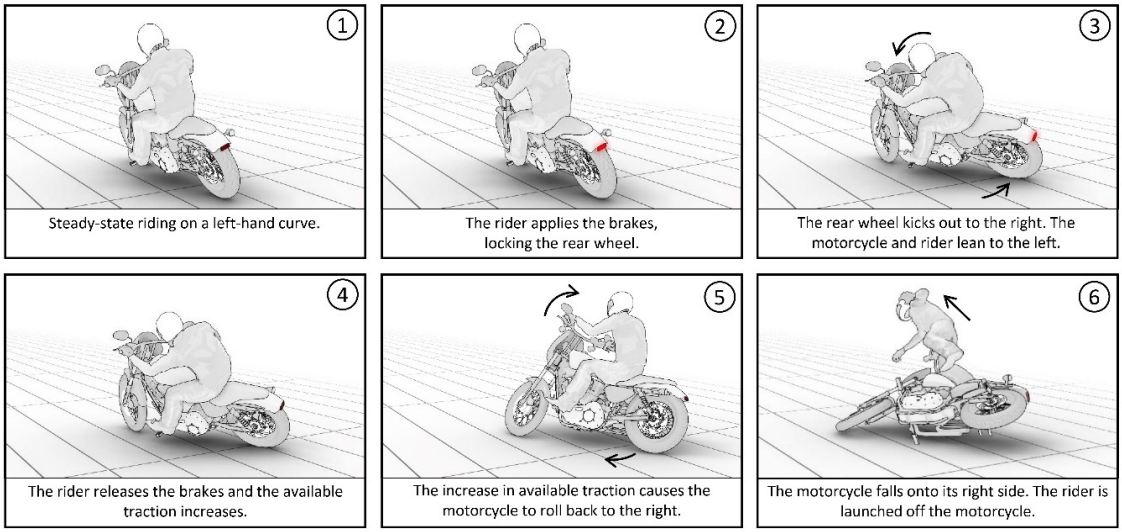
High-Side Fall



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Low-Side Fall

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Low-Side Fall

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High-Side Fall

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Physical Evidence from Low-Side Fall

- ☐ Little to no distance between tire mark and scrapes
- ☐ Motorcycle leads in the slide

Physical Evidence from High-Side Fall

- ☐ Gap between tire mark and scrapes
- ☐ Rider leads in the slide

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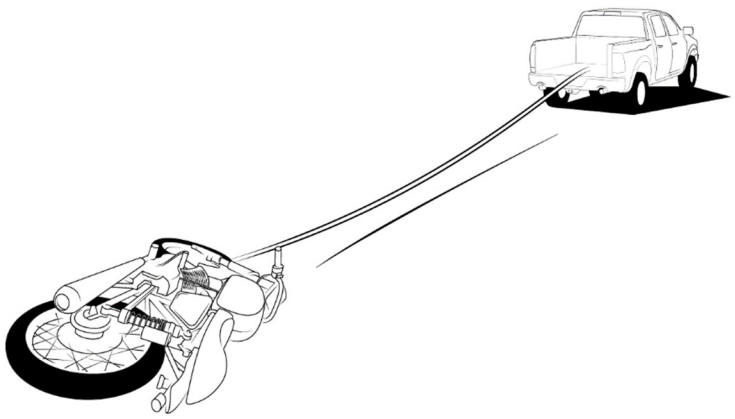

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TABLE 6.1 Average sliding deceleration for different motorcycle types (asphalt/concrete)

Motorcycle type	Average sliding deceleration on asphalt/concrete (g)
Standard	0.45 ± 0.15
Cruiser/touring	0.50 ± 0.11
Sport	0.47 ± 0.10
Dirt/enduro	0.50 ± 0.12
Scooter/moped	0.49 ± 0.11

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TABLE 6.2 Average sliding deceleration for different motorcycle characteristics (asphalt/concrete)

Motorcycle characteristics	Average sliding deceleration on asphalt/concrete (g)
No fairings or crash bars	0.54 ± 0.11
Partially faired	0.48 ± 0.11
Fully faired	0.46 ± 0.10
Crash bars	0.34 ± 0.07
Sports bikes with frame sliders	0.45 ± 0.10

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Collision

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Collision Physical Evidence

- Impact-Induced Yaw Rotation of the Struck Vehicle
- Reduction in Wheelbase of the Motorcycle
- Fractured Frame, Triple Clamp, or Fork of Motorcycle
- Frozen Speedometer
- Crush to the Car
  - Imprint on the Car from Motorcycle Components
- Tire Transfer from Motorcycle to Struck Car
- Tire Marks on the Road Surface from Impact
  - Crook in Tire Mark

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Tire Transfer and Wheel Imprint

Motorcycle Axle/Bottom of Right Fork

Counterclockwise Torque Applied to Wheel, Forks, and Handlebars From Tire Contact with Van

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Upside Down Forks (some bending)

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WREX 2000, Struck Car Damage

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WREX 2000, Wheel and Fork Deformation

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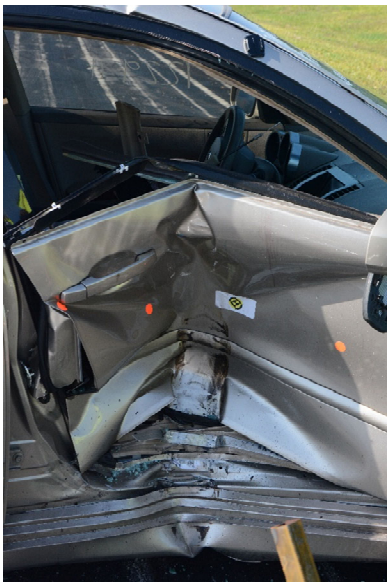
WREX 2016, Struck Car Damage



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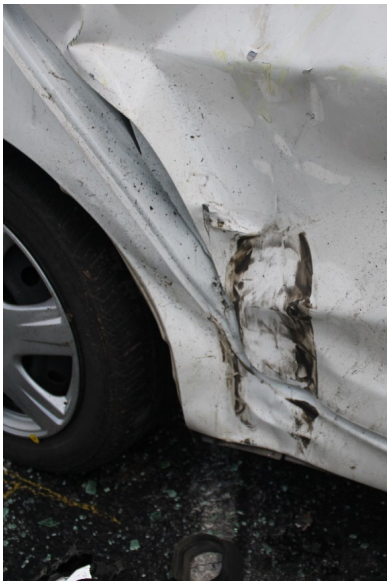
WREX 2016, Struck Car Damage



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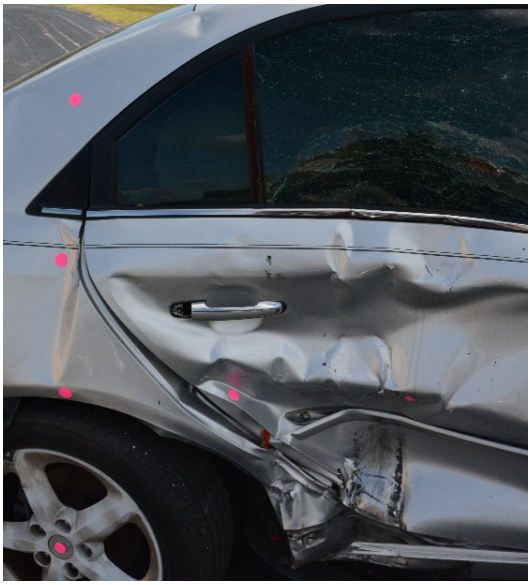

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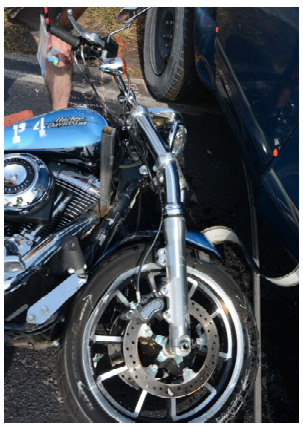
Wheel and Fork Deformation (Conventional Forks)



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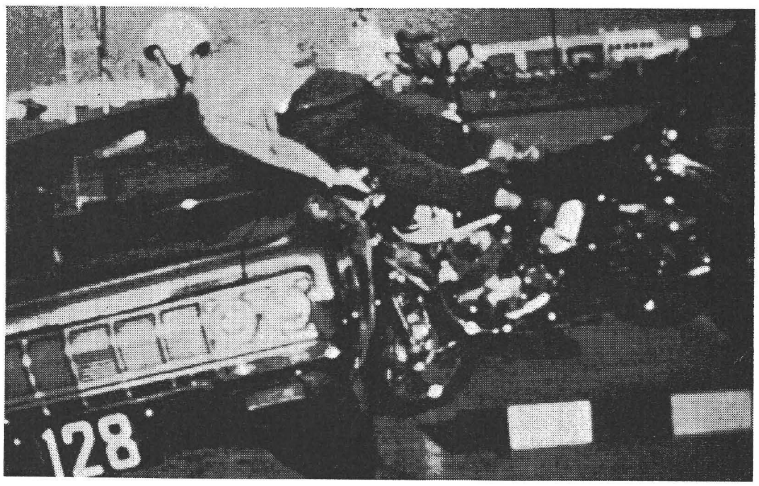
Severy et al. (1970)



Severy et al. (1970)

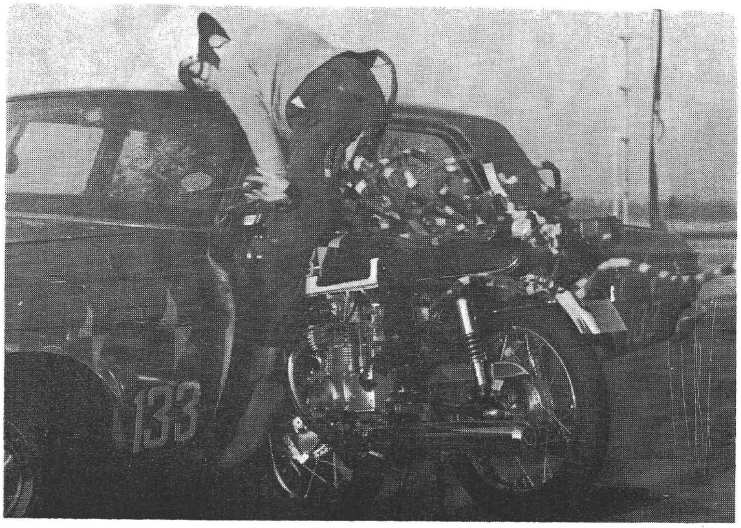


Fig. 32—Honda 350, 40 mph, Experiment 133

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Options for Motorcycle Impact Speed Analysis

- Motorcycle Wheelbase Reduction + Car Crush
- Struck Vehicle Translation and Rotation
  - McNally and Bartlett
  - Ogden
  - Simulation (PC-Crash, V-Crash, HVE)
- Struck Vehicle EDR Data
- Motorcycle EDR Data (Some Kawasaki Motorcycles)
- A Frozen Speedometer Reading on the Motorcycle
- Analysis of Video from a Motorcyclist's GoPro
- Analysis of Surveillance Footage

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Wheelbase Reduction + Car Crush

Eubanks Equation, Modified by Peck

Axle

$$S = 2.16(L + C) + 17.33$$

Eubanks Equation, Modified by Peck

Bumper/Pillar

$$S = 1.36(L + C) + 19.50$$

Eubanks Equation, Modified by Peck

Door

$$S = 1.50(L + C) + 9.27$$

Eubanks Equation, Modified by Peck

Fender

$$S = 1.26(L + C) + 22.95$$

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McNally and Bartlett

Equation for Calculating the Post-Impact Yaw Rotation Rate
Based on the Resulting Angular Displacement

$$\omega = \sqrt{\frac{2\tau_{rot}\theta}{I_{rot}}}$$

ω = Post−Collision Yaw Rotational Velocity

θ = Angular Displacement from Impact to Rest

τ_{rot} = Torque Applied to Vehicle During the Rotation

I_{rot} = Yaw Moment of Inertia about the Point of Rotation

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McNally and Bartlett

$$\tau_{rot} = f W_{rot} d_3$$

f = Coefficient of Friction

W_{rot} = Weight on the Sliding/Rotating Axle

d_3 = Distance from Point of Rotation to Sliding/Rotating Axle

$$I_{rot} = I + m d_1^2$$

I = Yaw Moment of Inertia about the CG

m = Mass of the Vehicle

d_1 = Distance from Point of Rotation to the CG

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McNally and Bartlett

$$\Delta v_{mc} = \frac{I_{rot} g \omega}{W_{mc} d_2}$$

Δv_{mc} = Change in Velocity Experienced by Motorcycle

g = Gravitational Constant

W_{mc} = Weight of the Motorcycle + Portion of Rider

d_2 = Collision Force Moment Arm about the Point of Rotation

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McNally and Bartlett

$$v_{mc} = v_{mc,post}\cos(\psi - \phi) + \sqrt{\Delta v_{mc}^2 - \left(v_{mc,post}\sin(\psi - \phi)\right)^2}$$

$v_{mc} =$ Motorcycle Impact Speed

$v_{mc,post} =$ Motorcycle Post−Impact Speed

$d_2 =$ Collision Force Moment Arm about the Point of Rotation

$\psi =$ Approach Angle of the Motorcycle

$\phi =$ Departure Angle of the Motorcycle

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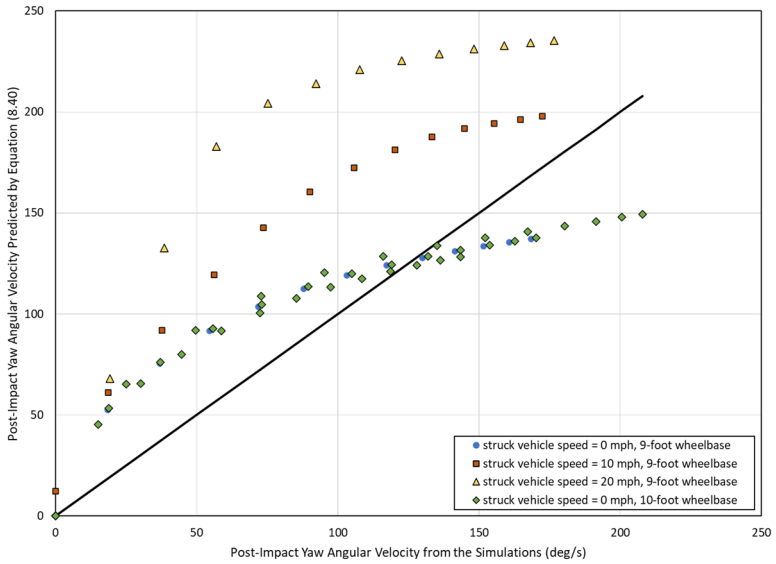
McNally and Bartlett



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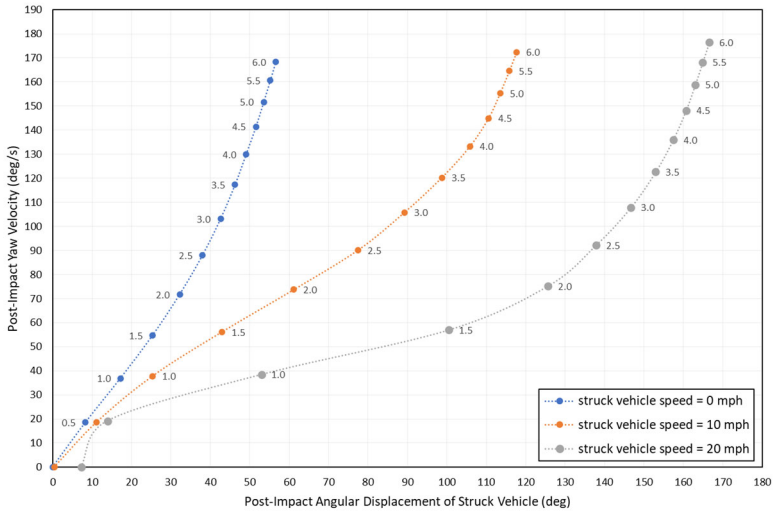
McNally and Bartlett

PC-Crash Simulations (Planar): Motorcycle Impact Behind Struck Vehicle CG, No Wheel Damage to Struck Vehicle, No Impact Induced Steering, 9-foot Wheelbase for Struck Vehicle



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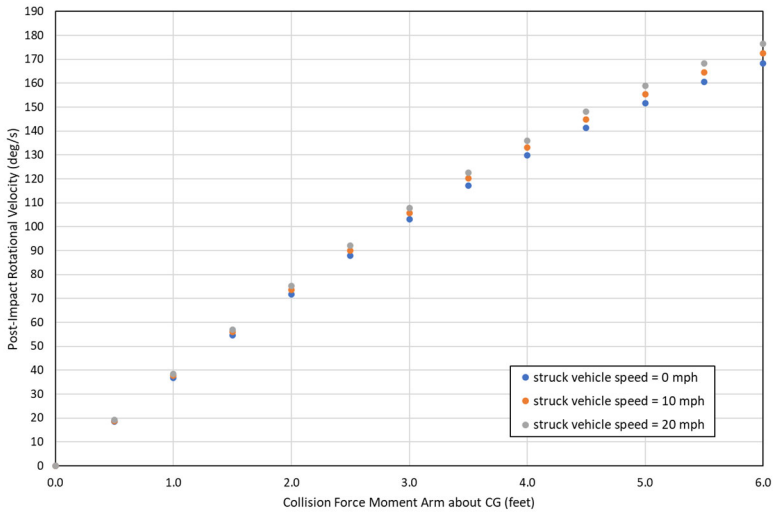
McNally and Bartlett

PC-Crash Simulations (Planar): Motorcycle Impact Behind Struck Vehicle CG, No Wheel Damage to Struck Vehicle, No Impact Induced Steering, 9-foot Wheelbase for Struck Vehicle



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Ogden

$$V_{mc} = \left(\frac{g}{W_{mc} d_i} + \frac{g}{W_{car} d_i} + \frac{d_i}{I_{y,car}} \right) \sqrt{I_{y,car} \theta \cdot \eta \mu W_{car} l_{w,car}}$$

V_{mc} = Impact Speed of the Motorcycle

g = Gravitational Constant

W_{mc} = Weight of Motorcycle + Portion of Rider

W_{car} = Weight of Struck Vehicle

$I_{y,car}$ = Struck Vehicle Yaw Moment of Inertia about CG

d_i = Moment Arm of Collision Force about the CG

θ = Angular Displacement in Radians

μ = Roadway Friction Coefficient

η = Efficiency Factor

$l_{w,car}$ = Struck Vehicle Wheelbase

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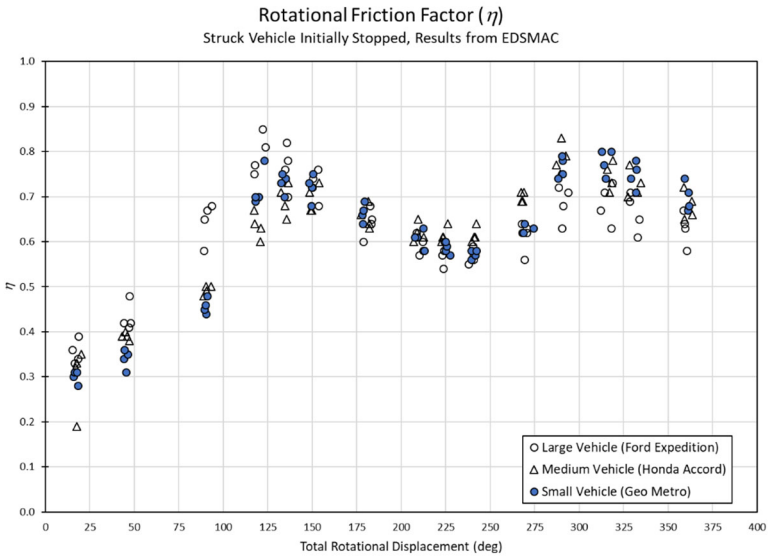
Keifer Data



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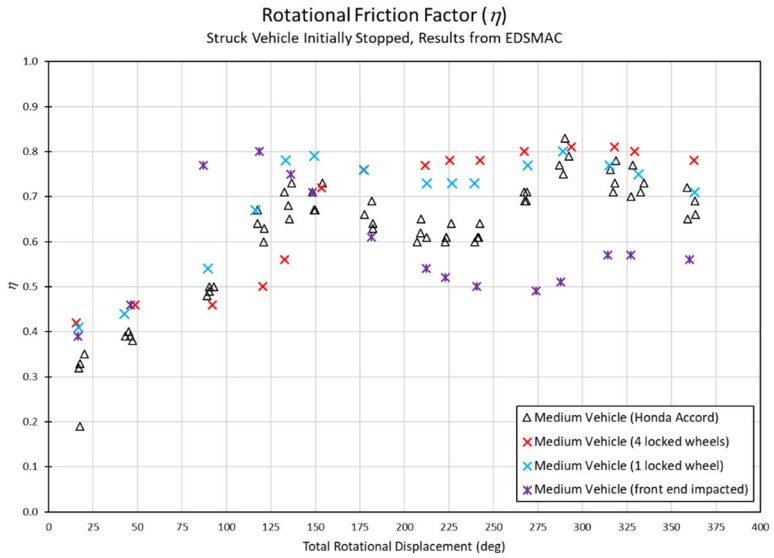

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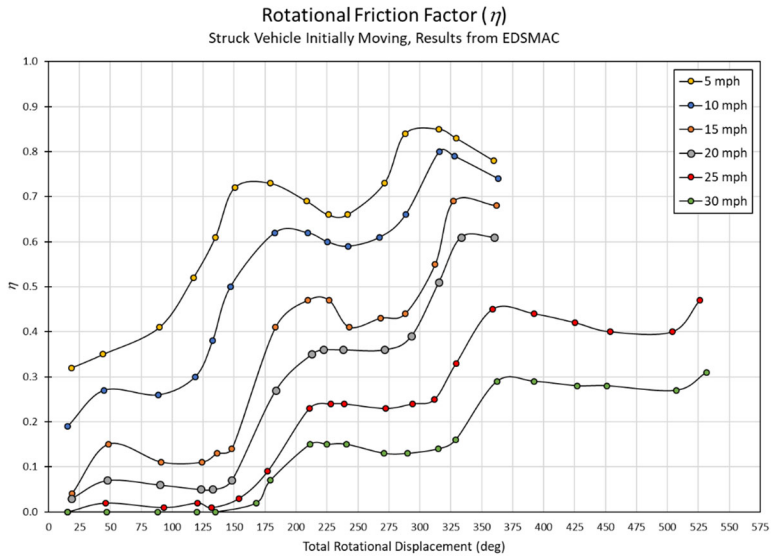

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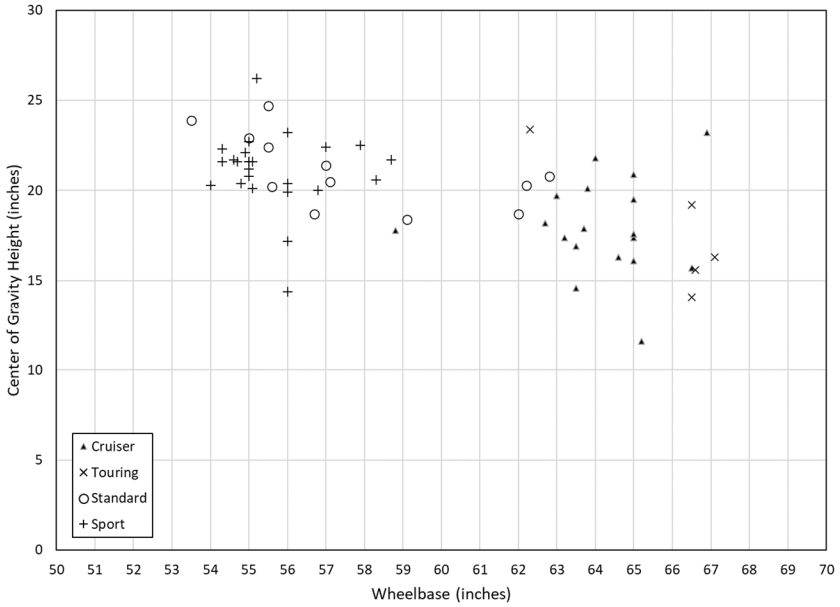
Motorcycle Center of Gravity Height



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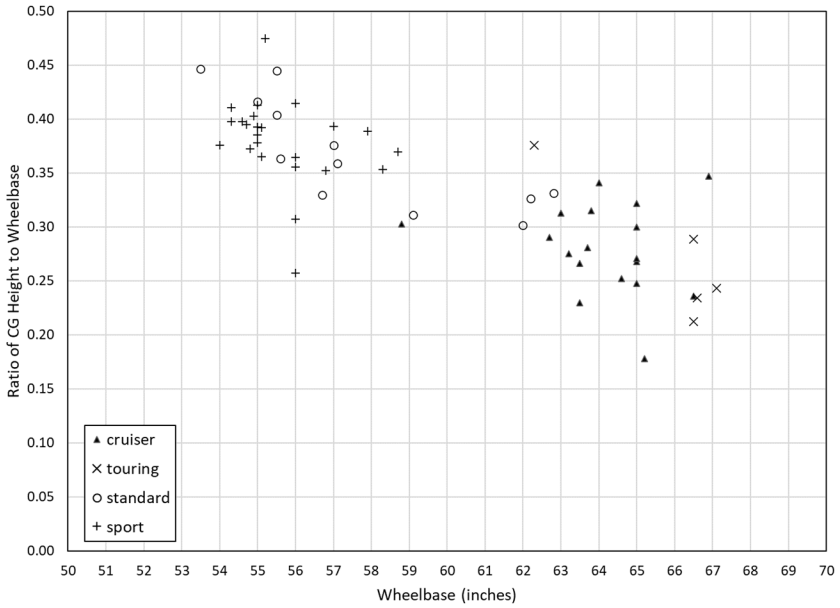
Motorcycle Center of Gravity Height



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Simulation

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PC-Crash Simulation (Impulse-Momentum)

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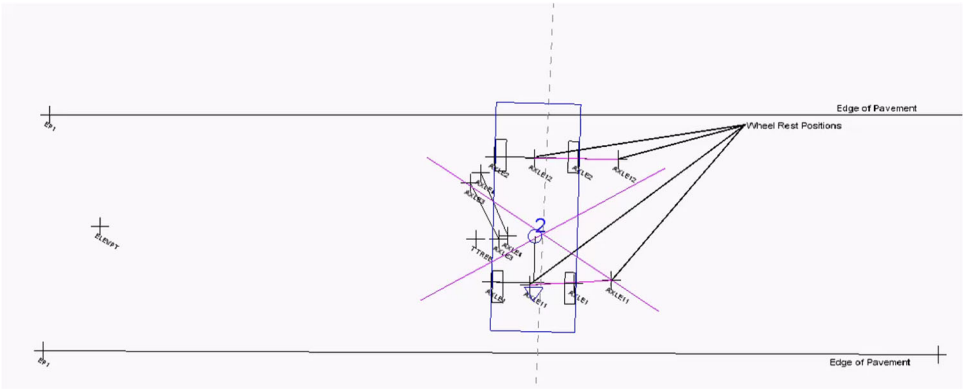

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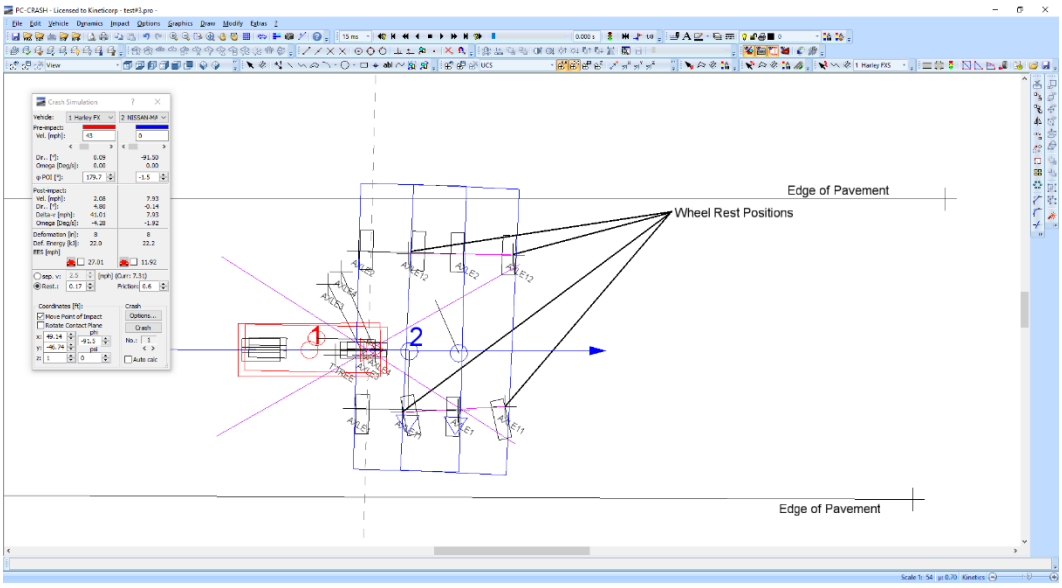

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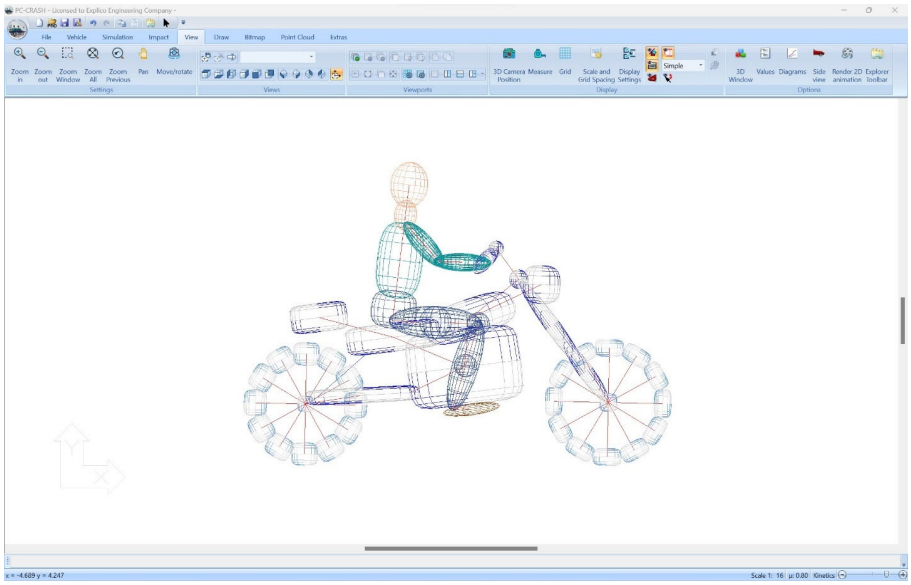

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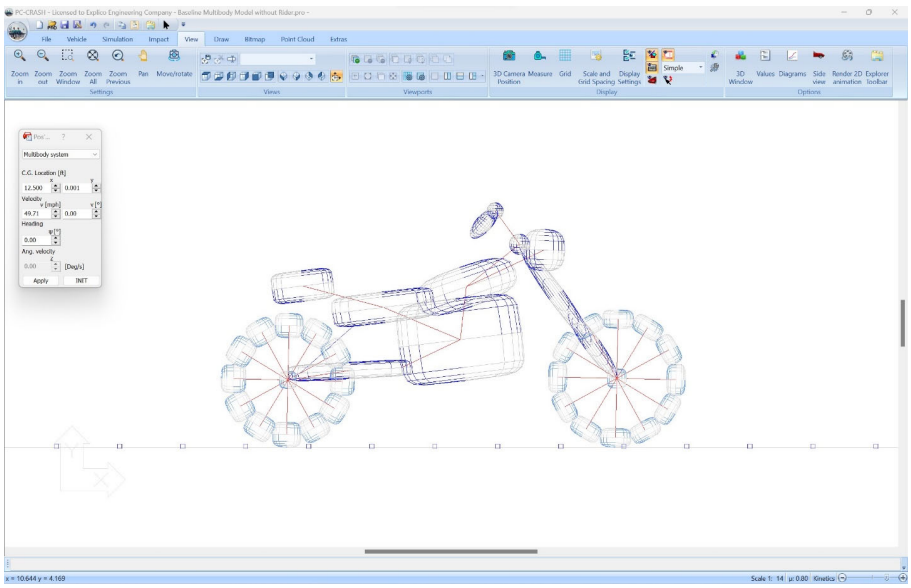

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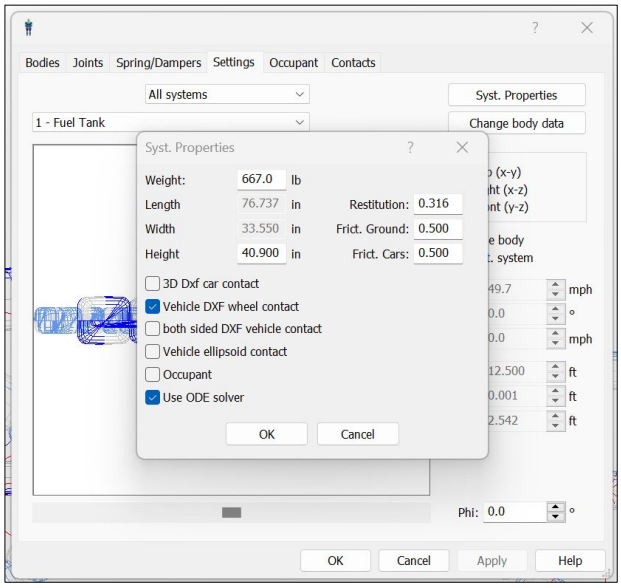

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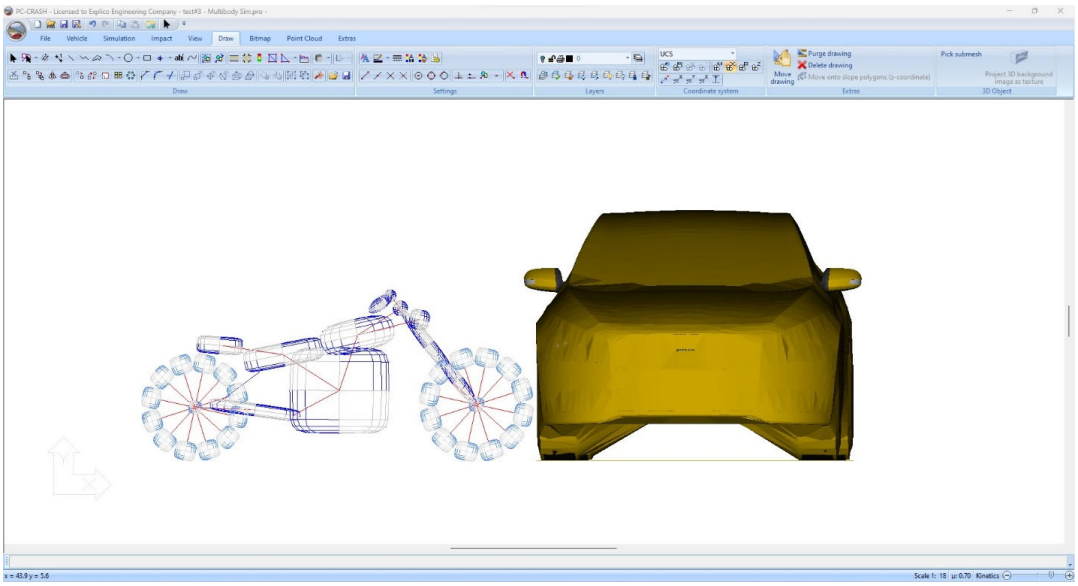

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PC-Crash Simulation (Multibody)

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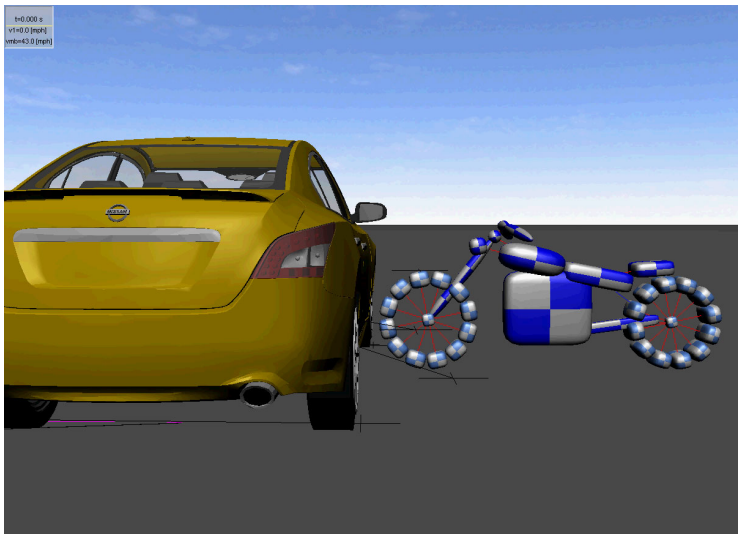

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PC-Crash Simulation (Multibody)

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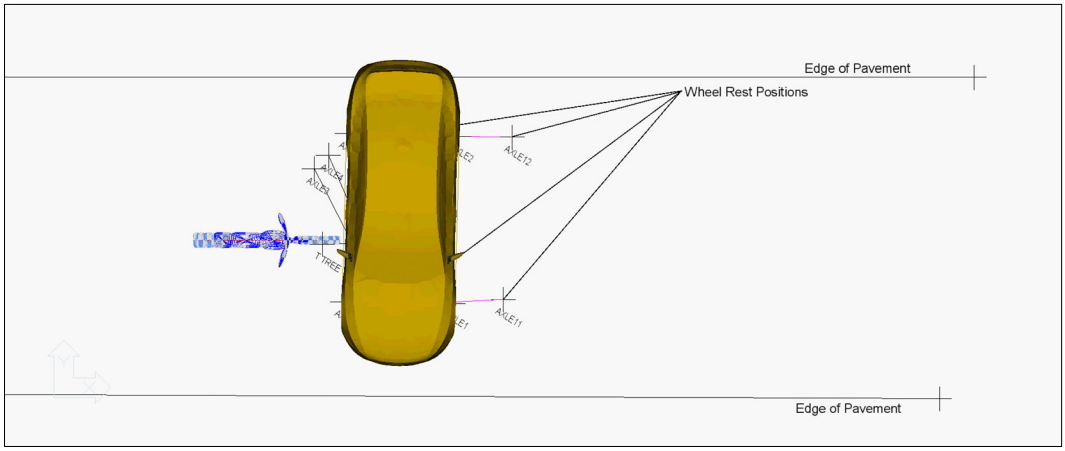

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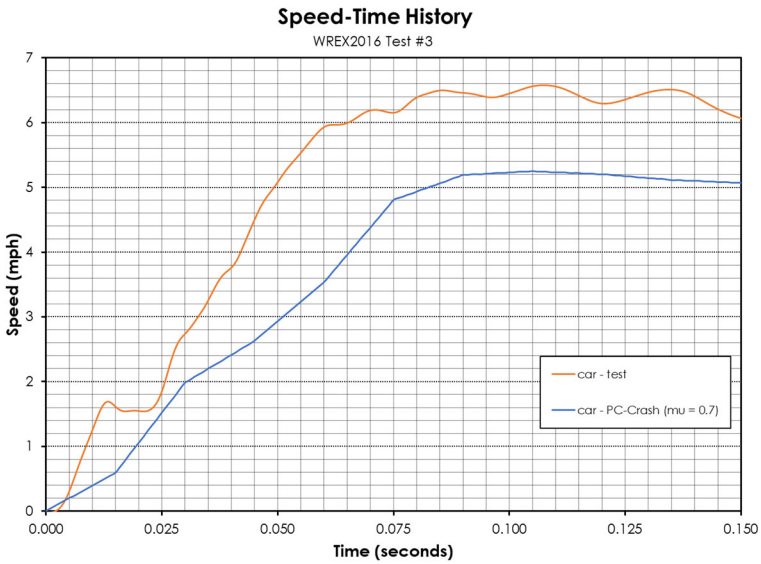

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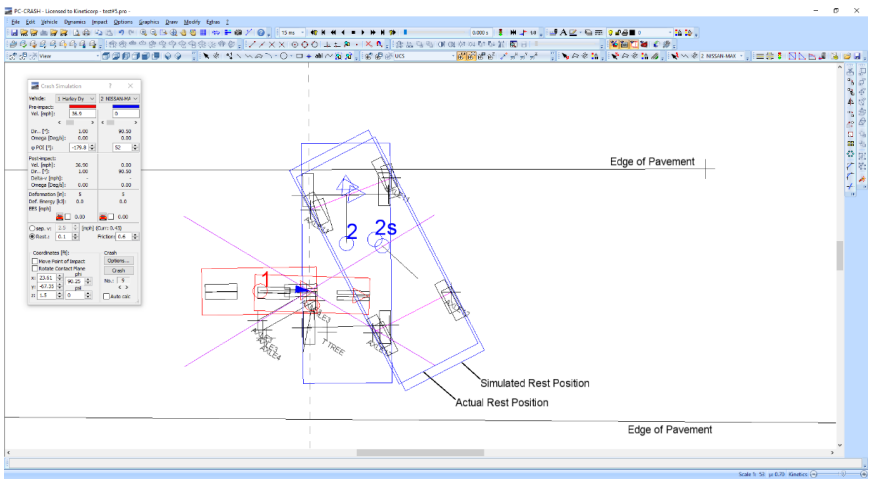

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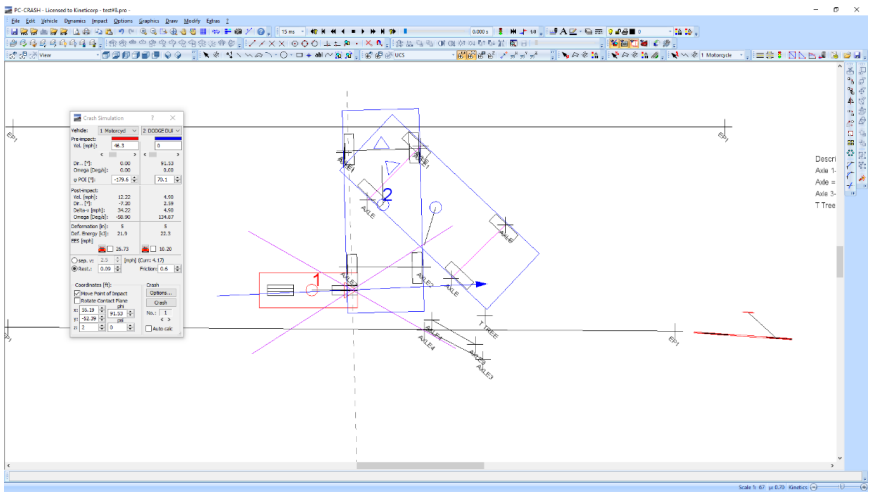

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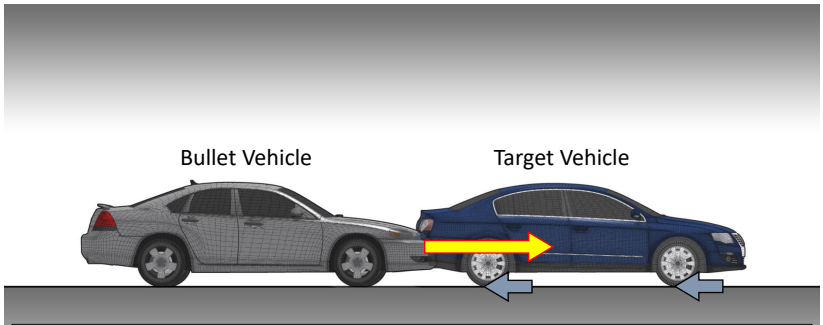
The Effect of Tire Forces



Assume the drivers of both vehicles are braking during the collision…

For the target vehicle in the depicted collision, the tire forces act opposite the collision force. For this vehicle, the tire forces will decrease the ΔV sensed by the ACM.

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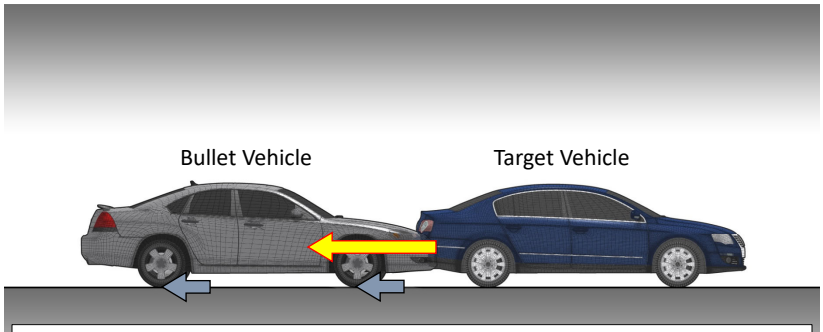
The Effect of Tire Forces



Assume the drivers of both vehicles are braking during the collision…

For the bullet vehicle in the depicted collision, the tire forces act in the same direction as the collision force. For this vehicle, the tire forces will increase the ΔV sensed by the ACM.

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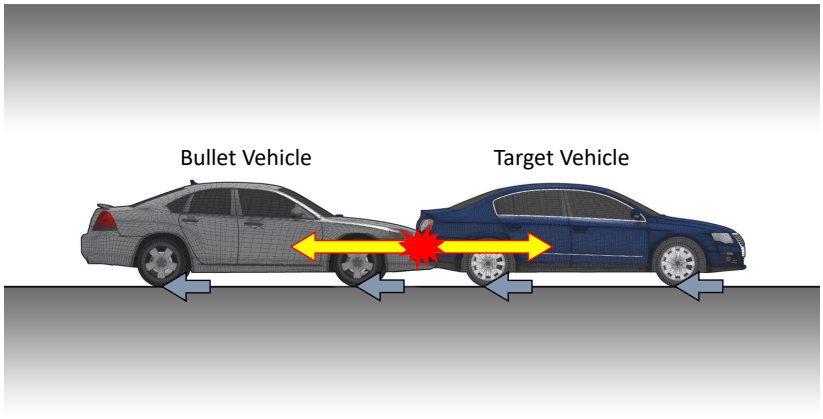
The Effect of Tire Forces



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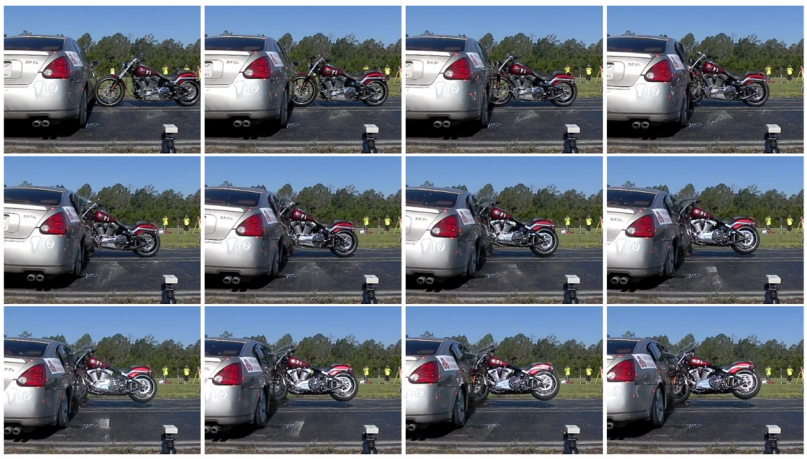
WREX 2016 Test #3 (43.0 mph)



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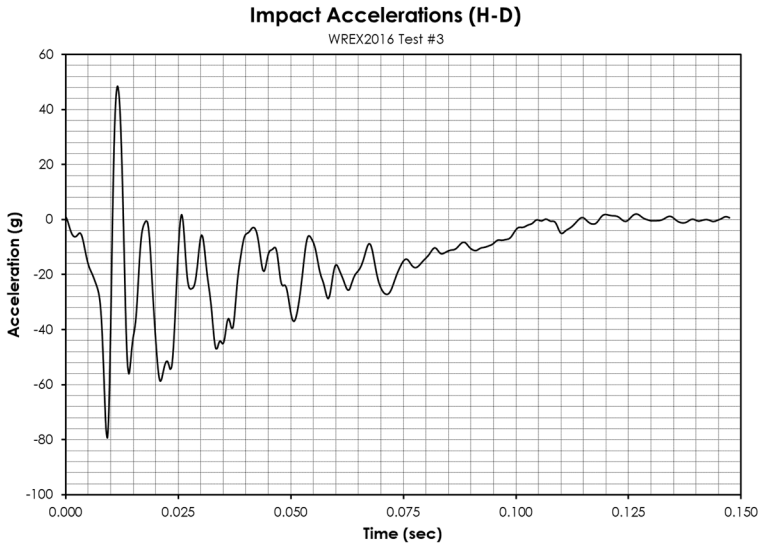
WREX 2016 Test #3 (43.0 mph)



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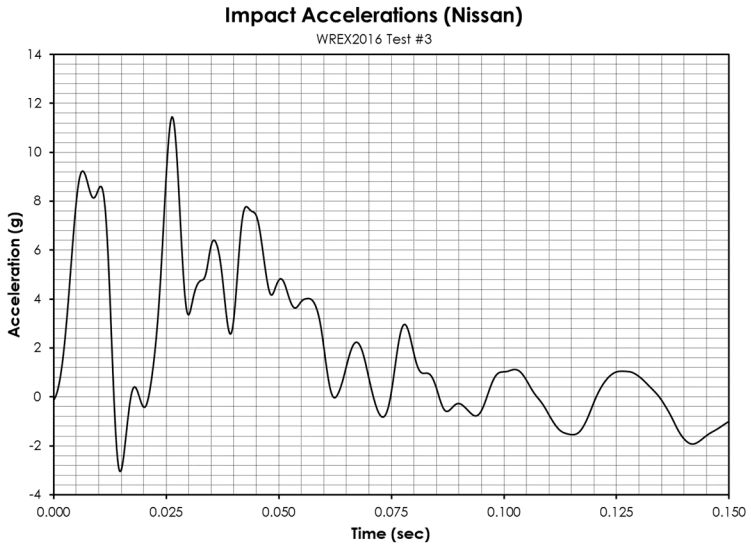

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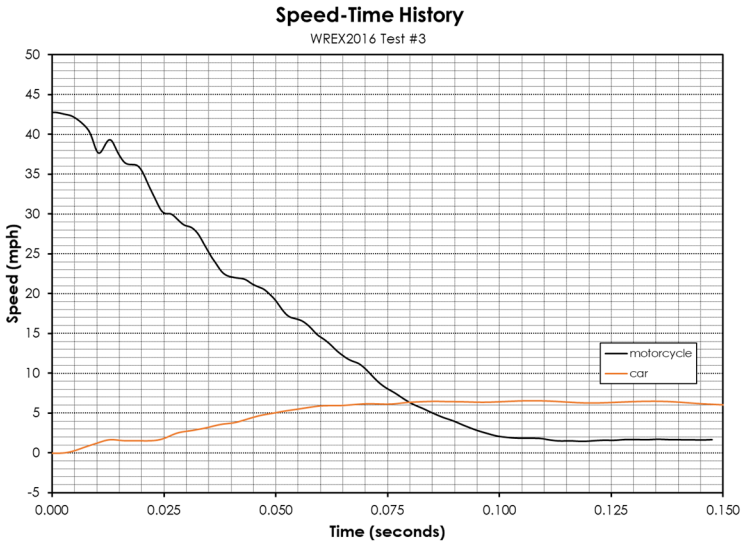
WREX 2016 Test #3 (43.0 mph)



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WREX 2016 Test #3 (43.0 mph)

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WREX 2016 Test #3 (43.0 mph)

Conservation of Energy

$$v_{post,nissan} = \Delta V_{nissan} = \sqrt{2\eta_{trans}\mu g d_{post}}$$

$$v_{post,nissan} = \sqrt{2 \times 0.7 \times 32.2 \times 2.08} = 9.68\ fps = 6.6\ mph$$

Conservation of Momentum

$$v_{pre,mc} = \frac{1}{1+e}\left(1 + \frac{W_{nissan}}{W_{mc}}\right) v_{post,nissan}$$

$$v_{pre,mc} = \frac{1}{1+0.09}\left(1 + \frac{3449}{667}\right) 6.6 = 37.4\ mph$$

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WREX 2016 Test #3 (43.0 mph)

Conservation of Energy

$$v_{post,nissan} = \Delta V_{nissan} = \sqrt{2\eta_{trans}\mu g d_{post}}$$

$$v_{post,nissan} = \sqrt{2 \times 0.7 \times 32.2 \times 2.08} = 9.68\ fps = 6.6\ mph$$

Conservation of Momentum

$$v_{pre,mc} = \frac{1}{1+e}\left(1 + \frac{W_{nissan}}{W_{mc}}\right) v_{post,nissan}$$

Underestimate because we didn't account for tire force.

$$v_{pre,mc} = \frac{1}{1+0.09}\left(1 + \frac{3449}{667}\right) 6.6 = \boxed{37.4\ mph}$$

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Important Concepts related to Tire Forces

- For many common motorcycle-to-car collision scenarios, the tire forces will act opposite the collision force.
- When this is the case, they will lower the measured ΔV.
- In these cases, the EDR-reported ΔV can be used without adjusting for tire forces and the calculated motorcycle impact speed will be conservative (low).
- This is assuming that other error sources are dealt with adequately.

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WREX 2016 Test #3 (43.0 mph)

Estimate the Distance Traveled by the Vehicle during Impact

$$d_{impact} = 0.636 \times \Delta V \times \Delta t$$

$$d_{impact} = 0.636 \times 6.6 \times 1.4667 \times 0.105 = 0.65\ feet$$

Correct the ΔV based on this Distance

$$\Delta V_{corrected} = \sqrt{\Delta V_{edr}^2 + 2\eta_{trans}\mu g d_{impact}}$$

$$\Delta V_{corrected} = \frac{\sqrt{(6.6 \times 1.4667)^2 + 2 \times 1 \times 0.7 \times 32.2 \times 0.65}}{1.4667} = 7.6\ mph$$

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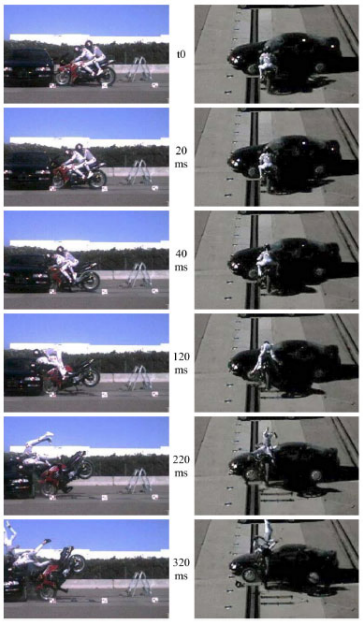
Motorcycles Striking Moving Vehicles



Smith, Frank, Bosch, Fowler, Carter, 2012

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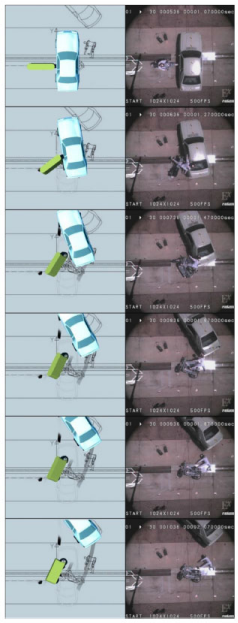
Motorcycles Striking Moving Vehicles



Frank, Smith, Fowler, Carter, Bosch, 2012

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Incorporating Struck Vehicle EDR Data

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The Issue…Significant Weight Discrepancy

Honda CBR1000RR

BMW X3

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Conservation of Momentum

$$\Delta V_{motorcycle} = \Delta V_{car} \times \frac{W_{car}}{W_{motorcycle}}$$

W_{car} = weight of struck car

$W_{motorcycle}$ = weight of motorcycle

ΔV_{car} = change in velocity of car

$\Delta V_{motorcycle}$ = change in velocity of motorcycle

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Conservation of Momentum

$$\Delta V_{motorcycle} = \Delta V_{car} \frac{W_{car}}{W_{motorcycle}}$$

$$\Delta V_{motorcycle} = 12.1\ mph \frac{4{,}370\ lbs}{441\ lbs + 170\ lbs}$$

Assumes the rider strikes the car.

$$\Delta V_{motorcycle} = 12.1\ mph \times 7.15 = 86.5\ mph$$

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Conservation of Momentum

$$\Delta V_{motorcycle} = \Delta V_{car} \frac{W_{car}}{W_{motorcycle}}$$

Every 1 mph of potential error in the BMW ΔV, translates to 7.15 mph of uncertainty in the motorcycle ΔV.

$$\Delta V_{motorcycle} = 12.1\ mph \times 7.15 = 86.5\ mph$$

This goes to 9.91 if we assumes the rider does not strike the car.

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How much of the rider's weight participated?

The answer to this question will affect the weight ratio used for the calculation…

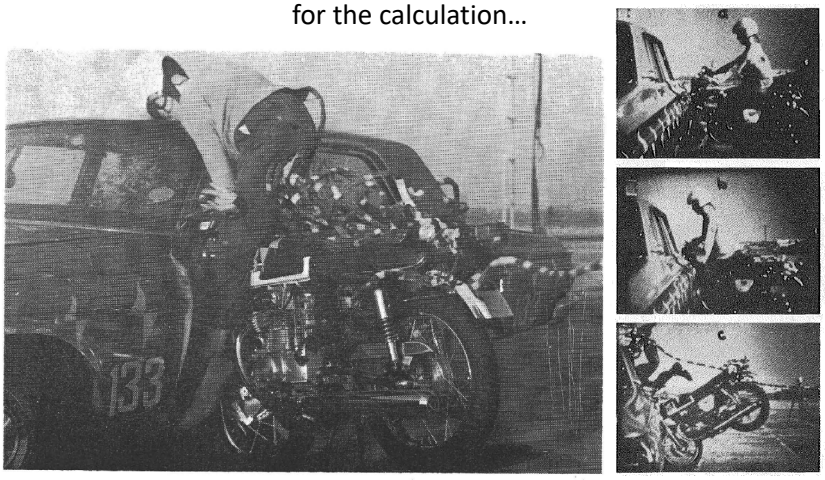

Fig. 32—Honda 350, 40 mph, Experiment 133

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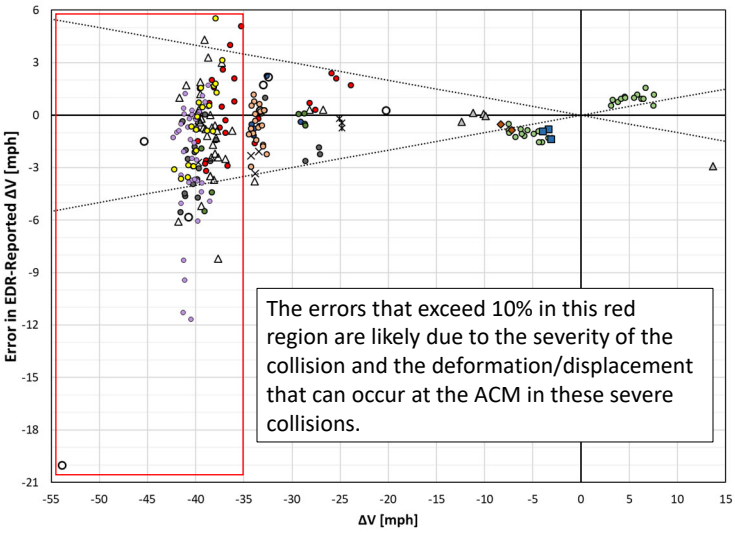

The errors that exceed 10% in this red region are likely due to the severity of the collision and the deformation/displacement that can occur at the ACM in these severe collisions.

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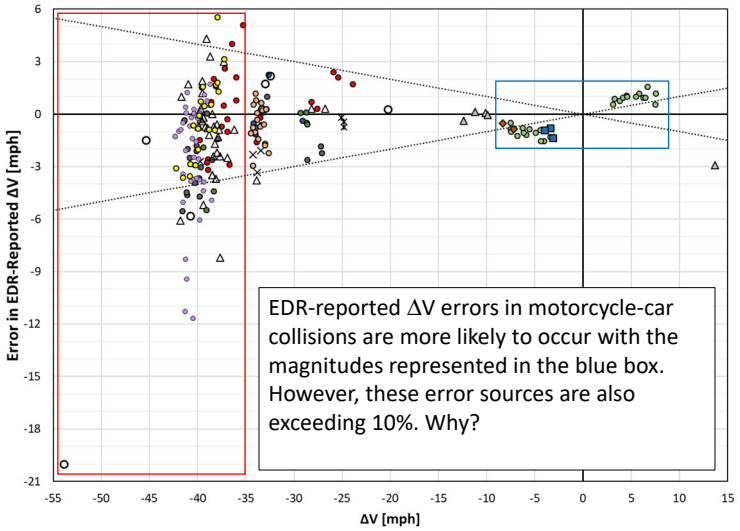

EDR-reported ΔV errors in motorcycle-car collisions are more likely to occur with the magnitudes represented in the blue box. However, these error sources are also exceeding 10%. Why?

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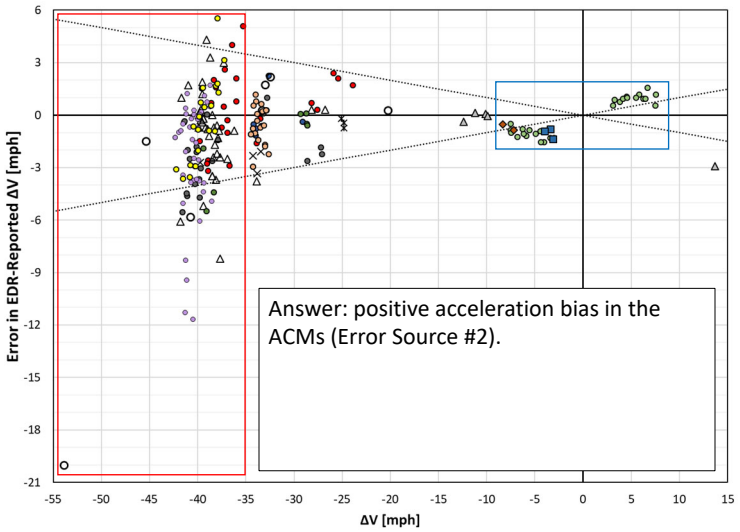

Answer: positive acceleration bias in the ACMs (Error Source #2).

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Beck, R., Casteel, D., Phillips, E., et al., "Motorcycle Collinear Collisions Involving Motor Vehicles Equipped with Event Data Recorders," Collision Magazine 1(1): 82-96, 2006.

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Beck, R., Casteel, D., Phillips, E., et al., "Motorcycle Collinear Collisions Involving Motor Vehicles Equipped with Event Data Recorders," Collision Magazine 1(1): 82-96, 2006.

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	Test 1	Test 2	Test 3
Farmtek Impact velocity (mph)	11.96	27.21	37.10
Stalker Impact velocity (mph)	12.06	27.79	37.24
IST Average *fx* during impact (g)	-0.75	-1.82	-3.93
IST Peak *fx* during impact (g)	-1.70	-6.57	-12.68
IST Peak *fy* during impact (g)	0.53	3.19	4.22
IST Impact duration (ms)	185.6	180.3	96.2
IST Δv (mph)	-3.05	-7.22	-8.31
Separation velocity (mph)	8.91	19.99	28.79
SDM Δv (mph)	n/a	-6.36	-7.78
SDM under-reporting (mph)	n/a	-0.86	-0.53
SDM under-reporting (%)	n/a	11.94	6.35
SDM time from algorithm enable to maximum Δv (ms)	n/a	180.0	97.5
Velocity at about -1 second before algorithm enable (mph)	n/a	26	42

Table 1. Summary of crash test data.

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Error Range for EDR-Reported ΔV

Motorcycle-Passenger Vehicle Collisions

- Longitudinal ΔV: As-Reported to As-Reported Magnitude + 1.5 mph
  - For example, if the EDR reported a longitudinal ΔV of -5 mph, the range would be -5 to -6.5 mph.
- Lateral ΔV: As-Reported ± 1.0 mph (perhaps ± 1.5 mph)
- This assumes that any discrepancy between the ACM and CG locations is dealt with somewhere else in the analysis.
- This also assumes there is no deformation to the area where the ACM is mounted or to the ACM itself.
- This is for struck vehicle ΔVs in the range of ± 10 mph

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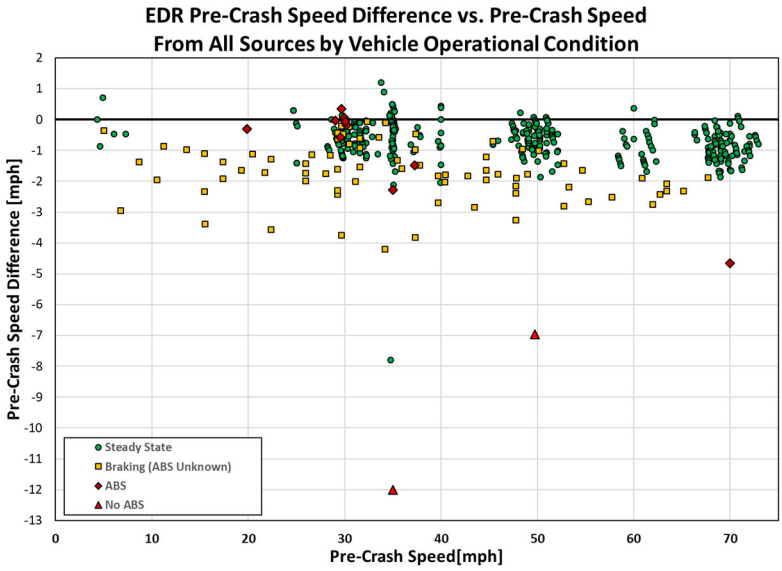

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Example
WREX 2016 Test #5

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WREX 2016 Test #5 (36.9 mph)

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WREX 2016 Test #5 (36.9 mph)

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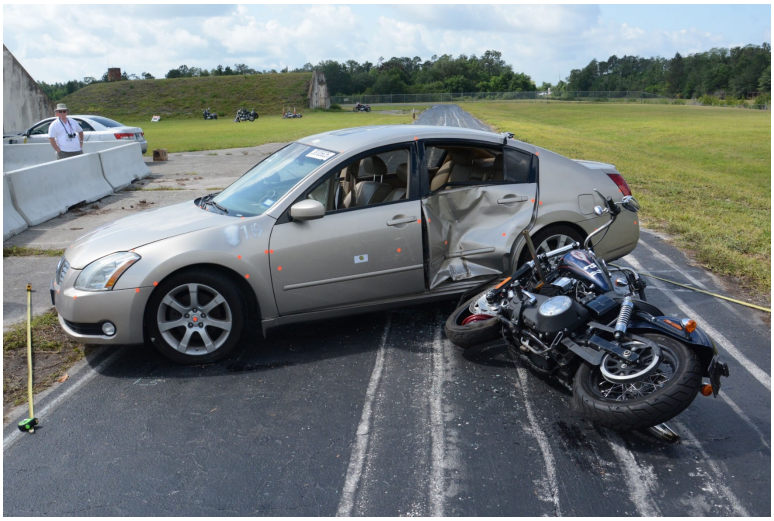
WREX 2016 Test #5 (36.9 mph)



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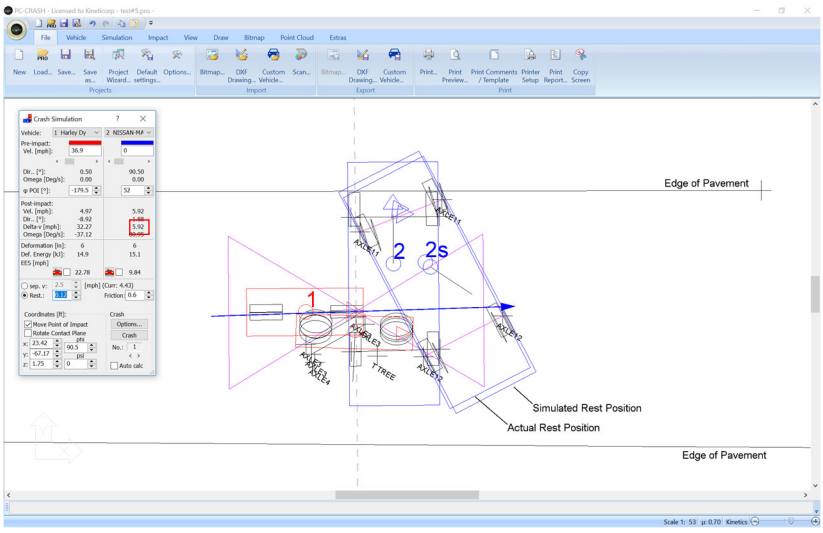
WREX 2016 Test #5 (36.9 mph)



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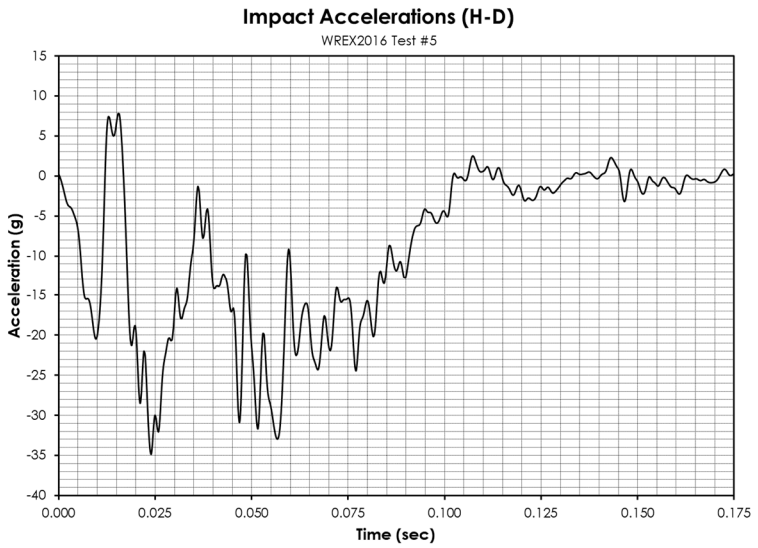
WREX 2016 Test #5 (36.9 mph)



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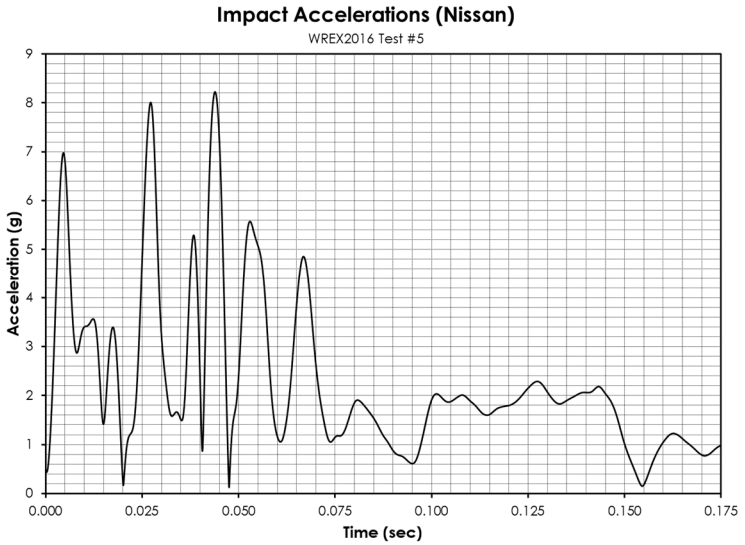
WREX 2016 Test #5 (36.9 mph)



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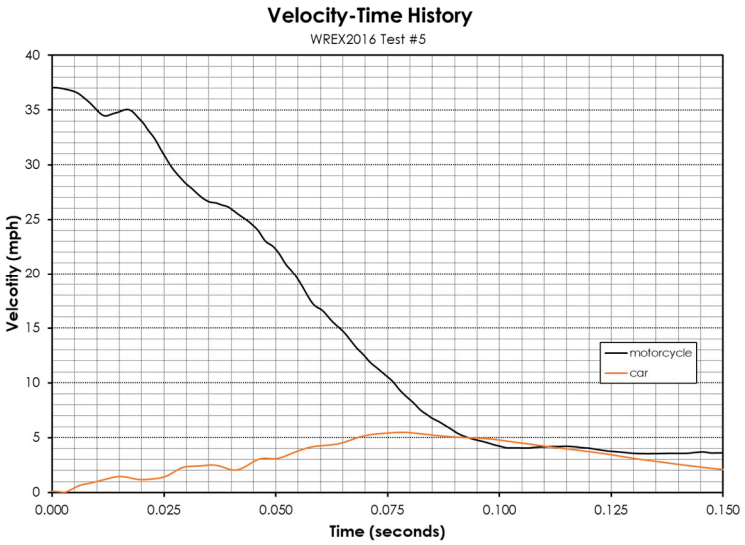
WREX 2016 Test #5 (36.9 mph)



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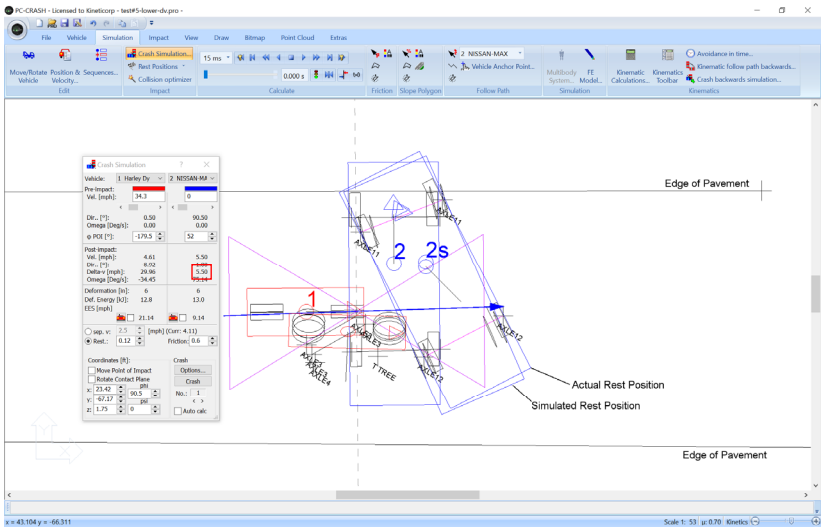
WREX 2016 Test #5 (36.9 mph)



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Post-Collision

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Post-Collision Physical Evidence

- Scrapes or Gouges on the Road Surface
  - Distinguish between Scrapes and Scuffs
- Scuffs on the Road Surface
  - Motorcycle
  - Rider Clothing and Gear
- Fluid on the Road Surface
- Rest Positions
  - Struck Vehicle
  - Motorcycle
  - Rider

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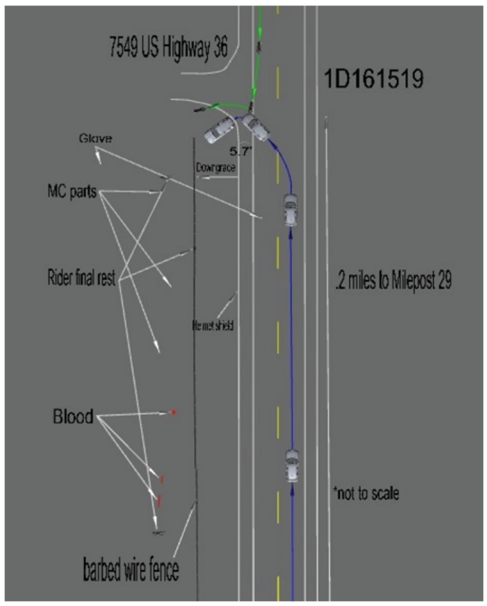

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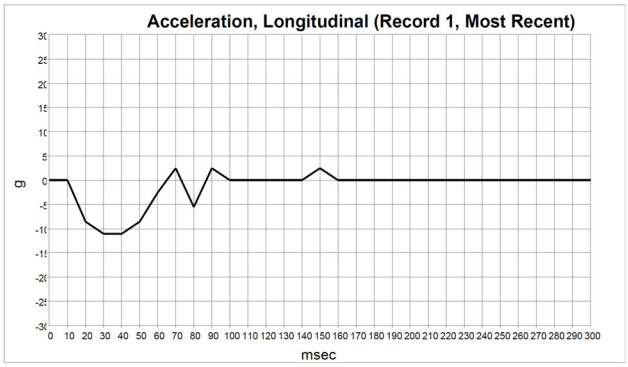

Error Sources #1, #2, #3, and #4 can be ruled out from this graph.

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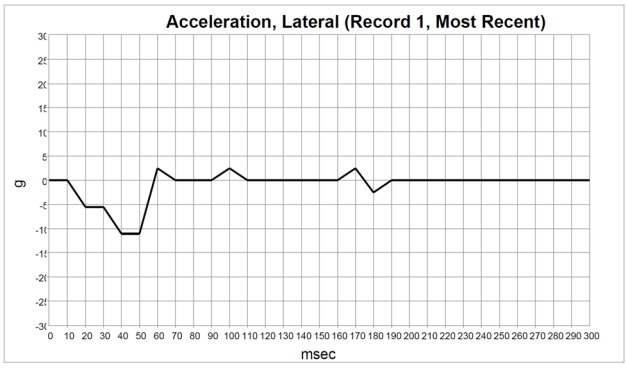

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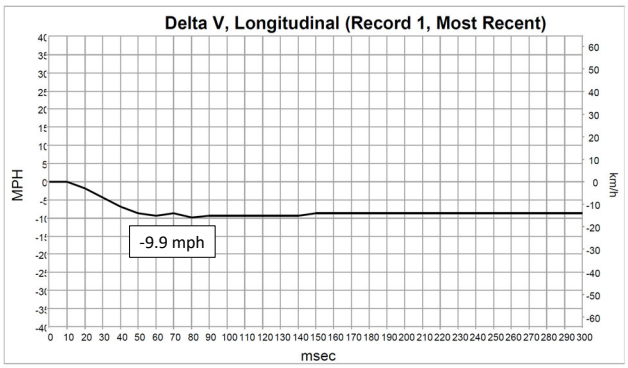

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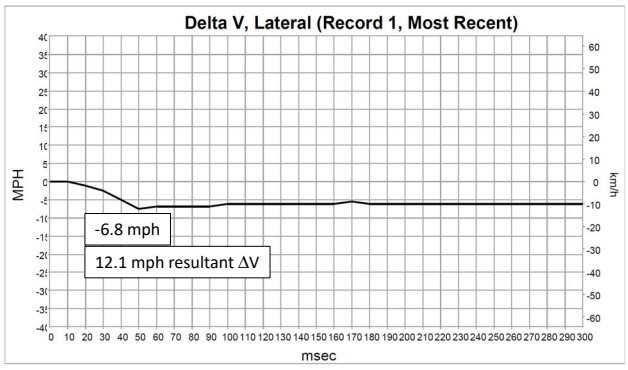

Error Sources #1, #2, #3, and #4 can be ruled out from this graph.

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Case Study #1 – Incorporating Struck Vehicle EDR Data

System Status at Retrieval

Ignition Cycle, Download (cycle)	2,303

System Status at Event (Record 1, Most Recent)

Event Type	Frontal
Ignition ON Timer, at Event (msec)	2,637,448,813
Time From Time Zero to Frontal Threshold (Beginning of Impact) (msec)	Not Recorded
Time From Time Zero to Side Threshold (Beginning of Impact) (msec)	Not Recorded
Time From Time Zero to Algorithm Wake-Up Start (Front) (msec)	0
Time From Time Zero to Algorithm Wake-Up Start (Side) (msec)	4
Time From Time Zero to Algorithm Wake-Up Start (Rear) (msec)	4
Time From Time Zero to Deployment (Rollover) (msec)	Not Recorded
Time From Time Zero to Deployment (Pitchover) (msec)	Not Recorded
Time From Time Zero to Algorithm Wake-Up Start (Pedestrian Protection) (msec)	Not Recorded
Event Counter (counts)	1
Complete File Recorded (Yes, No)	Yes
Multi-Event, Number of Events	1
Time From Previous Event to Current Event (msec)	0
Maximum Delta-V, Longitudinal (MPH [km/h])	-9.9 [-16.0]
Maximum Delta-V, Lateral (MPH [km/h])	-8.1 [-13.0]
Time, Maximum Delta-V, Longitudinal (msec)	80
Time, Maximum Delta-V, Lateral (msec)	48
Time, Maximum Delta-V, Resultant (msec)	82

Error Source #6 can be ruled out because the CDR report indicates that a complete file was recorded. The -8.1 mph maximum lateral ΔV is inconsistent with the tabular and graphical cumulative ΔV data.

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Case Study #1 – Incorporating Struck Vehicle EDR Data

The ACM was under the center console. Error Source #7 was ruled out during inspection of the vehicle by confirming that the center console area was not deformed from the collision.

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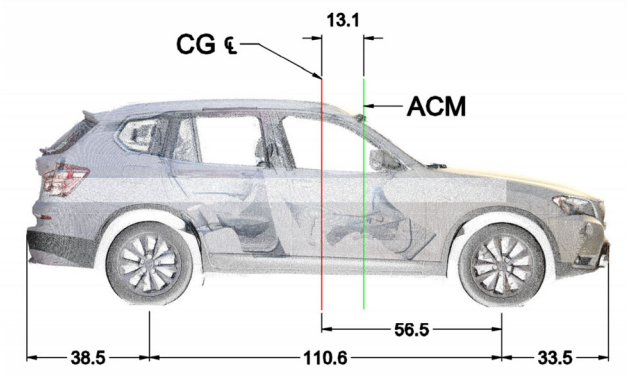

There is a discrepancy between the ACM and CG locations and significant post-impact rotation of the BMW. Error source #5 will need to be considered.

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Case Study #1 – Incorporating Struck Vehicle EDR Data

$$\Delta V_{CG,long} = \Delta V_{ACM,long} + r_y \Delta \omega_{yaw}$$

$$\Delta V_{CG,lat} = \Delta V_{ACM,lat} - r_x \Delta \omega_{yaw}$$

These equations can be used to calculate the ΔV components at the CG based on the ΔV components at the ACM. $\Delta\omega_{yaw}$ is the change in yaw velocity from the impact. r_x and r_y are the coordinates of the ACM relative to the CG.

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Case Study #1 – Incorporating Struck Vehicle EDR Data

$$\Delta V_{CG,long} = \Delta V_{ACM,long} + r_y \Delta \omega_{yaw}$$

$$\Delta V_{CG,lat} = \Delta V_{ACM,lat} - r_x \Delta \omega_{yaw}$$

A positive value of r_x is used if the ACM is located forward of the center of gravity. A positive value of r_y is used if the ACM is situated to the right of the CG. Clockwise rotation is positive. Counter-clockwise rotation is negative.

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Case Study #1 – Incorporating Struck Vehicle EDR Data

$$\Delta V_{CG,long} = \Delta V_{ACM,long} + r_y \Delta \omega_{yaw}$$

$$\Delta V_{CG,lat} = \Delta V_{ACM,lat} - r_x \Delta \omega_{yaw}$$

In this case, r_y is zero, so the longitudinal component of the ΔV does not need corrected.

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Case Study #1 – Incorporating Struck Vehicle EDR Data

$$\Delta V_{CG,lat} = \Delta V_{ACM,lat} - r_x \Delta\omega_{yaw}$$

$$\Delta V_{CG,lat} = -6.8 \text{ mph} - \left(\frac{13.1}{12}\text{ft}\right) \times \left(-100\frac{\text{deg}}{\text{sec}}\right) \times \left(\frac{\pi}{180 \text{ deg}}\right) \times \frac{1 \text{ mph}}{1.4667 \text{ ft/s}}$$

$$\Delta V_{CG,lat} = -5.5 \text{ mph}$$

Resultant ΔV drops from 12.1 mph to 11.3 mph.

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Case Study #1 – Incorporating Struck Vehicle EDR Data

$$\Delta V_2 = \frac{W_{bmw}}{W_{honda}} \Delta V_{bmw} = 7.15 \times 12.1\ mph = 86.5\ mph$$

$$\Delta V_2 = \frac{W_{bmw}}{W_{honda}} \Delta V_{bmw} = 7.15 \times 11.3\ mph = 80.8\ mph$$

Resultant ΔV drops from 12.1 mph to 11.3 mph.

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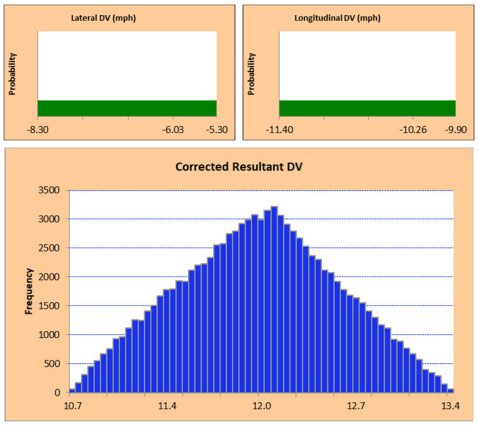

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Case Study #1 – Incorporating Struck Vehicle EDR Data

Pre-Crash Data -5 to 0 sec (Record 1, Most Recent)

Time (sec)	Speed, Vehicle Indicated (MPH [km/h])	Accelerator Pedal, % Full (%)	Engine RPM	Steering Input (deg)	Service Brake, On/Off	ABS Activity (Engaged, Non-engaged)	Stability Control (On Engaged, Non-engaged)
-5.0	35 [56]	0	1200	0	On	No ABS Activity	Non-engaged
-4.5	34 [55]	0	1200	0	On	No ABS Activity	Non-engaged
-4.0	33 [53]	0	1300	0	On	No ABS Activity	Non-engaged
-3.5	31 [50]	0	1300	0	On	No ABS Activity	Non-engaged
-3.0	29 [47]	0	1300	0	On	No ABS Activity	Non-engaged
-2.5	27 [43]	0	1400	0	On	No ABS Activity	Non-engaged
-2.0	25 [40]	0	1300	0	On	No ABS Activity	Non-engaged
-1.5	24 [38]	0	1300	-20	On	No ABS Activity	Non-engaged
-1.0	22 [36]	0	1500	-45	On	No ABS Activity	Non-engaged
-0.5	20 [32]	0	1400	-80	On	No ABS Activity	Non-engaged
0.0	19 [31]	100	1500	-270	Off	No ABS Activity	Non-engaged

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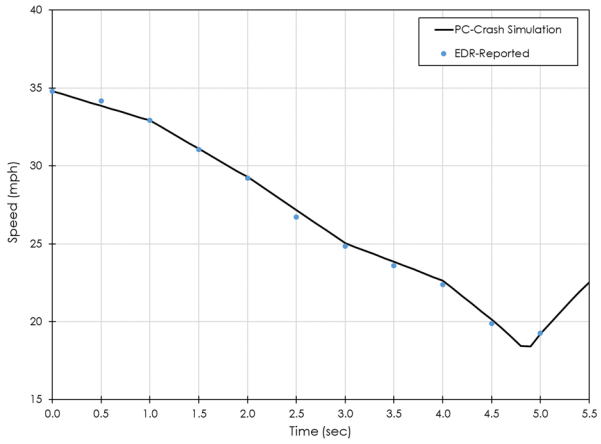

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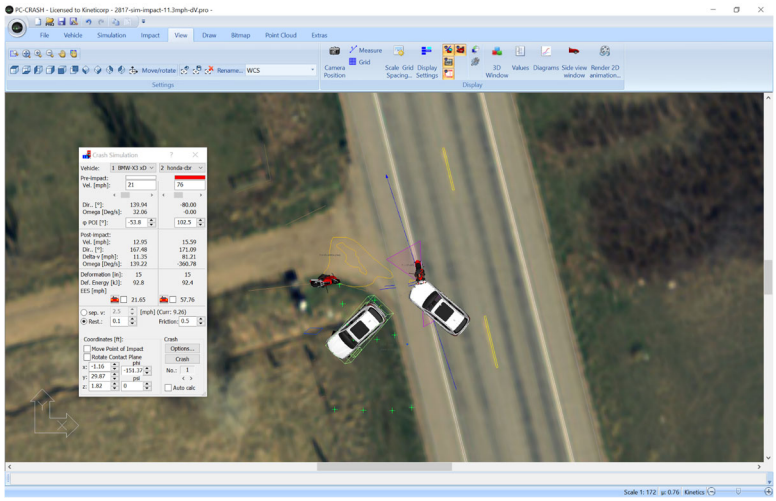

Impact simulation using PC-Crash (low-end speed = 76 mph)

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Case Study #1 – Incorporating Struck Vehicle EDR Data

Impact simulation using PC-Crash (low-end speed = 76 mph)

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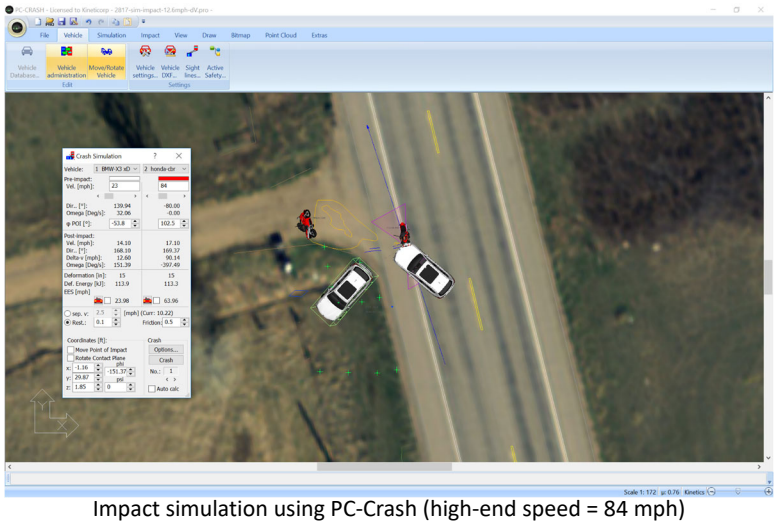

Impact simulation using PC-Crash (high-end speed = 84 mph)

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Case Study #1 – Incorporating Struck Vehicle EDR Data

Impact simulation using PC-Crash (high-end speed = 84 mph)

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Case Study #2 – Incorporating Struck Vehicle EDR Data

Pre-Crash Data (First Record)

Time (sec)	-4	-3	-2	-1	0
Accelerator Pedal Position (%)	0	0	22	40	99
Vehicle Speed (MPH [km/h])	6.0 [9.7]	3.2 [5.1]	4.1 [6.6]	10.8 [17.4]	17.3 [27.9]
ABS Event in Progress	No	No	No	No	No
ESP Event in Progress	No	No	No	No	No
TCS Event in Progress	No	No	No	No	No
Brake Lamp Switch Depressed (from PCM)	Yes	Yes	No	No	No
RCM Serial Number Received by OCS	No	No	No	No	No
OCS Sensor Status	Empty	Empty	Empty	Empty	Empty
OCS System Level 1 Fault	No	No	No	No	No
OCS System Level 2 Fault	No	No	No	No	No
Vehicle Calibration ID	56	56	56	56	56
Vehicle Model Year Calibration ID	07	07	07	07	07

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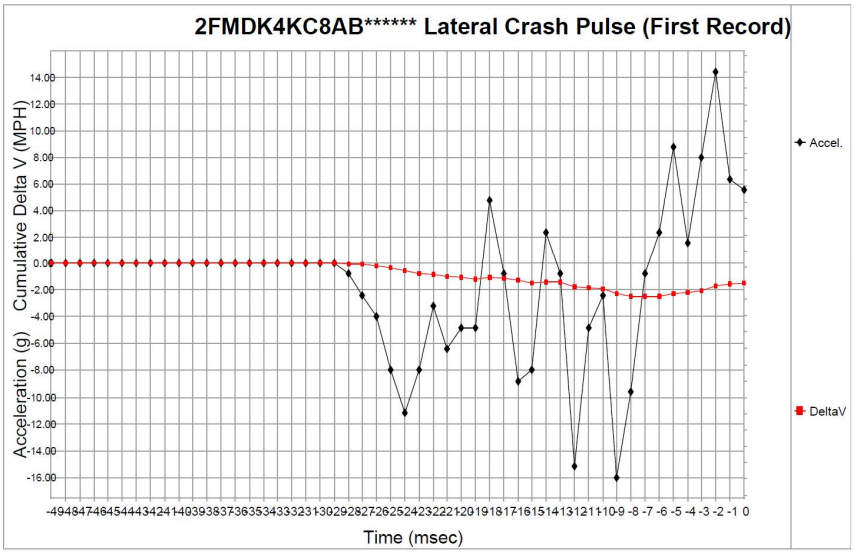
Case Study #2 – Incorporating Struck Vehicle EDR Data



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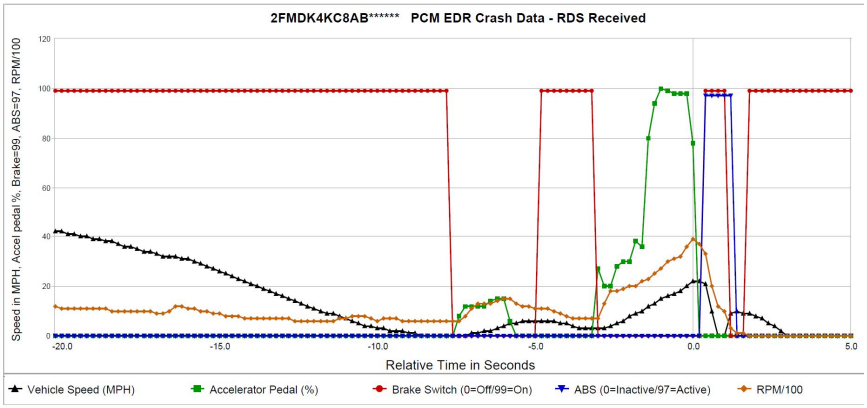
Case Study #2 – Incorporating Struck Vehicle EDR Data



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Case Study #2 – Incorporating Struck Vehicle EDR Data

Buffer Address (Hex)	Relative Time (calc.) (Seconds)	Transmission - Reverse (Reverse / Not Reverse)	Speed Control (On / Off)	Engine RPM (RPM)	Engine Output Torque Calculated (N-m)	Driveline Output Torque Calculated (N-m)	Traction Control (Active / Inactive)	Stability Control (Active / Inactive)	Key On Timer 63.75 Max (sec) (Seconds)
EA000290	-7.2	Not Reverse	OFF	768	51	300	Not Active	Not Active	63.75
EA0002A0	-7.0	Not Reverse	OFF	1113	71	434	Not Active	Not Active	63.75
EA0002B0	-6.8	Not Reverse	OFF	1287	69	401	Not Active	Not Active	63.75
EA0002C0	-6.6	Not Reverse	OFF	1301	58	316	Not Active	Not Active	63.75
EA0002D0	-6.4	Not Reverse	OFF	1289	64	342	Not Active	Not Active	63.75
EA0002E0	-6.2	Not Reverse	OFF	1385	84	448	Not Active	Not Active	63.75
EA0002F0	-6.0	Not Reverse	OFF	1507	90	476	Not Active	Not Active	63.75
EA000300	-5.8	Not Reverse	OFF	1485	81	386	Not Active	Not Active	63.75
EA000310	-5.6	Not Reverse	OFF	1330	42	156	Not Active	Not Active	63.75
EA000320	-5.4	Not Reverse	OFF	1205	36	124	Not Active	Not Active	63.75
EA000330	-5.2	Not Reverse	OFF	1152	25	69	Not Active	Not Active	63.75
EA000340	-5.0	Not Reverse	OFF	1138	25	69	Not Active	Not Active	63.75
EA000350	-4.8	Not Reverse	OFF	1120	-1	-31	Not Active	Not Active	63.75
EA000360	-4.6	Not Reverse	OFF	1061	-5	-63	Not Active	Not Active	63.75
EA000370	-4.4	Not Reverse	OFF	970	-8	-77	Not Active	Not Active	63.75
EA000380	-4.2	Not Reverse	OFF	872	-2	-56	Not Active	Not Active	63.75
EA000390	-4.0	Not Reverse	OFF	818	2	-33	Not Active	Not Active	63.75
EA0003A0	-3.8	Not Reverse	OFF	728	5	-25	Not Active	Not Active	63.75
EA0003B0	-3.6	Not Reverse	OFF	662	20	37	Not Active	Not Active	63.75
EA0003C0	-3.4	Not Reverse	OFF	729	28	89	Not Active	Not Active	63.75
EA0003D0	-3.2	Not Reverse	OFF	714	13	14	Not Active	Not Active	63.75
EA0003E0	-3.0	Not Reverse	OFF	728	5	-37	Not Active	Not Active	63.75
EA0003F0	-2.8	Not Reverse	OFF	1258	121	626	Not Active	Not Active	63.75
EA000400	-2.6	Not Reverse	OFF	1765	133	752	Not Active	Not Active	63.75
EA000410	-2.4	Not Reverse	OFF	1772	122	637	Not Active	Not Active	63.75
EA000420	-2.2	Not Reverse	OFF	1946	173	893	Not Active	Not Active	63.75
EA000430	-2.0	Not Reverse	OFF	1967	158	748	Not Active	Not Active	63.75
EA000440	-1.8	Not Reverse	OFF	2015	166	751	Not Active	Not Active	63.75
EA000450	-1.6	Not Reverse	OFF	2206	187	817	Not Active	Not Active	63.75
EA000460	-1.4	Not Reverse	OFF	2348	191	801	Not Active	Not Active	63.75
EA000470	-1.2	Not Reverse	OFF	2535	214	878	Not Active	Not Active	63.75
EA000480	-1.0	Not Reverse	OFF	2744	216	884	Not Active	Not Active	63.75
EA000490	-0.8	Not Reverse	OFF	2958	225	920	Not Active	Not Active	63.75
EA0004A0	-0.6	Not Reverse	OFF	3086	226	927	Not Active	Not Active	63.75
EA0004B0	-0.4	Not Reverse	OFF	3212	228	933	Not Active	Not Active	63.75
EA0004C0	-0.2	Not Reverse	OFF	3629	234	961	Not Active	Not Active	63.75
EA0004D0	0.0	Not Reverse	OFF	3858	242	991	Not Active	Not Active	63.75
EA0004E0	0.2	Not Reverse	OFF	3725	218	889	Not Active	Active	63.75
EA0004F0	0.4	Not Reverse	OFF	3301	49	159	Not Active	Active	63.75
EA000500	0.6	Not Reverse	OFF	2039	62	241	Not Active	Active	63.75
EA000510	0.8	Not Reverse	OFF	1232	61	355	Not Active	Active	63.75
EA000520	1.0	Not Reverse	OFF	955	60	348	Not Active	Active	63.75
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EA000540	1.4	Not Reverse	OFF	105	89	135	Not Active	Active	63.75
EA000550	1.6	Not Reverse	OFF	105	114	448	Not Active	Not Active	63.75
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EA000570	2.0	Not Reverse	OFF	0	117	828	Not Active	Not Active	63.75
EA000580	2.2	Not Reverse	OFF	0	117	825	Not Active	Not Active	63.75
EA000590	2.4	Not Reverse	OFF	0	117	841	Not Active	Not Active	63.75
EA0005A0	2.6	Not Reverse	OFF	0	117	833	Not Active	Not Active	63.75
EA0005B0	2.8	Not Reverse	OFF	0	117	824	Not Active	Not Active	63.75
EA0005C0	3.0	Not Reverse	OFF	0	117	813	Not Active	Not Active	63.75
EA0005D0	3.2	Not Reverse	OFF	0	117	813	Not Active	Not Active	63.75
EA0005E0	3.4	Not Reverse	OFF	0	117	813	Not Active	Not Active	63.75
EA0005F0	3.6	Not Reverse	OFF	0	117	813	Not Active	Not Active	63.75
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EA000640	4.6	Not Reverse	OFF	0	117	813	Not Active	Not Active	63.75
EA000650	4.8	Not Reverse	OFF	0	117	813	Not Active	Not Active	63.75
EA000660	5.0	Not Reverse	OFF	0	117	813	Not Active	Not Active	63.75

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Case Study #2 – Incorporating Struck Vehicle EDR Data

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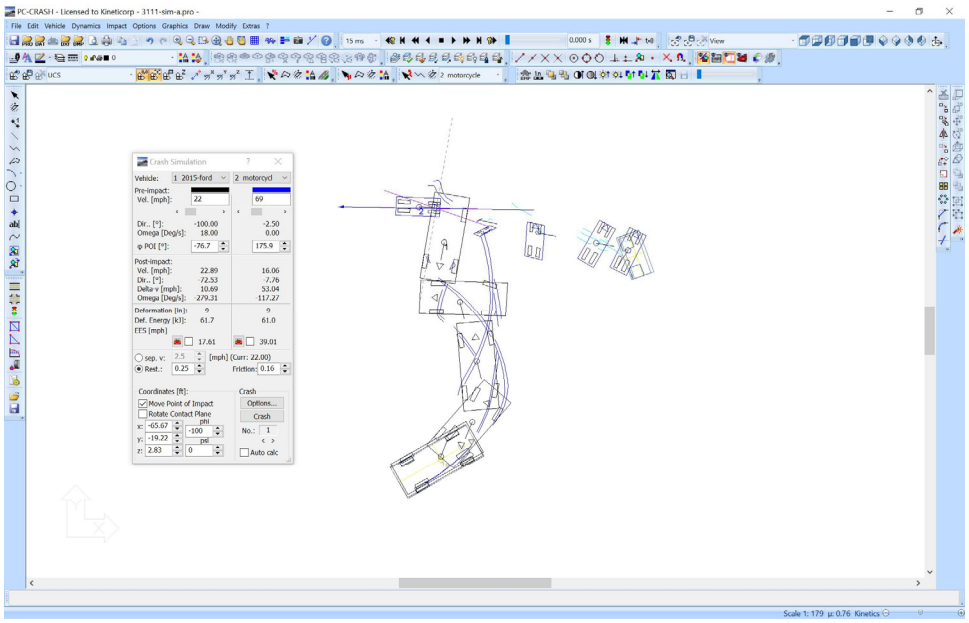

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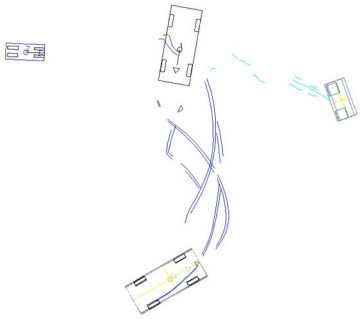

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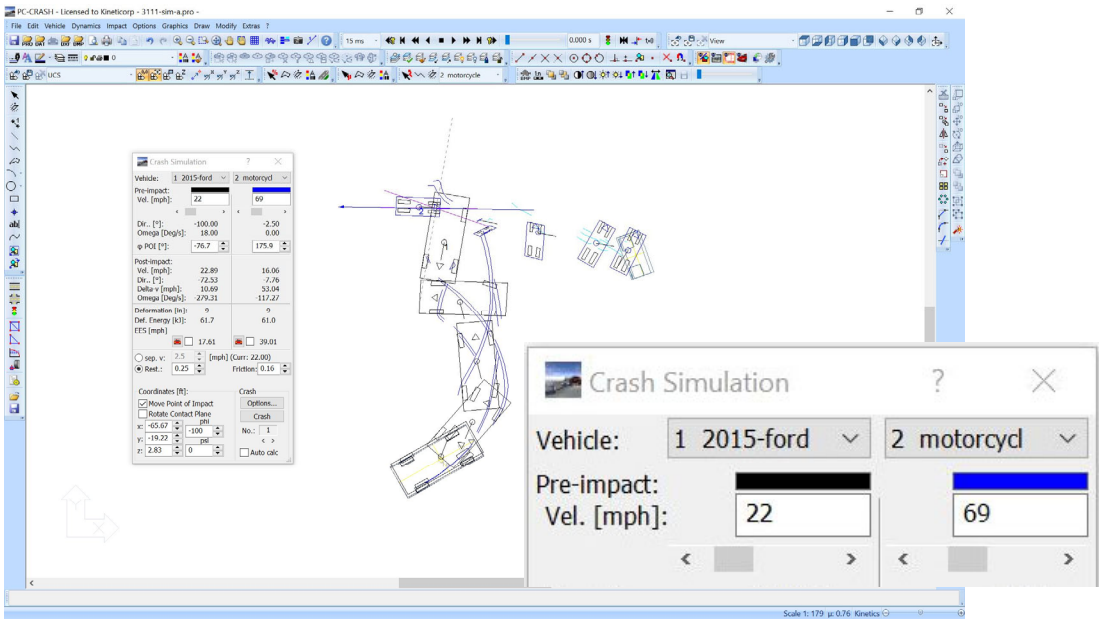

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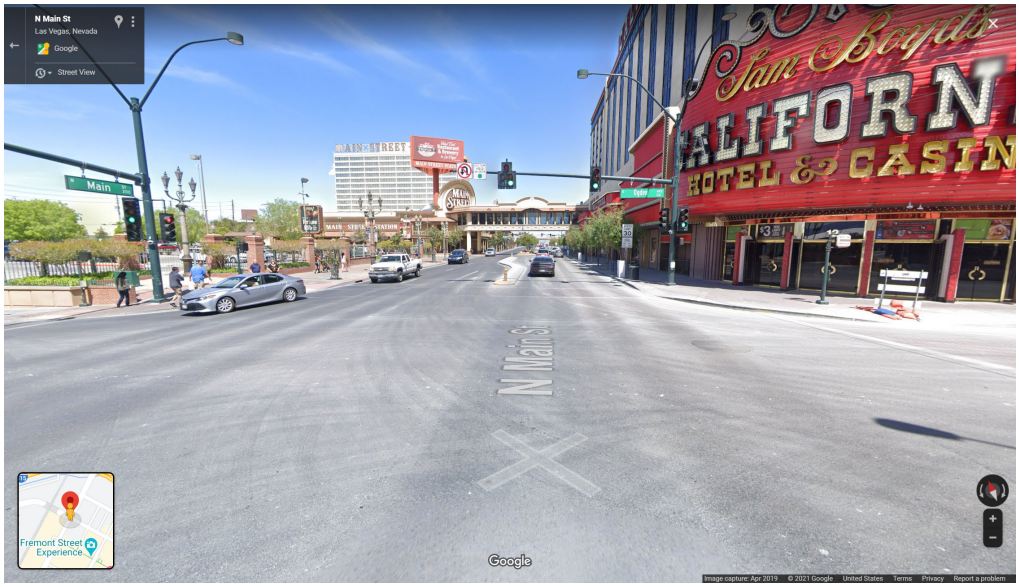

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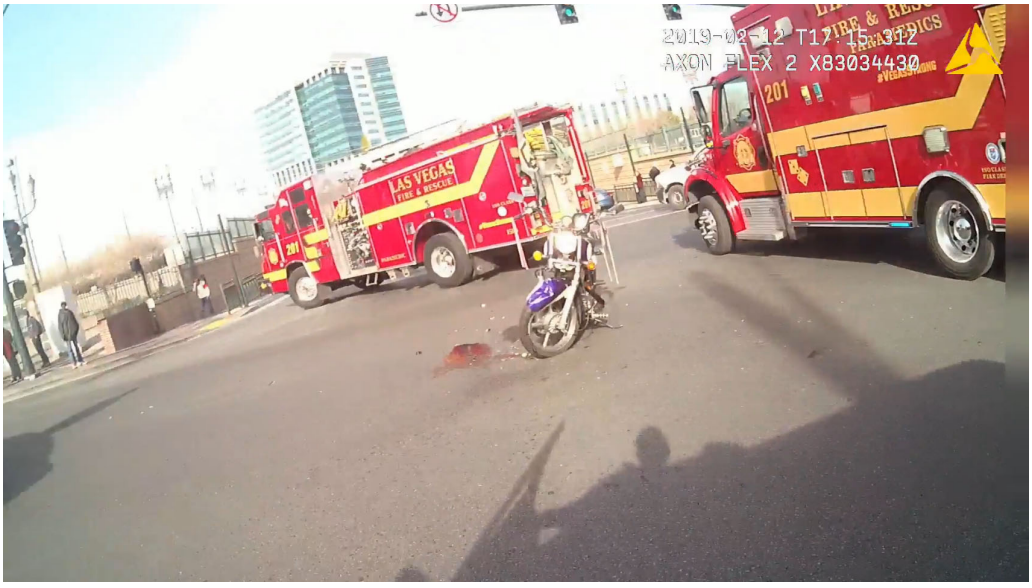

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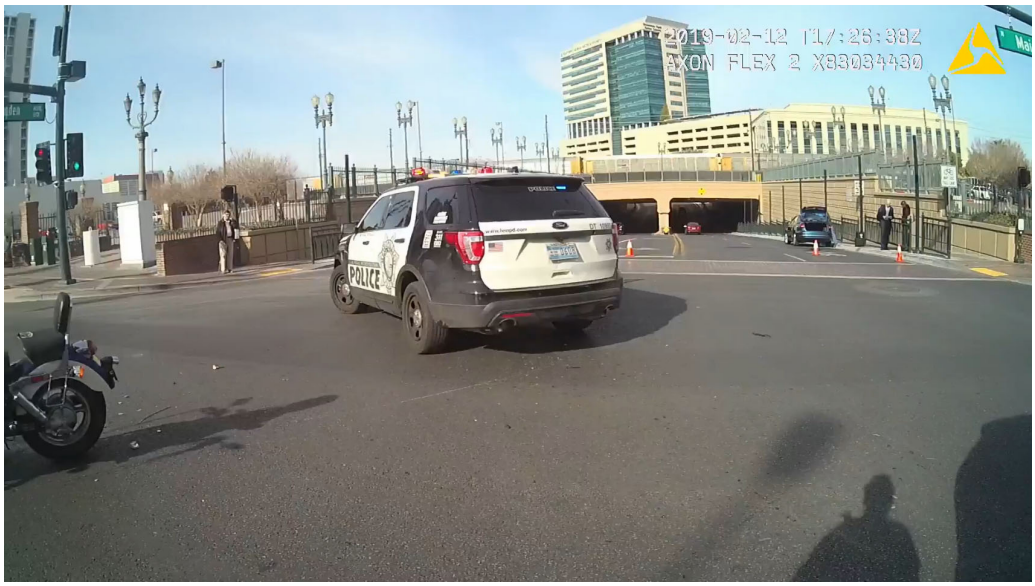

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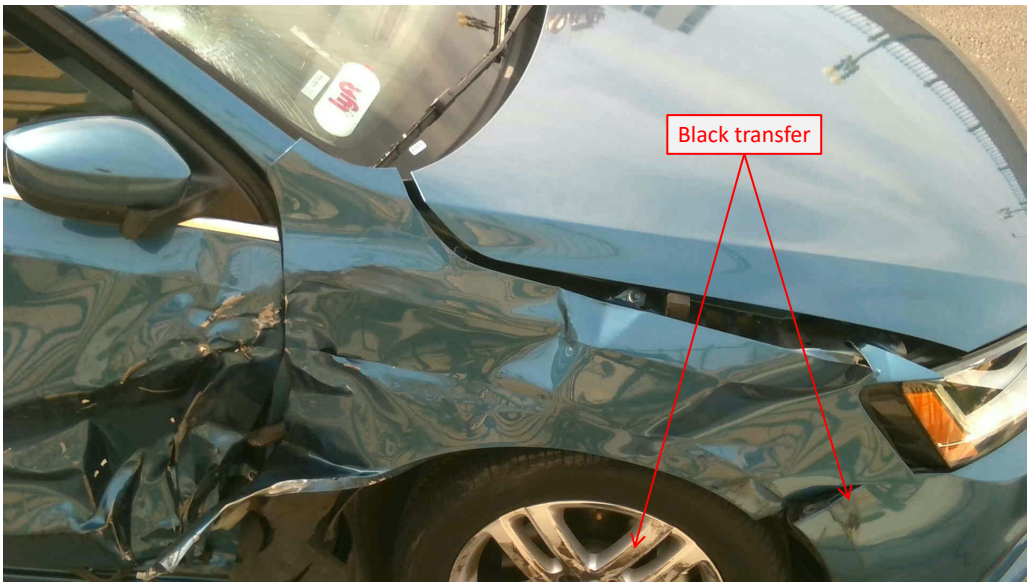

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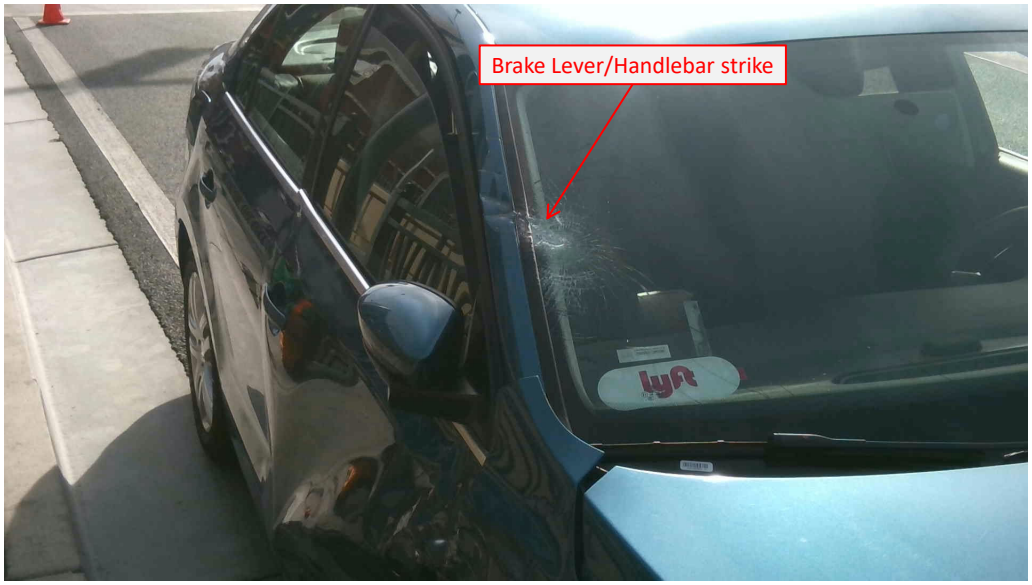

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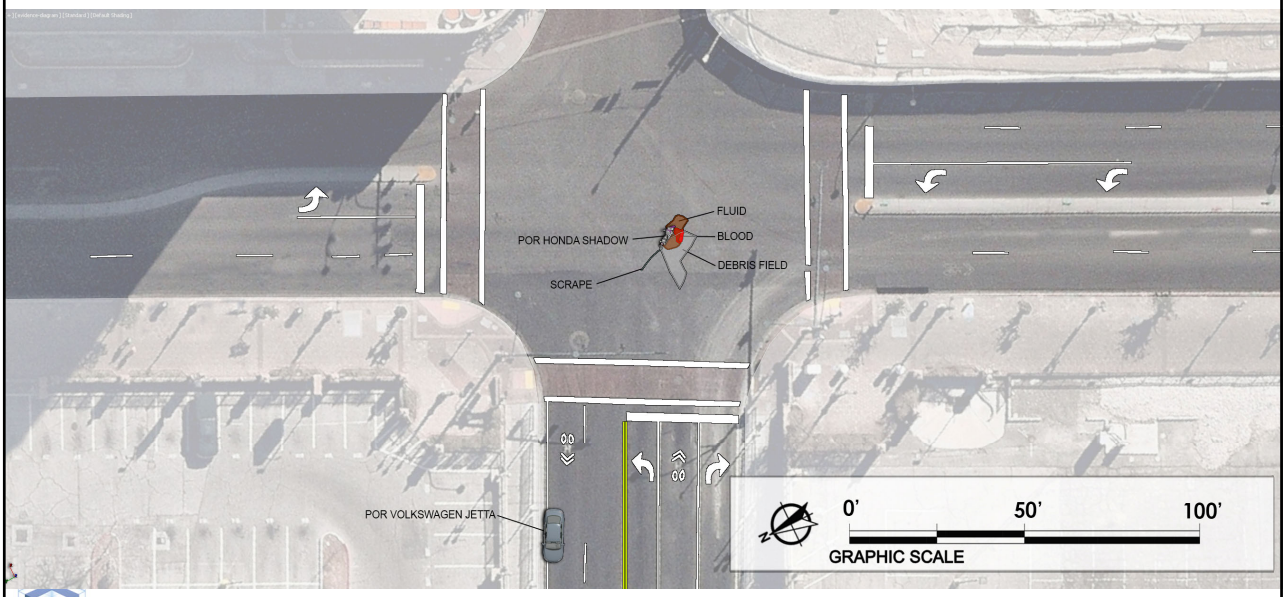

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Possible Analysis Methods

1. Damage analysis/wheelbase reduction methods
   a. Not applicable when the motorcycle forks are broken from twisting/torquing.
   b. Majority of the test data was collected with a stationary struck vehicle (moving struck vehicle is often what creates the torquing of the motorcycle forks).

2. Post-impact translation/rotation of struck vehicle
   a. Subject collision did not cause appreciable post-impact rotation of the struck vehicle.

3. Rigid body simulation
   a. Cannot consider rotation of the motorcycle front wheel, forks, and handlebars relative to the rest of the motorcycle.
   b. Cannot consider individual contact points on the struck vehicle.
   c. These are not significant limitations if the struck vehicle rotates significantly, but it didn't in this case.

4. Multibody simulation
   a. Experimental method that can incorporate rotation of front wheel, forks, and handlebars, and can incorporate individual contact points.

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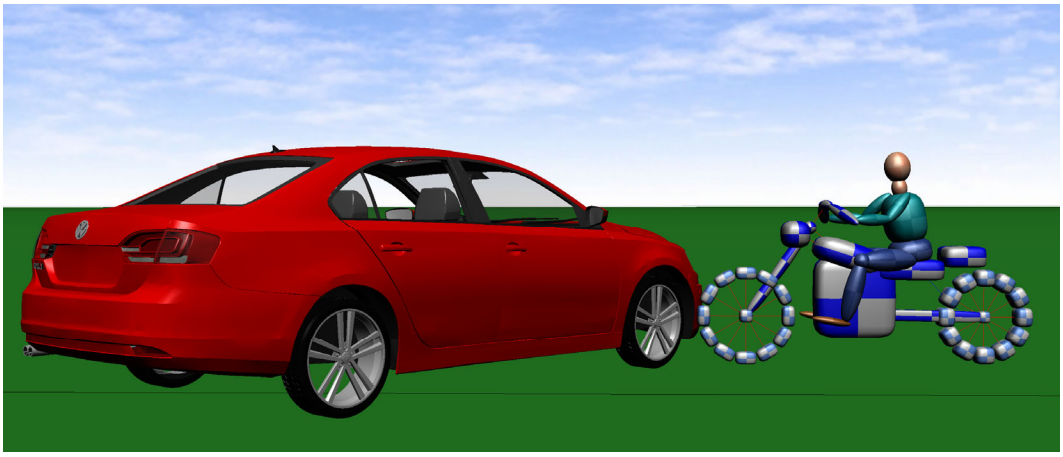
Multibody Simulation



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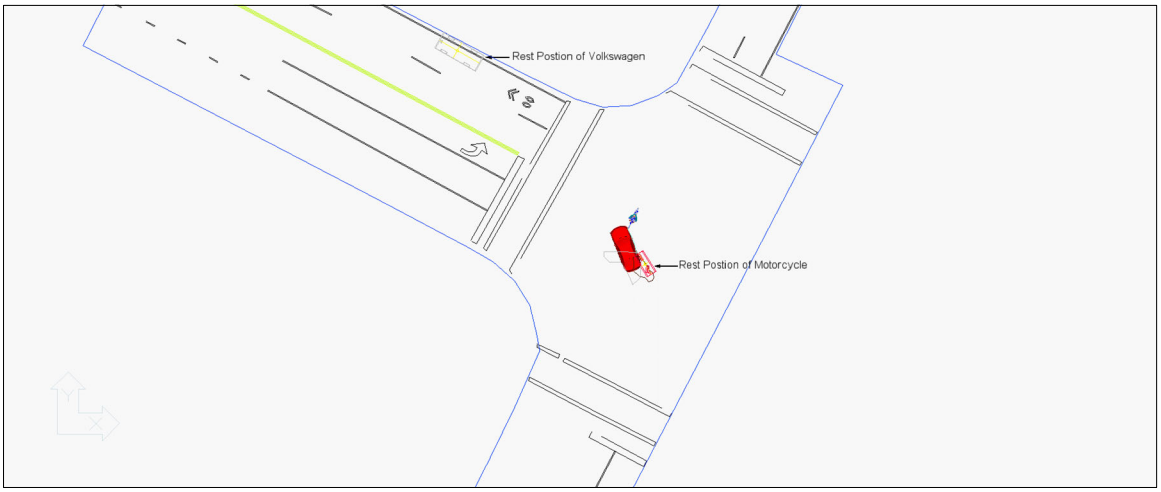

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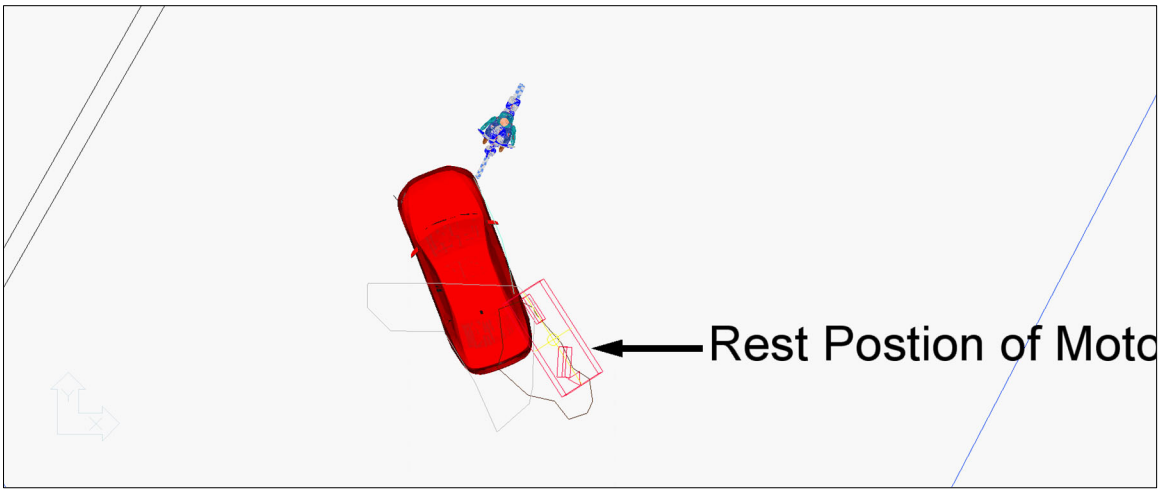

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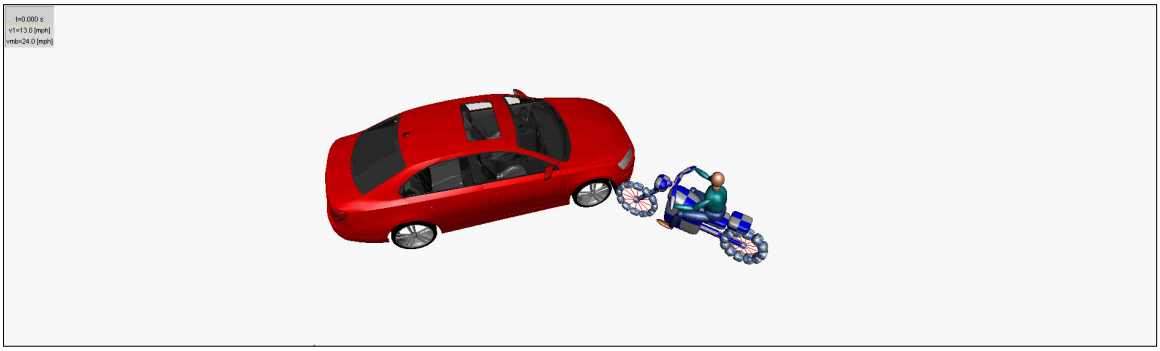

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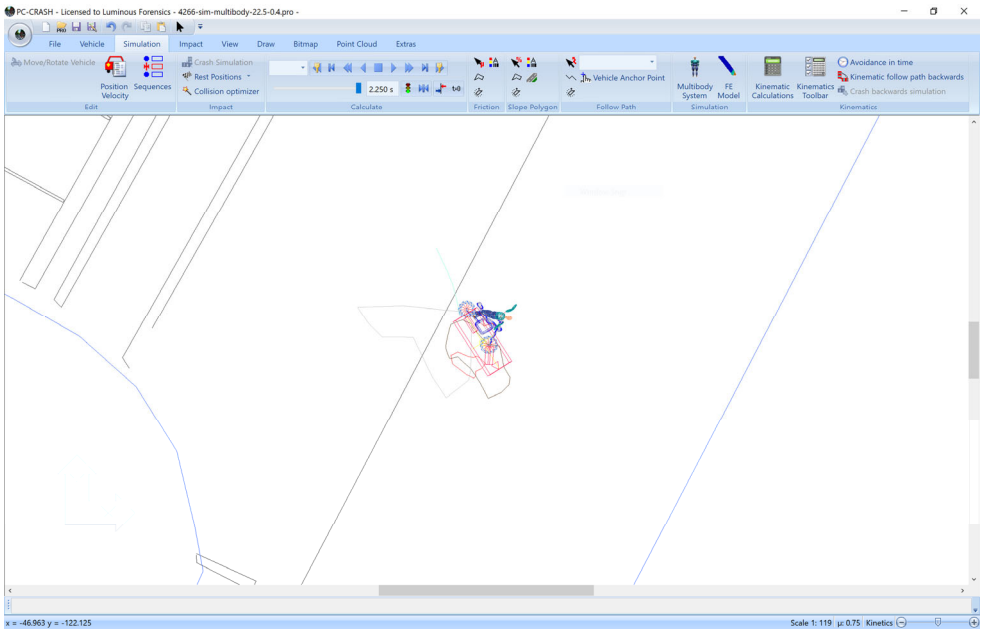
Multibody Simulation



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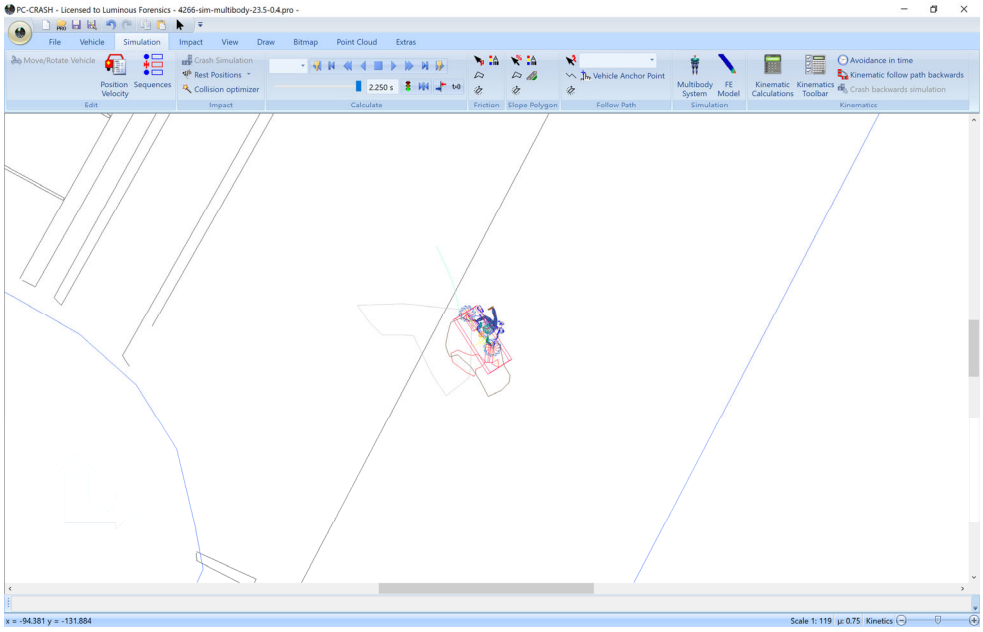

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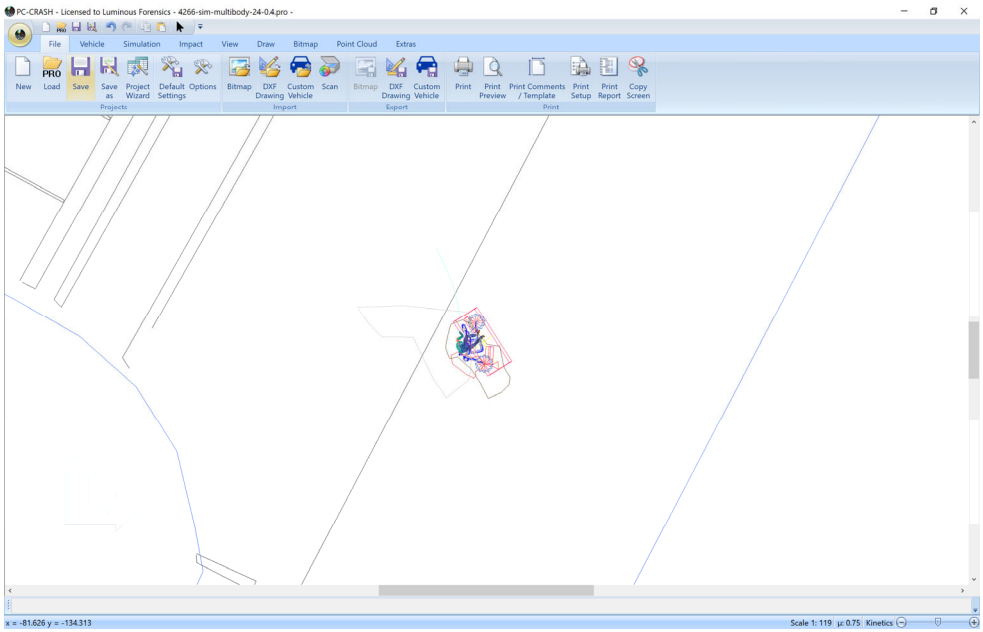

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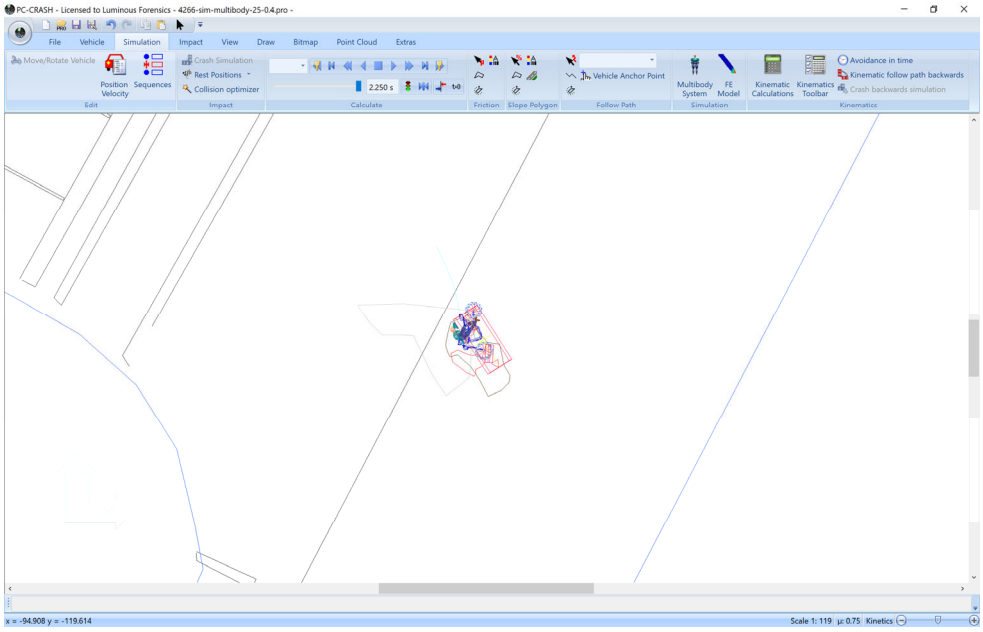

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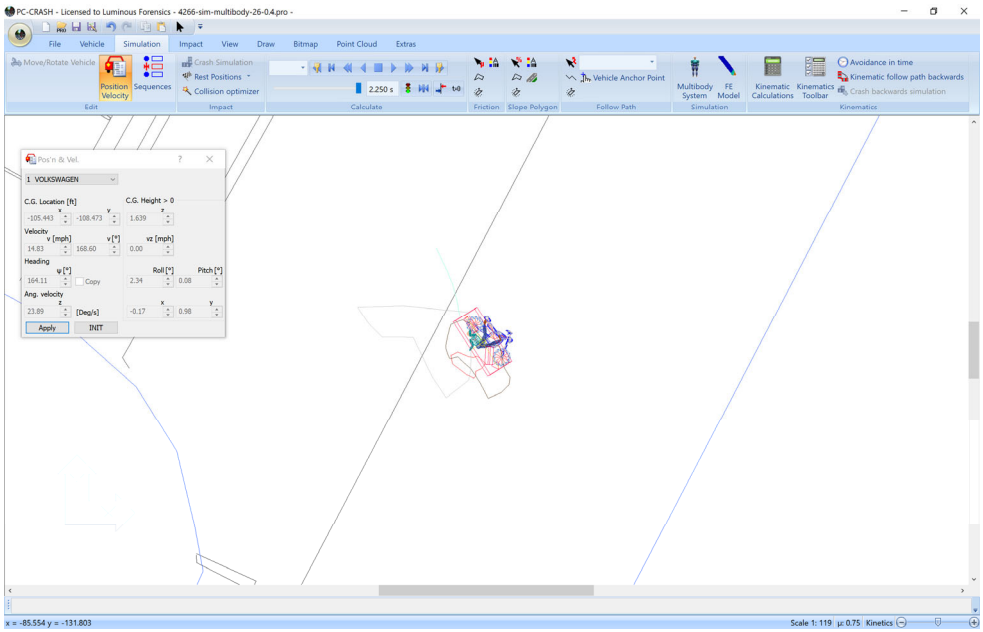

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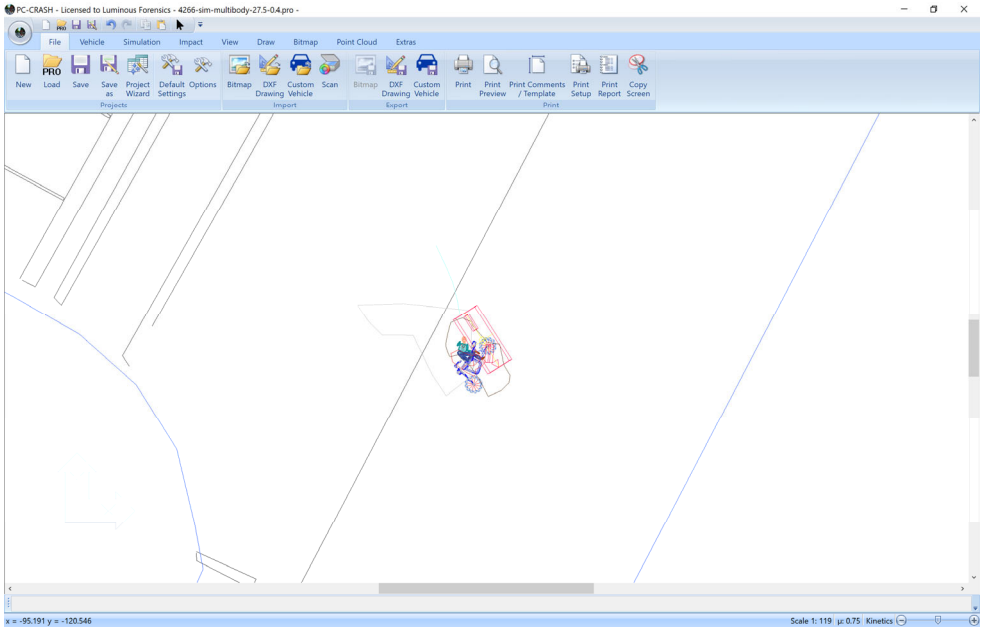

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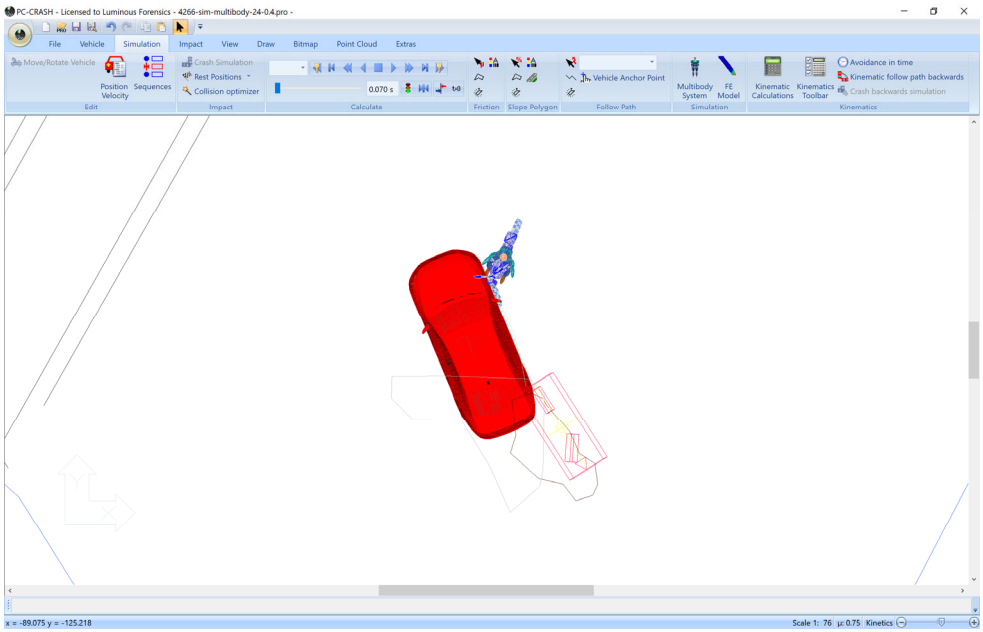

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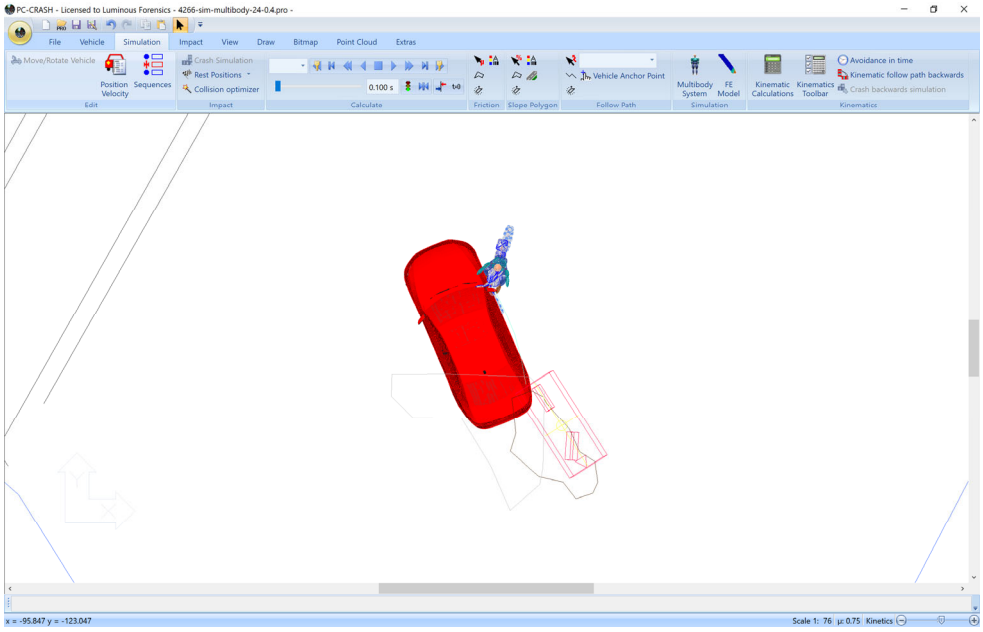

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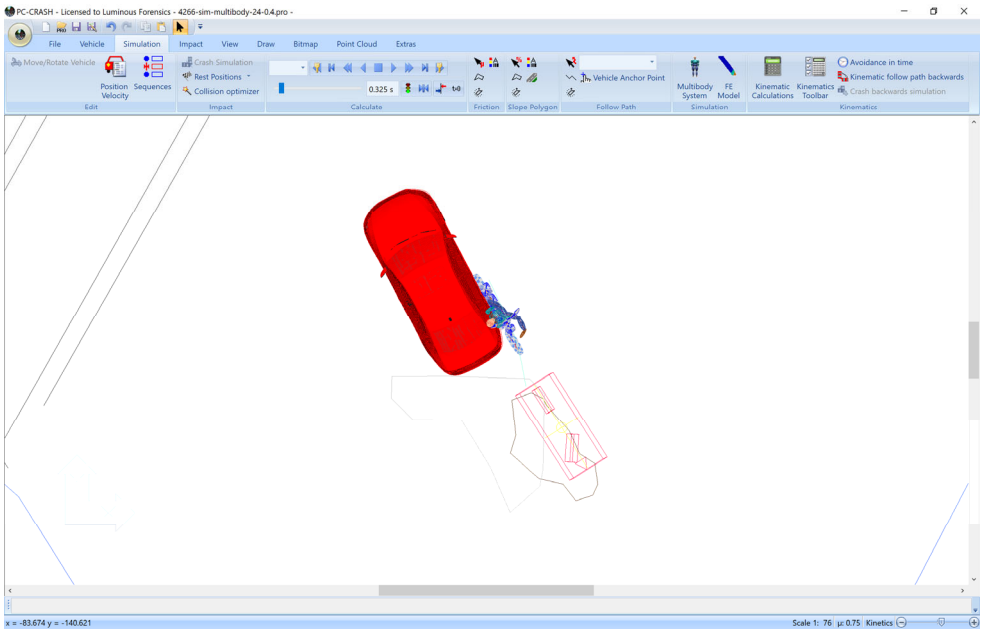

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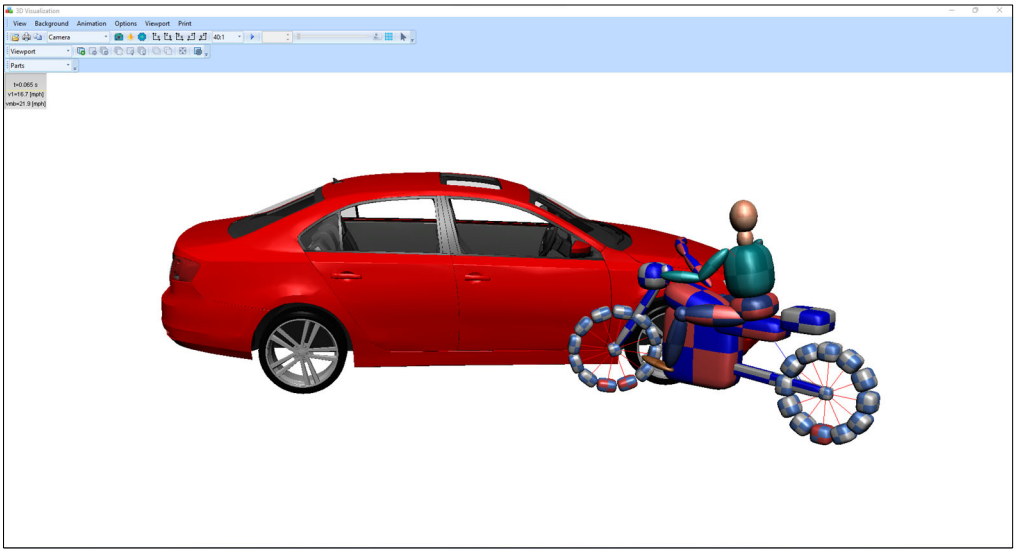

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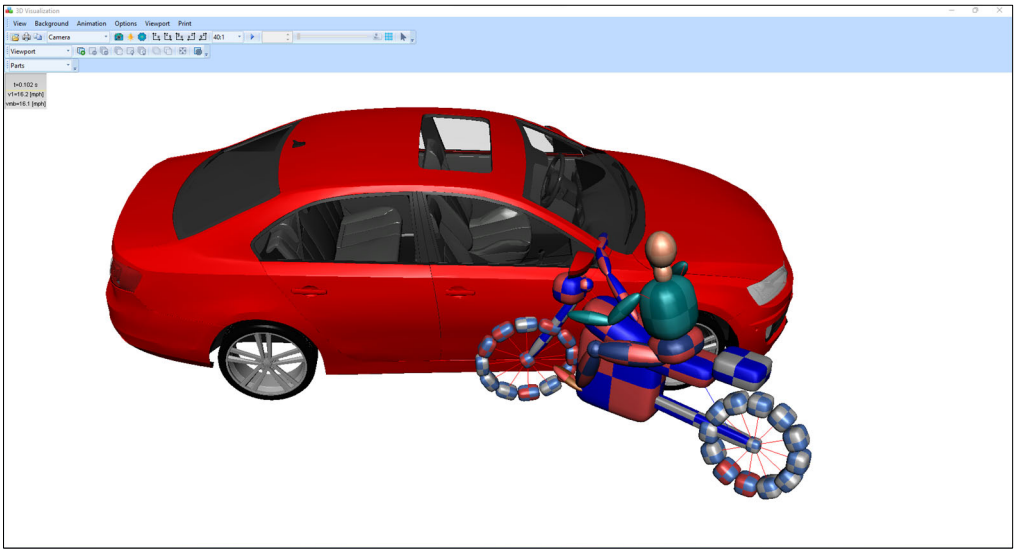

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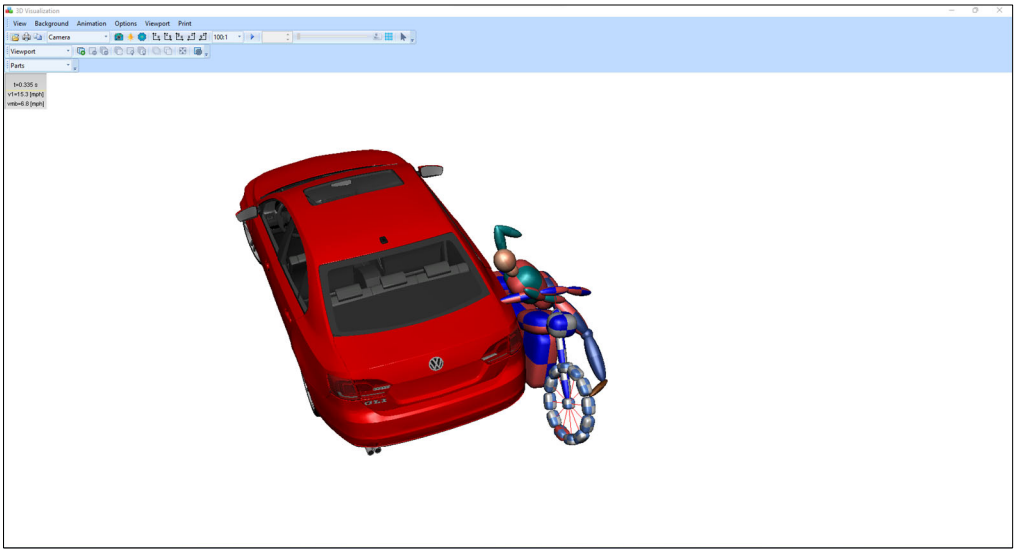

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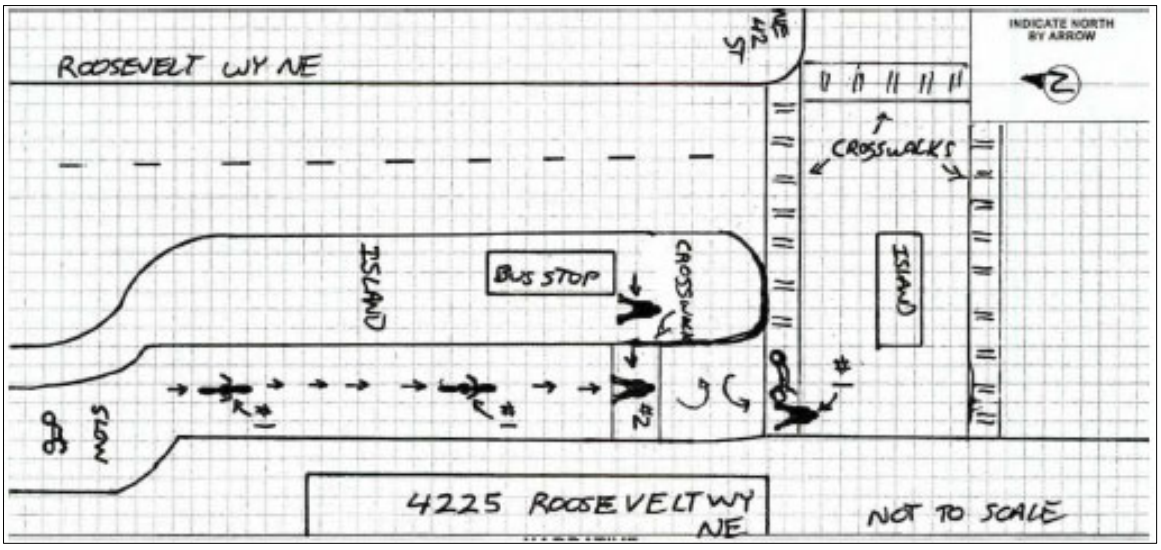

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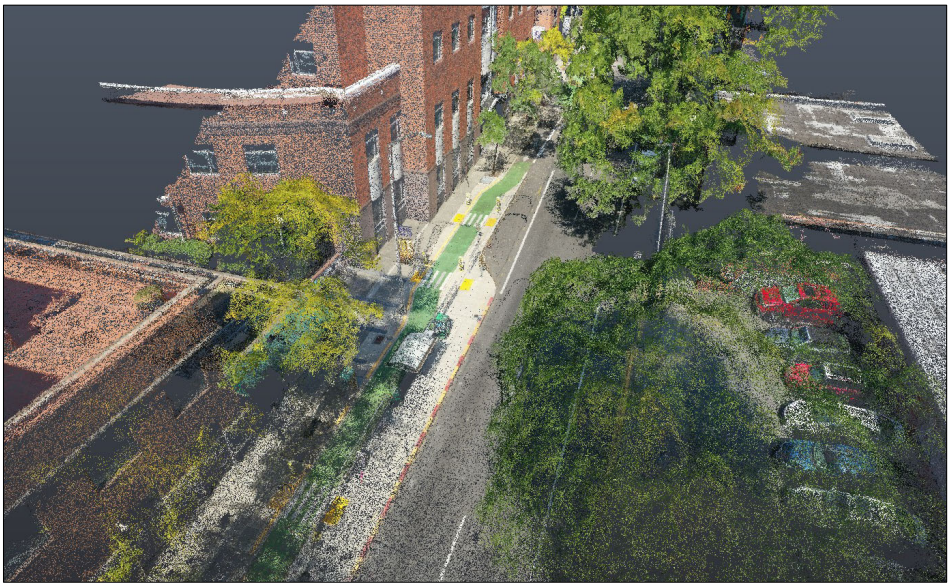

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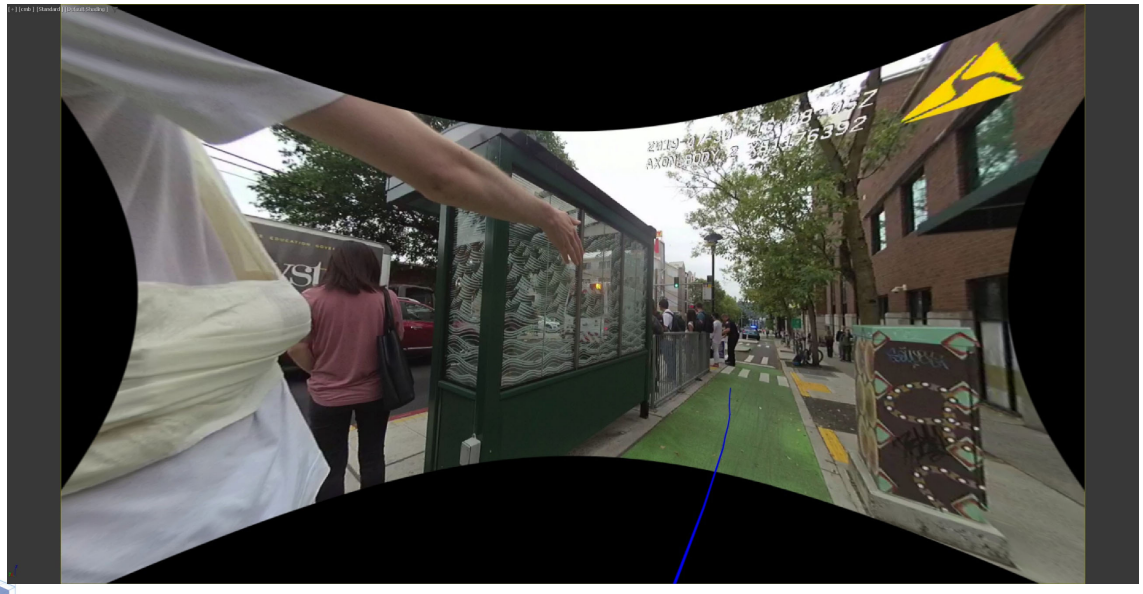

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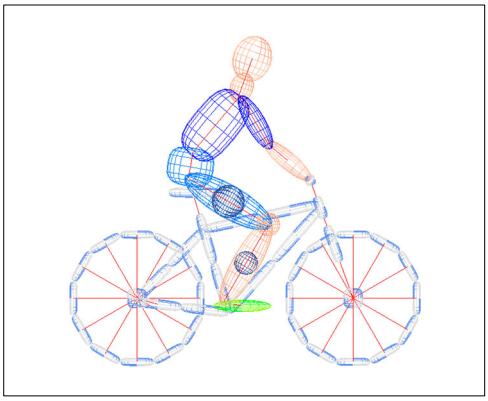

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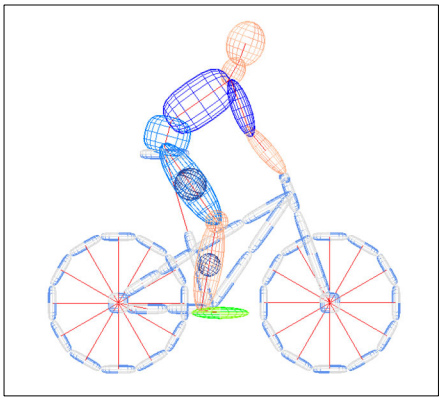

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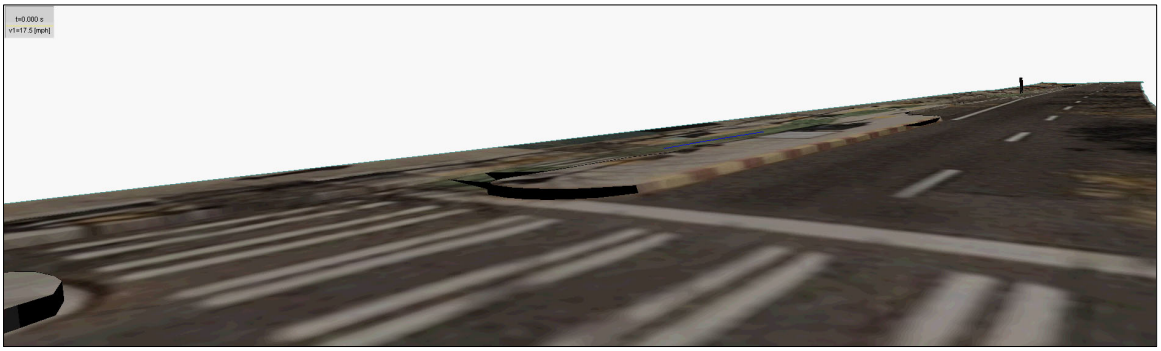

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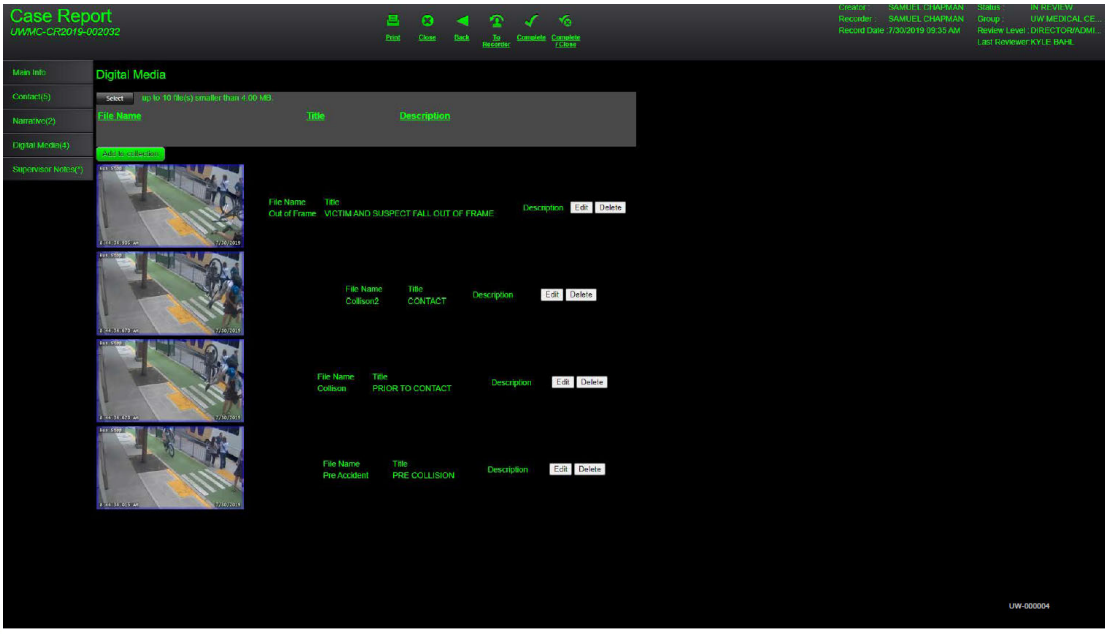

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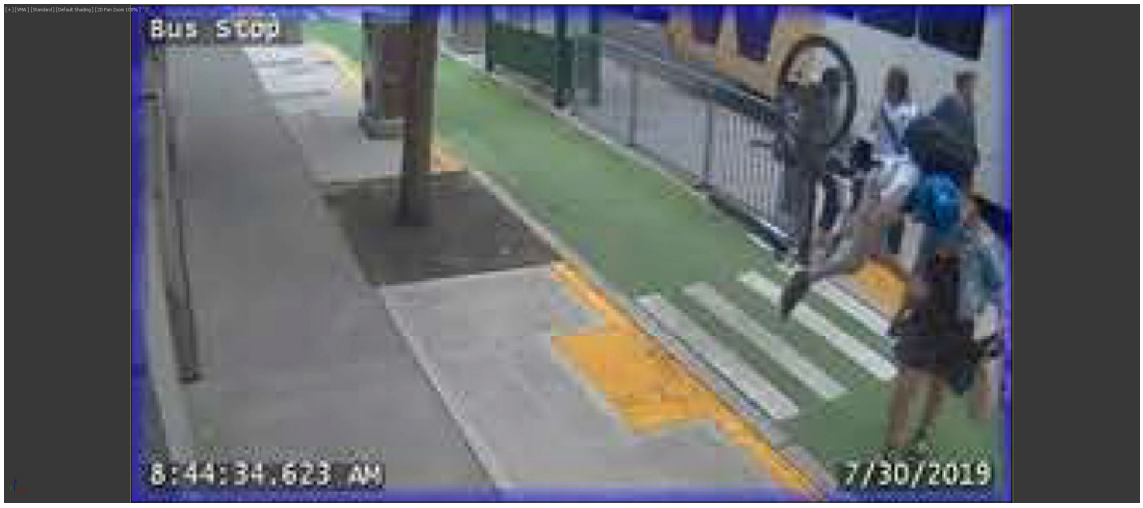

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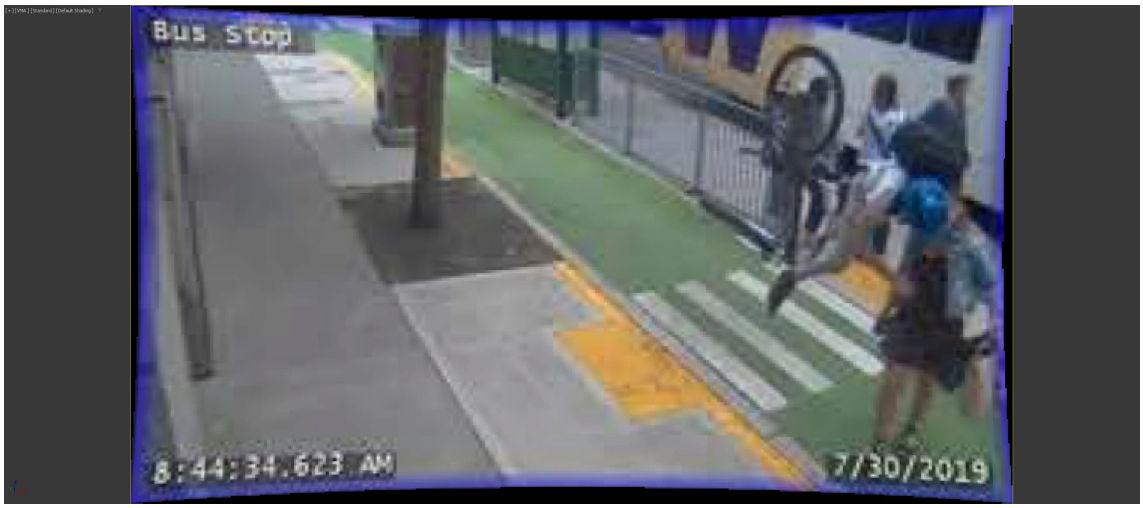

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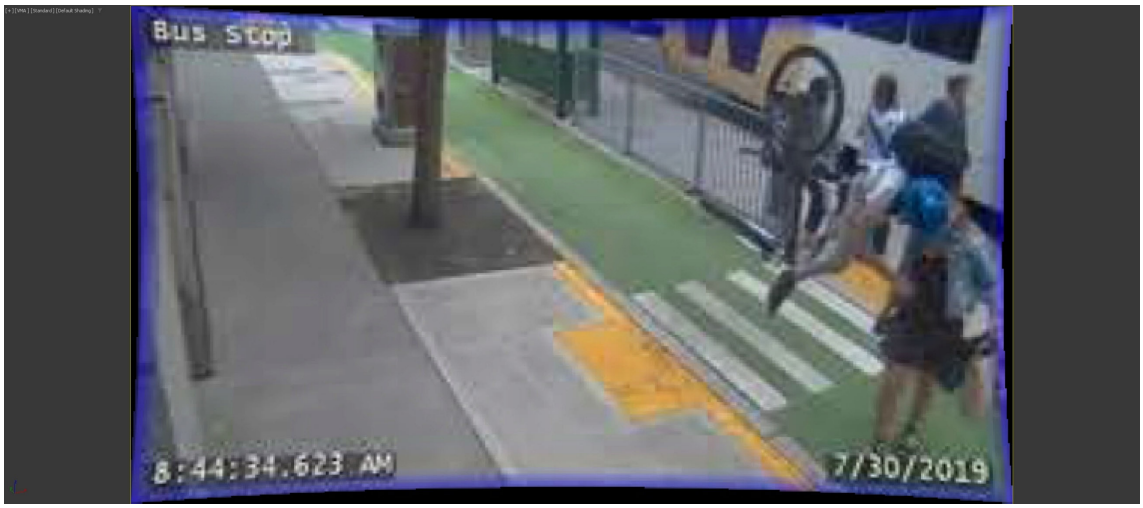

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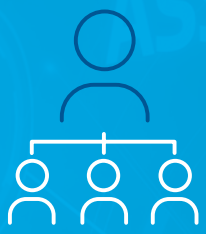

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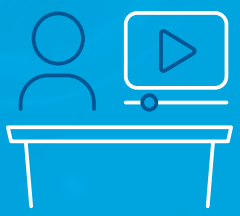

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